THROUGH THE BULKHEAD - every wire that lands on the DRB102 (signals, sensor rails, feeds) (part 1 of 4)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

ECU-A - Link G4X plug A (centre dash)

housing 4-1437290-1 | contacts 3-1447221-4 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w111_c	20	White		Injector 4 drive (-)	DRB102 receptacle (RH firewall) pin 24	630	ST185-AB-cabin	INJ4
2	w110_c	20	White		Injector 3 drive (-)	DRB102 receptacle (RH firewall) pin 23	630	ST185-AB-cabin	INJ3
3	w109_c	20	White		Injector 2 drive (-)	DRB102 receptacle (RH firewall) pin 22	630	ST185-AB-cabin	INJ2
4	w108_c	20	White		Injector 1 drive (-)	DRB102 receptacle (RH firewall) pin 21	630	ST185-AB-cabin	INJ1
8	w200_c	20	White		Crank sensor signal	DRB102 receptacle (RH firewall) pin 74	630	ST185-AB-cabin	CRANKC
10	w123_c	20	White		Coil 4 trigger (IGT)	DRB102 receptacle (RH firewall) pin 12	630	ST185-AB-cabin	COP4
11	w122_c	20	White		Coil 3 trigger (IGT)	DRB102 receptacle (RH firewall) pin 11	630	ST185-AB-cabin	COP3
12	w121_c	20	White		Coil 2 trigger (IGT)	DRB102 receptacle (RH firewall) pin 10	630	ST185-AB-cabin	COP2
13	w120_c	20	White		Coil 1 trigger (IGT)	DRB102 receptacle (RH firewall) pin 9	630	ST185-AB-cabin	COP1
15	w201_c	20	White		Coolant temp signal	DRB102 receptacle (RH firewall) pin 70	630	ST185-AB-cabin	ECT
16	w202_c	20	White		Intake temp signal	DRB102 receptacle (RH firewall) pin 33	630	ST185-AB-cabin	IATC
17	w26_c	20	White		MAP sensor signal	DRB102 receptacle (RH firewall) pin 17	630	ST185-AB-cabin	MAPC
21	w103_c	20	White		Boost solenoid (-)	DRB102 receptacle (RH firewall) pin 58	630	ST185-AB-cabin	BOOST
22	w128_c	20	White		Throttle pos 1 main	DRB102 receptacle (RH firewall) pin 42	630	ST185-AB-cabin	ETBS
27	w_mrs_c	20	White		EHPS pump speed	DRB102 receptacle (RH firewall) pin 84	630	ST185-AB-cabin	MRSSPD
30	w16_c	20	White		Turbo speed signal	DRB102 receptacle (RH firewall) pin 53	630	ST185-AB-cabin	TURBO
31	w48_c	20	White		Flex fuel signal	DRB102 receptacle (RH firewall) pin 31	630	ST185-AB-cabin	FLEX
33	w129_c	20	White		Throttle pos 2 sub	DRB102 receptacle (RH firewall) pin 44	630	ST185-AB-cabin	ETBS

ECU-B (Link G4X plug B, centre dash)

housing 4-1437290-0 | contacts 3-1447221-4 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
6	w203_c	20	White		Oil temp signal	DRB102 receptacle (RH firewall) pin 61	630	ST185-AB-cabin	OILT
7	w204_c	20	White		Charge temp signal	DRB102 receptacle (RH firewall) pin 59	630	ST185-AB-cabin	CIAT
9	w205_c	20	White		Knock sensor signal	DRB102 receptacle (RH firewall) pin 71	630	ST185-AB-cabin	KNOCK
15	w29_c	20	White		Oil pressure signal	DRB102 receptacle (RH firewall) pin 28	630	ST185-AB-cabin	OILPC
16	w32_c	20	White		Fuel pressure signal	DRB102 receptacle (RH firewall) pin 14	630	ST185-AB-cabin	FUELP
18	w124_c	18	White		ETB motor +	DRB102 receptacle (RH firewall) pin 45	630	ST185-AB-cabin	ETBM
23	w35_c	20	White		Coolant press signal	DRB102 receptacle (RH firewall) pin 26	630	ST185-AB-cabin	CLNTP
26	w215_c	18	White		ETB motor -	DRB102 receptacle (RH firewall) pin 46	630	ST185-AB-cabin	ETBM
29	w_vss_sig_c	20	White		Gearbox speed signal	DRB102 receptacle (RH firewall) pin 68	630	ST185-AB-cabin	VSS

BUS-5V - DT06-12 plug (ECU splice point)

housing DT06-12SA + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_rail_sp_5v_c	20	Orange		+5 V throttle sensor	DRB102 receptacle (RH firewall) pin 41	520	ST185-AB-cabin	5V
3	w_5v_a_c39	20	Orange		+5 V MAP sensor	DRB102 receptacle (RH firewall) pin 16	520	ST185-AB-cabin	5V
4	w_5v_a_c40	20	Orange		+5 V turbo speed	DRB102 receptacle (RH firewall) pin 52	520	ST185-AB-cabin	5V
5	w_5v_brail_c45	20	Orange		+5 V oil pressure	DRB102 receptacle (RH firewall) pin 27	520	ST185-AB-cabin	5V
6	w_5v_brail_c46	20	Orange		+5 V fuel pressure	DRB102 receptacle (RH firewall) pin 13	520	ST185-AB-cabin	5V
7	w_5v_brail_c47	20	Orange		+5 V coolant press	DRB102 receptacle (RH firewall) pin 25	520	ST185-AB-cabin	5V

BUS-GNDOUT-A - DT06-12 plug (ECU splice point)

housing DT06-12SA + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_rail_sp_gndout_c	20	Blue		Gnd IAT/TPS/coolant	DRB102 receptacle (RH firewall) pin 34	520	ST185-AB-cabin	SGND
3	w_gnd_a_c41	20	Blue		Gnd crank sensor	DRB102 receptacle (RH firewall) pin 75	520	ST185-AB-cabin	SGND
4	w_gnd_a_c42	20	Blue		Gnd cam sensor	DRB102 receptacle (RH firewall) pin 65	520	ST185-AB-cabin	SGND
5	w_gnd_a_c43	20	Blue		Gnd turbo speed	DRB102 receptacle (RH firewall) pin 56	520	ST185-AB-cabin	SGND
6	w_gnd_a_c44	20	Blue		Gnd flex + MAP	DRB102 receptacle (RH firewall) pin 18	520	ST185-AB-cabin	SGND

BUS-GNDOUT-B - DT06-12 plug (ECU splice point)

housing DT06-12SA + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
6	w12_c	20	Blue		Knock sensor (-)	DRB102 receptacle (RH firewall) pin 72	520	ST185-AB-cabin	KNOCK
7	w_gnd_b_c19	20	Blue		Gnd oil/clnt/temps	DRB102 receptacle (RH firewall) pin 29	520	ST185-AB-cabin	SGND
8	w_gnd_b_c21	20	Blue		Gnd fuel press + VSS	DRB102 receptacle (RH firewall) pin 15	520	ST185-AB-cabin	SGND

REVISIONS

REV	DATE	BY	DESCRIPTION
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J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

THROUGH THE BULKHEAD - every wire that lands on the DRB102 (signals, sensor rails, feeds) (part 1 of 4)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	1 OF 51	432 x 279 mm	DG	2026-10-29

THROUGH THE BULKHEAD - every wire that lands on the DRB102 (signals, sensor rails, feeds) (part 2 of 4)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

SP-8V - ECU splice point - SP-8V: +8 V cam sensor supply (ECU A6)

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_rail_sp_8v_c	20	Orange		+8 V to cam sensor	DRB102 receptacle (RH firewall) pin 63	520	ST185-AB-cabin	8V

SP-CAM - At the ECU plugs - SP-CAM: cam signal + 1.8k pull-up to +8V

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_cam_sig_c	20	White		Cam sensor signal	DRB102 receptacle (RH firewall) pin 64	520	ST185-AB-cabin	CAM

SP-CANH - CAN H join - ECU, CSB3, cluster, Pi, bulkhead

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	wc_h_bh_bc	20	White		CAN H to CAN-Lambda	DRB102 receptacle (RH firewall) pin 79	680	ST185-AB-cabin	CAN

SP-CANL - CAN L join - ECU, CSB3, cluster, Pi, bulkhead

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	wc_l_bh_bc	20	Green		CAN L to CAN-Lambda	DRB102 receptacle (RH firewall) pin 80	680	ST185-AB-cabin	CAN

SP-REV - reverse switch out to CSB3 opto IN2 and IG2-13 lamps

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_rev_sw_c	16	Yellow		Reverse sw +12 V out	DRB102 receptacle (RH firewall) pin 78	520	ST185-AB-cabin	REV

SP-SW12 - ECU splice point - SP-SW12: switched +12 V (LOGIC SW12)

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_rail_sp_sw12_c	18	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	DRB102 receptacle (RH firewall) pin 30	520	ST185-AB-cabin	SW12
	w_rail_sw12_b_c_ac	20	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	DRB102 receptacle (RH firewall) pin 67	520	ST185-AB-cabin	SW12

REVISIONS

REV	DATE	BY	DESCRIPTION
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ST185-harness - every wiring harness of the car, one drawing set

THROUGH THE BULKHEAD - every wire that lands on the DRB102 (signals, sensor rails, feeds) (part 2 of 4)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	2 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

THROUGH THE BULKHEAD - every wire that lands on the DRB102 (signals, sensor rails, feeds) (part 3 of 4)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

DRB102-CABIN - DRB102 receptacle (RH firewall)

housing DRB12-102PAE-L018 + DRBF-1A | contacts 0460-220-1231 | pins per TE drawing DRB12-102PXE-L018 rev C3

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
4	w_cop_gnd_ac	14	Black		Coil ground to the POWER box NEG bar	POWER box NEG bar M8 stud (glove box)	450	ST185-AB-cabin	COPG
9	w120_c	20	White		Coil 1 trigger (IGT)	Link G4X plug A (centre dash) pin 13	630	ST185-AB-cabin	COP1
10	w121_c	20	White		Coil 2 trigger (IGT)	Link G4X plug A (centre dash) pin 12	630	ST185-AB-cabin	COP2
11	w122_c	20	White		Coil 3 trigger (IGT)	Link G4X plug A (centre dash) pin 11	630	ST185-AB-cabin	COP3
12	w123_c	20	White		Coil 4 trigger (IGT)	Link G4X plug A (centre dash) pin 10	630	ST185-AB-cabin	COP4
13	w_5v_brail_c46	20	Orange		+5 V fuel pressure	DT06-12 plug (ECU splice point) pin 6	520	ST185-AB-cabin	5V
14	w32_c	20	White		Fuel pressure signal	ECU-B (Link G4X plug B, centre dash) pin 16	630	ST185-AB-cabin	FUELP
15	w_gnd_b_c21	20	Blue		Gnd fuel press + VSS	DT06-12 plug (ECU splice point) pin 8	520	ST185-AB-cabin	SGND
16	w_5v_a_c39	20	Orange		+5 V MAP sensor	DT06-12 plug (ECU splice point) pin 3	520	ST185-AB-cabin	5V
17	w26_c	20	White		MAP sensor signal	Link G4X plug A (centre dash) pin 17	630	ST185-AB-cabin	MAPC
18	w_gnd_a_c44	20	Blue		Gnd flex + MAP	DT06-12 plug (ECU splice point) pin 6	520	ST185-AB-cabin	SGND
21	w108_c	20	White		Injector 1 drive (-)	Link G4X plug A (centre dash) pin 4	630	ST185-AB-cabin	INJ1
22	w109_c	20	White		Injector 2 drive (-)	Link G4X plug A (centre dash) pin 3	630	ST185-AB-cabin	INJ2
23	w110_c	20	White		Injector 3 drive (-)	Link G4X plug A (centre dash) pin 2	630	ST185-AB-cabin	INJ3
24	w111_c	20	White		Injector 4 drive (-)	Link G4X plug A (centre dash) pin 1	630	ST185-AB-cabin	INJ4
25	w_5v_brail_c47	20	Orange		+5 V coolant press	DT06-12 plug (ECU splice point) pin 7	520	ST185-AB-cabin	5V
26	w35_c	20	White		Coolant press signal	ECU-B (Link G4X plug B, centre dash) pin 23	630	ST185-AB-cabin	CLNTP
27	w_5v_brail_c45	20	Orange		+5 V oil pressure	DT06-12 plug (ECU splice point) pin 5	520	ST185-AB-cabin	5V
28	w29_c	20	White		Oil pressure signal	ECU-B (Link G4X plug B, centre dash) pin 15	630	ST185-AB-cabin	OILPC
29	w_gnd_b_c19	20	Blue		Gnd oil/clnt/temps	DT06-12 plug (ECU splice point) pin 7	520	ST185-AB-cabin	SGND
30	w_rail_sp_sw12_c	18	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	ECU splice point - SP-SW12: switched +12 V (LOGIC SW12)	520	ST185-AB-cabin	SW12
31	w48_c	20	White		Flex fuel signal	Link G4X plug A (centre dash) pin 31	630	ST185-AB-cabin	FLEX
33	w202_c	20	White		Intake temp signal	Link G4X plug A (centre dash) pin 16	630	ST185-AB-cabin	IATC
34	w_rail_sp_gndout_c	20	Blue		Gnd IAT/TPS/coolant	DT06-12 plug (ECU splice point) pin 2	520	ST185-AB-cabin	SGND
41	w_rail_sp_5v_c	20	Orange		+5 V throttle sensor	DT06-12 plug (ECU splice point) pin 2	520	ST185-AB-cabin	5V
42	w128_c	20	White		Throttle pos 1 main	Link G4X plug A (centre dash) pin 22	630	ST185-AB-cabin	ETBS
44	w129_c	20	White		Throttle pos 2 sub	Link G4X plug A (centre dash) pin 33	630	ST185-AB-cabin	ETBS
45	w124_c	18	White		ETB motor +	ECU-B (Link G4X plug B, centre dash) pin 18	630	ST185-AB-cabin	ETBM
46	w215_c	18	White		ETB motor -	ECU-B (Link G4X plug B, centre dash) pin 26	630	ST185-AB-cabin	ETBM
52	w_5v_a_c40	20	Orange		+5 V turbo speed	DT06-12 plug (ECU splice point) pin 4	520	ST185-AB-cabin	5V
53	w16_c	20	White		Turbo speed signal	Link G4X plug A (centre dash) pin 30	630	ST185-AB-cabin	TURBO
56	w_gnd_a_c43	20	Blue		Gnd turbo speed	DT06-12 plug (ECU splice point) pin 5	520	ST185-AB-cabin	SGND
58	w103_c	20	White		Boost solenoid (-)	Link G4X plug A (centre dash) pin 21	630	ST185-AB-cabin	BOOST
59	w204_c	20	White		Charge temp signal	ECU-B (Link G4X plug B, centre dash) pin 7	630	ST185-AB-cabin	CIAT
61	w203_c	20	White		Oil temp signal	ECU-B (Link G4X plug B, centre dash) pin 6	630	ST185-AB-cabin	OILT
63	w_rail_sp_8v_c	20	Orange		+8 V to cam sensor	ECU splice point - SP-8V: +8 V cam sensor supply (ECU A6)	520	ST185-AB-cabin	8V
64	w_cam_sig_c	20	White		Cam sensor signal	At the ECU plugs - SP-CAM: cam signal + 1.8k pull-up to +8V	520	ST185-AB-cabin	CAM
65	w_gnd_a_c42	20	Blue		Gnd cam sensor	DT06-12 plug (ECU splice point) pin 4	520	ST185-AB-cabin	SGND
67	w_rail_sw12_b_c_ac	20	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	ECU splice point - SP-SW12: switched +12 V (LOGIC SW12)	520	ST185-AB-cabin	SW12
68	w_vss_sig_c	20	White		Gearbox speed signal	ECU-B (Link G4X plug B, centre dash) pin 29	630	ST185-AB-cabin	VSS
70	w201_c	20	White		Coolant temp signal	Link G4X plug A (centre dash) pin 15	630	ST185-AB-cabin	ECT
71	w205_c	20	White		Knock sensor signal	ECU-B (Link G4X plug B, centre dash) pin 9	630	ST185-AB-cabin	KNOCK
72	w12_c	20	Blue		Knock sensor (-)	DT06-12 plug (ECU splice point) pin 6	520	ST185-AB-cabin	KNOCK
74	w200_c	20	White		Crank sensor signal	Link G4X plug A (centre dash) pin 8	630	ST185-AB-cabin	CRANKC
75	w_gnd_a_c41	20	Blue		Gnd crank sensor	DT06-12 plug (ECU splice point) pin 3	520	ST185-AB-cabin	SGND
77	w_rev_feed_c	16	Red		Back-up sw +12 V in	OEM IG2 pin 4 (under the ECU, cowl side) - factory back-up SW feed pin 4	630	ST185-AB-cabin	REV
78	w_rev_sw_c	16	Yellow		Reverse sw +12 V out	SP-REV - reverse switch out to CSB3 opto IN2 and IG2-13 lamps	520	ST185-AB-cabin	REV
79	wc_h_bh_bc	20	White		CAN H to CAN-Lambda	CAN H join - ECU, CSB3, cluster, Pi, bulkhead	680	ST185-AB-cabin	CAN
80	wc_l_bh_bc	20	Green		CAN L to CAN-Lambda	CAN L join - ECU, CSB3, cluster, Pi, bulkhead	680	ST185-AB-cabin	CAN
83	w_fb_mrs_en	20	Red		EFI relay k_efi -> LOGIC MRS 7.5 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	450	ST185-AB-cabin	EPSEN
84	w_mrs_c	20	White		EHPS pump speed	Link G4X plug A (centre dash) pin 27	630	ST185-AB-cabin	MRSSPD
85	w_eps_rly	18	White		EHPS relay coil (-)	B4 EPS pump relay (HCR150, glove box, beside the POWER box) - coil- (85) switched to gro...	450	ST185-AB-cabin	EPS
96	w_lamg_ac	14	Black		CAN-Lambda ground	POWER box NEG bar M8 stud (glove box)	450	ST185-AB-cabin	LAM

K-EPS - B4 EPS pump relay (HCR150, glove box, beside the POWER box) - coil- (85) switched to ground by the MR-S pump pin B5 (R22); coil+ (86) switched 12 V - HCR 150 coil is a 2-pole plug (TE terminals 1/2, not marked 85/86), res...

housing V23132-A2001-B200 | contacts 282110-1 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
85	w_eps_rly	18	White		EHPS relay coil (-)	DRB102 receptacle (RH firewall) pin 85	450	ST185-AB-cabin	EPS

REVISIONS

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THROUGH THE BULKHEAD - every wire that lands on the DRB102 (signals, sensor rails, feeds) (part 3 of 4)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	3 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

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LOGIC - B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS R75): Bus 1 (permanent) and Bus 2 (switched, k_efi 30) mini fuse strips, micro relays M1-M10

housing ST185-FB-LOGIC rev 5 | contacts 1-968849-1 | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
BUS2-12	w_fb_mrs_en	20	Red		EFI relay k_efi -> LOGIC MRS 7.5 A fuse	DRB102 receptacle (RH firewall) pin 83	450	ST185-AB-cabin	EPSEN

POWER-NEG-BAR - POWER box NEG bar M8 stud (glove box)

terminal 320575

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_lamg_ac	14	Black		CAN-Lambda ground	DRB102 receptacle (RH firewall) pin 96	450	ST185-AB-cabin	LAM
	w_cop_gnd_ac	14	Black		Coil ground	DRB102 receptacle (RH firewall) pin 4	450	ST185-AB-cabin	COPG

OEM IG2 pin 4 (under the ECU, cowl side) - factory back-up SW feed

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
4	w_rev_feed_c	16	Red		Back-up sw +12 V in	DRB102 receptacle (RH firewall) pin 77	630	ST185-AB-cabin	REV

REVISIONS

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L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

THROUGH THE BULKHEAD - every wire that lands on the DRB102 (signals, sensor rails, feeds) (part 4 of 4)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	4 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

ECU SPLICE POINT - power, grounds and sensor rails (bussed DT plugs, soldered splices, NEG bar) (part 1 of 3)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

ECU-A - Link G4X plug A (centre dash)

housing 4-1437290-1 | contacts 3-1447221-3 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
5	w_a5_pin	18	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse (+ IG2 wak...	ECU splice point - SP-A5: ECU main feed (SW12 + IG2 diode)	310	ST185-AB-cabin	SW12
6	w19	20	Orange		+8 V sensor supply	ECU splice point - SP-8V: +8 V cam sensor supply (ECU A6)	310	ST185-AB-cabin	8V
24	w68	18	Blue		Sensor ground	DT06-12 plug (ECU splice point) pin 1	310	ST185-AB-cabin	SGND
25	w70	18	Black		ECU power ground	ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar	310	ST185-AB-cabin	GND
32	w67	18	Orange		+5 V sensor supply	DT06-12 plug (ECU splice point) pin 1	310	ST185-AB-cabin	5V
34	w_a34	18	Black		ECU power ground	ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar	310	ST185-AB-cabin	GND

ECU-B (Link G4X plug B, centre dash)

housing 4-1437290-0 | contacts 3-1447221-3 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
22	w69	18	Blue		Sensor ground B	DT06-12 plug (ECU splice point) pin 1	310	ST185-AB-cabin	SGND
25	w_b25_ac	18	Black		ECU power ground	ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar	310	ST185-AB-cabin	GND
34	w_b34_ac	18	Black		ECU power ground	ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar	310	ST185-AB-cabin	GND

BUS-5V - DT06-12 plug (ECU splice point)

housing DT06-12SA + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w67	18	Orange		+5 V from ECU A32	Link G4X plug A (centre dash) pin 32	310	ST185-AB-cabin	5V
8	w_aps_vc1	20	Red		+5 V pedal sensor 1	accelerator pedal sensor pin 4	1200	ST185-AB-cabin	5V
9	w210	20	Red		+5 V pedal sensor 2	accelerator pedal sensor pin 1	1200	ST185-AB-cabin	5V
10	cab_fspur_5v	20	Red		+5 V front VRC	to front VR conditioner pin 1	640	ST185-AB-cabin	5V
11	cab_rspur_5v	20	Red		+5 V rear VRC	IX-WS-R inline to rear VRC (dash above ECU) pin 1	640	ST185-AB-cabin	5V
12	w_5v_bus_link	20	Orange		+5 V to BUS-5V-2	DT06-8 plug (ECU splice point) pin 1	200	ST185-AB-cabin	5V

BUS-5V-2 - DT06-8 plug (ECU splice point)

housing DT06-08SA + W8S; DT04-08PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_5v_bus_link	20	Orange		+5 V from BUS-5V	DT06-12 plug (ECU splice point) pin 12	200	ST185-AB-cabin	5V
2	w_cru_5v_ac	20	Orange		+5 V cruise ladder	resistor r_cruise on CSB3 pin 1	680	ST185-AB-cabin	5V
4	w44	20	Orange		+5 V fuel lvl pullup	470R fuel level pull-up to +5V pin 2	200	ST185-AB-cabin	FLVL

BUS-GNDOUT-A - DT06-12 plug (ECU splice point)

housing DT06-12SA + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w68	18	Blue		Sensor gnd from A24	Link G4X plug A (centre dash) pin 24	310	ST185-AB-cabin	SGND
7	w40	20	Yellow		Pedal 1 ground	accelerator pedal sensor pin 5	1200	ST185-AB-cabin	SGND
8	w_aps_gnd2	20	Yellow		Pedal 2 ground	accelerator pedal sensor pin 2	1200	ST185-AB-cabin	SGND
9	cab_fspur_gnd	20	Yellow		Front VRC ground	to front VR conditioner pin 2	640	ST185-AB-cabin	SGND

BUS-GNDOUT-B - DT06-12 plug (ECU splice point)

housing DT06-12SA + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w69	18	Blue		Sensor gnd from B22	ECU-B (Link G4X plug B, centre dash) pin 22	310	ST185-AB-cabin	SGND
2	w45	20	Blue		Fuel level ground	IX-FUEL inline to fuel sender lead (trunk end) pin 2	2820	ST185-AB-cabin	SGND
3	cab_rspur_gnd	20	Yellow		Rear VRC ground	IX-WS-R inline to rear VRC (dash above ECU) pin 2	640	ST185-AB-cabin	SGND

BUS-SW12 - SW12 bus plug DT06-12 (glove box, by LOGIC box)

housing DT06-12SA + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_fb_sw12	18	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	ECU splice point - SP-SW12: switched +12 V (LOGIC SW12)	570	ST185-AB-cabin	SW12

SP-8V - ECU splice point - SP-8V: +8 V cam sensor supply (ECU A6)

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_cam_pullup_8v	20	Orange		+8 V to cam pull-up	Cam pull-up 1.8 k to +8 V pin 1	200	ST185-AB-cabin	CAM
	w19	20	Orange		+8 V sensor supply	Link G4X plug A (centre dash) pin 6	310	ST185-AB-cabin	8V

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

ECU SPLICE POINT - power, grounds and sensor rails (bussed DT plugs, soldered splices, NEG bar) (part 1 of 3)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	**L**	**5 OF 51**	**432 x 279 mm**	**DG**	**2026-10-29**

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

ECU SPLICE POINT - power, grounds and sensor rails (bussed DT plugs, soldered splices, NEG bar) (part 2 of 3)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

SP-A5 - ECU splice point - SP-A5: ECU main feed (SW12 + IG2 diode)

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w78	18	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse (+ IG2 wak...	ECU splice point - SP-SW12: switched +12 V (LOGIC SW12)	200	ST185-AB-cabin	SW12
	w_ig2_diode	20	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse (+ IG2 wak...	IG2 wake diode (1N5822) pin 2	200	ST185-AB-cabin	SW12
	w_a5_pin	18	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse (+ IG2 wak...	Link G4X plug A (centre dash) pin 5	310	ST185-AB-cabin	SW12

SP-CSB-GND - CSB3 ground splice - CSB3 pin 2, opto board COM and stalk ground

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_csbg_ac	20	Black		CSB3 + opto ground	POWER box NEG bar M8 stud (glove box)	1050	ST185-AB-cabin	CSB

SP-GND-ECU - ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w70	18	Black		ECU power ground	Link G4X plug A (centre dash) pin 25	310	ST185-AB-cabin	GND
	w_a34	18	Black		ECU power ground	Link G4X plug A (centre dash) pin 34	310	ST185-AB-cabin	GND
	w_b25_ac	18	Black		ECU power ground	ECU-B (Link G4X plug B, centre dash) pin 25	310	ST185-AB-cabin	GND
	w_b34_ac	18	Black		ECU power ground	ECU-B (Link G4X plug B, centre dash) pin 34	310	ST185-AB-cabin	GND
	w66	10	Black		ECU power ground	POWER box NEG bar M8 stud (glove box)	570	ST185-AB-cabin	GND

SP-SW12 - ECU splice point - SP-SW12: switched +12 V (LOGIC SW12)

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_fb_sw12	18	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	SW12 bus plug DT06-12 (glove box, by LOGIC box) pin 2	570	ST185-AB-cabin	SW12
	w78	18	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse (+ IG2 wak...	ECU splice point - SP-A5: ECU main feed (SW12 + IG2 diode)	200	ST185-AB-cabin	SW12

D-IG2-WAKE - IG2 wake diode (1N5822)

inline diode 1N5822

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_ig2_in	20	Yellow		IG2 wake feed	OEM J/B No.1 (LH kick panel) 1H pin 7, B-O pin 7	1210	ST185-AB-cabin	IG2WAKE
2	w_ig2_diode	20	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse (+ IG2 wak...	ECU splice point - SP-A5: ECU main feed (SW12 + IG2 diode)	200	ST185-AB-cabin	SW12

R-CAM-PULLUP - Cam pull-up 1.8 k to +8 V

inline resistor 1.8k 1/4W

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_cam_pullup_8v	20	Orange		+8 V to cam pull-up	ECU splice point - SP-8V: +8 V cam sensor supply (ECU A6)	200	ST185-AB-cabin	CAM

R-CRUISE - resistor r_cruise on CSB3

inline resistor 1k 1/4W

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_cru_5v_ac	20	Orange		+5 V cruise ladder	DT06-8 plug (ECU splice point) pin 2	680	ST185-AB-cabin	5V

R-FUELLVL - 470R fuel level pull-up to +5V

inline resistor 470R 1/4W

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w44	20	Orange		+5 V fuel lvl pullup	DT06-8 plug (ECU splice point) pin 4	200	ST185-AB-cabin	FLVL

APS-PEDAL - accelerator pedal sensor

housing TBD | contacts TBD | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w210	20	Red		+5 V pedal sensor 2	DT06-12 plug (ECU splice point) pin 9	1200	ST185-AB-cabin	5V
2	w_aps_gnd2	20	Yellow		Pedal 2 ground	DT06-12 plug (ECU splice point) pin 8	1200	ST185-AB-cabin	SGND
4	w_aps_vc1	20	Red		+5 V pedal sensor 1	DT06-12 plug (ECU splice point) pin 8	1200	ST185-AB-cabin	5V
5	w40	20	Yellow		Pedal 1 ground	DT06-12 plug (ECU splice point) pin 7	1200	ST185-AB-cabin	SGND

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

ECU SPLICE POINT - power, grounds and sensor rails (bussed DT plugs, soldered splices, NEG bar) (part 2 of 3)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	6 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

ECU SPLICE POINT - power, grounds and sensor rails (bussed DT plugs, soldered splices, NEG bar) (part 3 of 3)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

IX-FUEL-P - IX-FUEL inline to fuel sender lead (trunk end)

housing DT04-2P + W2P | contacts 0460-202-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w45	20	Blue		Fuel level ground	DT06-12 plug (ECU splice point) pin 2	2820	ST185-AB-cabin	SGND

IX-WS-F-S - to front VR conditioner

housing DT06-6S + W6S | contacts 0462-201-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	cab_fspur_5v	20	Red		VRC-F +5V	DT06-12 plug (ECU splice point) pin 10	640	ST185-AB-cabin	5V
2	cab_fspur_gnd	20	Yellow		VRC-F GND	DT06-12 plug (ECU splice point) pin 9	640	ST185-AB-cabin	SGND

IX-WS-R-S - IX-WS-R inline to rear VRC (dash above ECU)

housing DT06-6S + W6S | contacts 0462-201-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	cab_rspur_5v	20	Red		VRC-R +5V	DT06-12 plug (ECU splice point) pin 11	640	ST185-AB-cabin	5V
2	cab_rspur_gnd	20	Yellow		VRC-R GND	DT06-12 plug (ECU splice point) pin 3	640	ST185-AB-cabin	SGND

POWER-NEG-BAR - POWER box NEG bar M8 stud (glove box)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_csbg_ac	20	Black		CSB3 + opto ground	CSB3 ground splice - CSB3 pin 2, opto board COM and stalk ground	1050	ST185-AB-cabin	CSB
	w66	10	Black		ECU power ground	ECU splice point - SP-GND-ECU: ECU power ground (A25 A34 B25 B34) to the NEG bar	570	ST185-AB-cabin	GND
	w_ac_cool_gnd	20	Black		A/C coolant-OK ground	OEM A/C amplifier A18 (1993 book: A15) pin 2	860	ST185-AB-cabin	ACCGND

OEM A/C amplifier A18 (1993 book: A15)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_ac_cool_gnd	20	Black		A/C coolant-OK ground	POWER box NEG bar M8 stud (glove box)	860	ST185-AB-cabin	ACCGND

OEM J/B No.1 (LH kick panel) 1H pin 7, B-O

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
7	w_ig2_in	20	Yellow		IG2 wake feed	IG2 wake diode (1N5822) pin 1	1210	ST185-AB-cabin	IG2WAKE

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

ECU SPLICE POINT - power, grounds and sensor rails (bussed DT plugs, soldered splices, NEG bar) (part 3 of 3)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	7 OF 51	432 x 279 mm	DG	2026-10-29

CABIN BRANCHES - pedal, wheel-speed inlines, fuel level, A/C, IG2 / start taps, relay coils, other harnesses (part 1 of 5)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

ECU-A - Link G4X plug A (centre dash)

housing 4-1437290-1 | contacts 3-1447221-4 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
9	w_trig2_pin	20	White		Cam sensor signal	At the ECU plugs - SP-CAM: cam signal + 1.8k pull-up to +8V	310	ST185-AB-cabin	CAM
14	w38	20	Blue		Pedal 1 signal main	accelerator pedal sensor pin 6	1090	ST185-AB-cabin	APS
18	w_ac_kill	20	White		A/C cancel (ACT)	OEM A/C amplifier pin 11	820	ST185-AB-cabin	ACKILL
19	w_fp85	18	White		Fuel pump relay (-)	B3-K1 Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ...	3130	ST185-AB-cabin	FP
23	cab_fspur_l	20	Blue		FL conditioned out	to front VR conditioner pin 3	530	ST185-AB-cabin	WSSF
26					spare, not used				
29	w_fan85	18	White		Rad fan relay (-)	B1-R4 Radiator fan relay (seated in the POWER box R4) pin 85	680	ST185-AB-cabin	FAN

ECU-B (Link G4X plug B, centre dash)

housing 4-1437290-0 | contacts TBD | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1					spare, not used				
2					spare, not used				
3					spare, not used				
4					spare, not used				
8					spare, not used				
10	w_tach	20	White		Engine speed (tacho)	SP-TACH - Ign 8 tacho to A15 pin 18 and 4.7k lock-sensor tap	820	ST185-AB-cabin	TACH
11					spare, not used				
12	w_eps_trig	18	White		EPS enable relay (-)	B2-M6 EPS Enable Relay (seated in the LOGIC box M6) - ACTIVE-LOW - ECU sinks coil- (85...	680	ST185-AB-cabin	EPSEN
13	w_fan285	18	White		Cond fan relay (-)	B1-R5 Condenser fan relay (seated in the POWER box R5) pin 85	680	ST185-AB-cabin	FAN2
14					spare, not used				
19	cab_rspur_r	20	Blue		Rear right wheel spd	IX-WS-R inline to rear VRC (dash above ECU) pin 4	530	ST185-AB-cabin	WSSR
20	cab_rspur_l	20	Green		Rear left wheel spd	IX-WS-R inline to rear VRC (dash above ECU) pin 3	530	ST185-AB-cabin	WSSR
21	cab_fspur_r_ac	20	White		FR conditioned out	to front VR conditioner pin 4	530	ST185-AB-cabin	WSSF
24	w_fl_sig_pin	20	White		Fuel level signal	SP-FUEL-LVL - fuel sender signal + 470R pull-up to ECU-B B24	310	ST185-AB-cabin	FLVL
28	w_ign	20	Yellow		Ignition sw sense	OEM J/B No.1 (LH kick panel) 1H pin 7, B-O pin 7	1100	ST185-AB-cabin	IGNSW
30	w_start	20	Yellow		Start request	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	680	ST185-AB-cabin	B30
31					spare, not used				
32					spare, not used				
33	w_aps_vpa2	20	Blue		Pedal 2 signal sub	accelerator pedal sensor pin 3	1090	ST185-AB-cabin	APS

BUS-5V-2 - DT06-8 plug (ECU splice point)

housing DT06-08SA + W8S; DT04-08PA-P021 (bussed cap) | contacts 114017 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
5					spare, sealing plug				
6					spare, sealing plug				
7					spare, sealing plug				
8					spare, sealing plug				

BUS-GNDOUT-A - DT06-12 plug (ECU splice point)

housing DT06-12SA + W12S; DT04-12PA-P021 (bussed cap) | contacts 114017 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
10					spare, sealing plug				
11					spare, sealing plug				
12					spare, sealing plug				

BUS-GNDOUT-B - DT06-12 plug (ECU splice point)

housing DT06-12SA + W12S; DT04-12PA-P021 (bussed cap) | contacts 114017 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
9					spare, sealing plug				
10					spare, sealing plug				
11					spare, sealing plug				
12					spare, sealing plug				

REVISIONS

REV	DATE	BY	DESCRIPTION
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K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
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ST185-harness - every wiring harness of the car, one drawing set

CABIN BRANCHES - pedal, wheel-speed inlines, fuel level, A/C, IG2 / start taps, relay coils, other harnesses (part 1 of 5)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	8 OF 51	432 x 279 mm	DG	2026-10-29

CABIN BRANCHES - pedal, wheel-speed inlines, fuel level, A/C, IG2 / start taps, relay coils, other harnesses (part 2 of 5)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

SP-AC-PSW - At A/C amp - SP-AC-PSW: A/C pressure switch output

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_ac_psw_in	20	Yellow		A/C pressure sw out	OEM A5, factory Y-R lead pin 4	200	ST185-AB-cabin	ACC
	w_ac_psw_sw	20	Yellow		A/C pressure sw out	OEM A/C amplifier pin 13	200	ST185-AB-cabin	ACC

SP-AC-SG - cabin A/C sensor common (replaces E10; A32-15, A24-3, A21, A18-2 stay factory)

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_amb_rtn	20	Yellow		Ambient sensor return	OEM ambient air temp sensor A1 pin 1 (front, 6 in pigtail) pin 1	2010	ST185-AB-cabin	AMB
	w_amb_sg	20	Yellow		A/C sensor common	OEM auto A/C amplifier A32 pin 15 SG pin 15	200	ST185-AB-cabin	AMB

SP-ACC12 - B2-BUS2-8 ACC 10 A to k_acc 30 and 86

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_acc86	18	Red		A/C clutch relay coil+	B2-M5 A/C clutch relay (k_acc) - amp A15 pin 8 sinks coil- (85); coil+ (86) fused +12 V from B...	200	ST185-AB-cabin	ACC

SP-CAM - At the ECU plugs - SP-CAM: cam signal + 1.8k pull-up to +8V

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_trig2_pin	20	White		Cam sensor signal	Link G4X plug A (centre dash) pin 9	310	ST185-AB-cabin	CAM
	w_cam_pullup_sig	20	White		Cam pull-up 1.8k	Cam pull-up 1.8 k to +8 V pin 2	200	ST185-AB-cabin	CAM

SP-FUEL-LVL - fuel sender signal + 470R pull-up to ECU-B B24

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_fl_sig	20	White		Fuel level signal	IX-FUEL inline to fuel sender lead (trunk end) pin 1	2820	ST185-AB-cabin	FLVL
	w_fl_sig_pin	20	White		Fuel level signal	ECU-B (Link G4X plug B, centre dash) pin 24	310	ST185-AB-cabin	FLVL
	w43	20	White		Fuel lvl pull-up	470R fuel level pull-up to +5V pin 1	200	ST185-AB-cabin	FLVL

SP-REV - reverse switch out to CSB3 opto IN2 and IG2-13 lamps

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_sw7_c	16	Yellow		Reverse to opto in 2	Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3 pin IN2	680	ST185-AB-cabin	REV
	w_rev_lamps	16	Yellow		Back-up lamp +12 V	OEM IG2 pin 13 (under the ECU, cowl side) - factory back-up lamp output pin 13	310	ST185-AB-cabin	REV

SP-TACH - Ign 8 tacho to A15 pin 18 and 4.7k lock-sensor tap

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_tach	20	White		Engine speed (tacho)	ECU-B (Link G4X plug B, centre dash) pin 10	820	ST185-AB-cabin	TACH
	w_tach_18	20	White		Engine speed (tacho)	OEM A/C amplifier A18 (1993 book: A15) pin 18	200	ST185-AB-cabin	TACH
	w_tach_r	20	White		Tacho to lock-sensor 4.7k	Lock-sensor tap 4.7 k from tacho pin 1	200	ST185-AB-cabin	TACH

R-CAM-PULLUP - Cam pull-up 1.8 k to +8 V

inline resistor 1.8k 1/4W

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_cam_pullup_sig	20	White		Cam pull-up 1.8k	At the ECU plugs - SP-CAM: cam signal + 1.8k pull-up to +8V	200	ST185-AB-cabin	CAM

R-FUELLVL - 470R fuel level pull-up to +5V

inline resistor 470R 1/4W

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w43	20	White		Fuel lvl pull-up	SP-FUEL-LVL - fuel sender signal + 470R pull-up to ECU-B B24	200	ST185-AB-cabin	FLVL

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
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CABIN BRANCHES - pedal, wheel-speed inlines, fuel level, A/C, IG2 / start taps, relay coils, other harnesses (part 2 of 5)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	9 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

CABIN BRANCHES - pedal, wheel-speed inlines, fuel level, A/C, IG2 / start taps, relay coils, other harnesses (part 3 of 5)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

R-LOCK - Lock-sensor tap 4.7 k from tacho

inline resistor 4.7k 1/4W

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_tach_r	20	White		Tacho to lock-sensor 4.7k	SP-TACH - Ign 8 tacho to A15 pin 18 and 4.7k lock-sensor tap	200	ST185-AB-cabin	TACH
2	w_lock_9	20	White		Lock-sensor input (modified tacho)	OEM A/C amplifier A18 (1993 book: A15) pin 9	200	ST185-AB-cabin	TACH

DRB102-CABIN - DRB102 receptacle (RH firewall)

housing DRB12-102PAE-L018 + DRBF-1A | contacts 0413-214-1205 | pins per TE drawing DRB12-102PXE-L018 rev C3

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
5					spare, sealing plug				
6					spare, sealing plug				
7					spare, sealing plug				
8					spare, sealing plug				
19					spare, sealing plug				
20					spare, sealing plug				
35					spare, sealing plug				
36					spare, sealing plug				
37					spare, sealing plug				
38					spare, sealing plug				
39					spare, sealing plug				
40					spare, sealing plug				
43					spare, sealing plug				
48					spare, sealing plug				
49					spare, sealing plug				
51					spare, sealing plug				
54					spare, sealing plug				
55					spare, sealing plug				
60					spare, sealing plug				
62					spare, sealing plug				
82					spare, sealing plug				
86					spare, sealing plug				
87					spare, sealing plug				
88					spare, sealing plug				
89					spare, sealing plug				
90					spare, sealing plug				
91					spare, sealing plug				
92					spare, sealing plug				
93					spare, sealing plug				
94					spare, sealing plug				
97					spare, sealing plug				
98					spare, sealing plug				
99					spare, sealing plug				
100					spare, sealing plug				
101					spare, sealing plug				
102					spare, sealing plug				

APS-PEDAL - accelerator pedal sensor

housing TBD | contacts TBD | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
3	w_aps_vpa2	20	Blue		Pedal 2 signal sub	ECU-B (Link G4X plug B, centre dash) pin 33	1090	ST185-AB-cabin	APS
6	w38	20	Blue		Pedal 1 signal main	Link G4X plug A (centre dash) pin 14	1090	ST185-AB-cabin	APS

IX-FUEL-P - IX-FUEL inline to fuel sender lead (trunk end)

housing DT04-2P + W2P | contacts 0460-202-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_fl_sig	20	White		Fuel level signal	SP-FUEL-LVL - fuel sender signal + 470R pull-up to ECU-B B24	2820	ST185-AB-cabin	FLVL

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
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ST185-harness - every wiring harness of the car, one drawing set

CABIN BRANCHES - pedal, wheel-speed inlines, fuel level, A/C, IG2 / start taps, relay coils, other harnesses (part 3 of 5)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	10 OF 51	432 x 279 mm	DG	2026-10-29

CABIN BRANCHES - pedal, wheel-speed inlines, fuel level, A/C, IG2 / start taps, relay coils, other harnesses (part 4 of 5)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

IX-WS-F-S - to front VR conditioner

housing DT06-6S + W6S | contacts 0462-201-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
3	cab_fspur_l	20	Blue		FL conditioned out	Link G4X plug A (centre dash) pin 23	530	ST185-AB-cabin	WSSF
4	cab_fspur_r_ac	20	White		FR conditioned out	ECU-B (Link G4X plug B, centre dash) pin 21	530	ST185-AB-cabin	WSSF
6					spare, sealing plug				

IX-WS-R-S - IX-WS-R inline to rear VRC (dash above ECU)

housing DT06-6S + W6S | contacts 0462-201-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
3	cab_rspur_l	20	Green		Rear left wheel spd	ECU-B (Link G4X plug B, centre dash) pin 20	530	ST185-AB-cabin	WSSR
4	cab_rspur_r	20	Blue		Rear right wheel spd	ECU-B (Link G4X plug B, centre dash) pin 19	530	ST185-AB-cabin	WSSR
6					spare, sealing plug				

K-ACC - B2-M5 A/C clutch relay (k_acc) - amp A15 pin 8 sinks coil- (85); coil+ (86) fused +12 V from B2-BUS2-8

housing 6-1419137-4 | contacts 280919-2 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
85	w_acc_trig	18	White		A/C clutch relay coil (-)	OEM A/C amplifier A18 (1993 book: A15) pin 8	860	ST185-AB-cabin	ACC
86	w_acc86	18	Red		A/C clutch relay coil+	SP-ACC12 - B2-BUS2-8 ACC 10 A to k_acc 30 and 86	200	ST185-AB-cabin	ACC

K-EPS-EN - B2-M6 EPS Enable Relay (seated in the LOGIC box M6) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) switched 12 V

housing 6-1419137-4 | contacts 280919-2 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
85	w_eps_trig	18	White		EPS enable relay (-)	ECU-B (Link G4X plug B, centre dash) pin 12	680	ST185-AB-cabin	EPSEN

K-FAN - B1-R4 Radiator fan relay (seated in the POWER box R4)

housing 1-1414147-0 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
85	w_fan85	18	White		Rad fan relay (-)	Link G4X plug A (centre dash) pin 29	680	ST185-AB-cabin	FAN

K-FAN2 - B1-R5 Condenser fan relay (seated in the POWER box R5)

housing 1-1393304-0 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
85	w_fan285	18	White		Cond fan relay (-)	ECU-B (Link G4X plug B, centre dash) pin 13	680	ST185-AB-cabin	FAN2

K-FP - B3-K1 Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) switched 12 V - HCR 150 coil is a 2-pole Superseal plug (TE terminals 1/2, not marked 85/86), resistor-su...

housing V23132-A2001-B200 | contacts 282110-1 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
85	w_fp85	18	White		Fuel pump relay (-)	Link G4X plug A (centre dash) pin 19	3130	ST185-AB-cabin	FP

LOGIC - B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS R75): Bus 1 (permanent) and Bus 2 (switched, k_efi 30) mini fuse strips, micro relays M1-M10

housing ST185-FB-LOGIC rev 5 | contacts 1-968849-1 | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
BUS2-13	w_start	20	Yellow		Start request	ECU-B (Link G4X plug B, centre dash) pin 30	680	ST185-AB-cabin	B30

OPTO-8 - Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3

housing TBD | contacts screw terminal (on the board) | pins per terminal names printed on the board

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
IN2	w_sw7_c	16	Yellow		Reverse to opto in 2	SP-REV - reverse switch out to CSB3 opto IN2 and IG2-13 lamps	680	ST185-AB-cabin	REV

OEM A/C amplifier

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
11	w_ac_kill	20	White		A/C cancel (ACT)	Link G4X plug A (centre dash) pin 18	820	ST185-AB-cabin	ACKILL
13	w_ac_psw_sw	20	Yellow		A/C pressure sw out	At A/C amp - SP-AC-PSW: A/C pressure switch output	200	ST185-AB-cabin	ACC

REVISIONS

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CABIN BRANCHES - pedal, wheel-speed inlines, fuel level, A/C, IG2 / start taps, relay coils, other harnesses (part 4 of 5)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	11 OF 51	432 x 279 mm	DG	2026-10-29

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OEM A/C amplifier A18 (1993 book: A15)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
18	w_tach_18	20	White		Engine speed (tacho)	SP-TACH - Ign 8 tacho to A15 pin 18 and 4.7k lock-sensor tap	200	ST185-AB-cabin	TACH
8	w_acc_trig	18	White		A/C clutch relay coil (-)	B2-M5 A/C clutch relay (k_acc) - amp A15 pin 8 sinks coil- (85); coil+ (86) fused +12 V from B...	860	ST185-AB-cabin	ACC
9	w_lock_9	20	White		Lock-sensor input (modified tacho)	Lock-sensor tap 4.7 k from tacho pin 2	200	ST185-AB-cabin	TACH

OEM ambient air temp sensor A1 pin 1 (front, 6 in pigtail)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_amb_rtn	20	Yellow		Ambient sensor return	SP-AC-SG - cabin A/C sensor common (replaces E10; A32-15, A24-3, A21, A18-2 stay factory)	2010	ST185-AB-cabin	AMB

OEM ambient air temp sensor A1 pin 2 (front, 6 in pigtail)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_amb_sig	20	Yellow		Ambient temp signal	OEM auto A/C amplifier A32 pin 6 TAM pin 6	2010	ST185-AB-cabin	AMB

OEM auto A/C amplifier A32 pin 15 SG

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
15	w_amb_sg	20	Yellow		A/C sensor common	SP-AC-SG - cabin A/C sensor common (replaces E10; A32-15, A24-3, A21, A18-2 stay factory)	200	ST185-AB-cabin	AMB

OEM auto A/C amplifier A32 pin 6 TAM

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
6	w_amb_sig	20	Yellow		Ambient temp signal	OEM ambient air temp sensor A1 pin 2 (front, 6 in pigtail) pin 2	2010	ST185-AB-cabin	AMB

OEM A5, factory Y-R lead

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
4	w_ac_psw_in	20	Yellow		A/C pressure sw out	At A/C amp - SP-AC-PSW: A/C pressure switch output	200	ST185-AB-cabin	ACC

OEM IG2 pin 13 (under the ECU, cowl side) - factory back-up lamp output

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
13	w_rev_lamps	16	Yellow		Back-up lamp +12 V	SP-REV - reverse switch out to CSB3 opto IN2 and IG2-13 lamps	310	ST185-AB-cabin	REV

OEM J/B No.1 (LH kick panel) 1H pin 7, B-O

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
7	w_ign	20	Yellow		Ignition sw sense	ECU-B (Link G4X plug B, centre dash) pin 28	1100	ST185-AB-cabin	IGNSW

REVISIONS

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SHIELD DRAINS - screens to A7 / B17 (R207), CAN screen to the CAN trunk (part 1 of 2)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

ECU-A - Link G4X plug A (centre dash)

housing 4-1437290-1 | contacts 3-1447221-4 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
7	w_drain_ecu_a	20	Black		Shield drain	6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7	200	ST185-AB-cabin	SHLD

ECU-B (Link G4X plug B, centre dash)

housing 4-1437290-0 | contacts 3-1447221-4 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
17	w_drain_ecu_b	20	Black		Shield drain B	6 mm behind the ECU-B plug - SP-SHIELD-B: drains to B17	200	ST185-AB-cabin	SHLD

SP-CAN-SH-BH - CAN trunk, bulkhead B branch - SP-CAN-SH-BH: CAN screen to CAN-Lambda

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_canbh_bc_sh		Shield		CAN-Lambda shield	DRB102 receptacle (RH firewall) pin 81	680	ST185-AB-cabin	CAN

SP-SHIELD-A - 6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_drain_ecu_a	20	Black		Shield drain	Link G4X plug A (centre dash) pin 7	200	ST185-AB-cabin	SHLD
	w_shd_a_aps	20	Black		Shield drain	At the pedal branch - SP-SHIELD-A-APS: pedal drains	450	ST185-AB-cabin	SHLD
	w_shd_a_wsf	20	Black		Shield drain	At the front WS inline - SP-SHIELD-A-WSF: front wheel-speed drains	230	ST185-AB-cabin	SHLD
	w_shc_crank	20	Black		Crank sensor shield	DRB102 receptacle (RH firewall) pin 76	630	ST185-AB-cabin	CRANKC
	w_shc_cam	20	Black		Cam sensor shield	DRB102 receptacle (RH firewall) pin 66	630	ST185-AB-cabin	CAM
	w_shc_turbo	20	Black		Turbo speed shield	DRB102 receptacle (RH firewall) pin 57	630	ST185-AB-cabin	TURBO
	w_shc_flex	20	Black		Flex fuel shield	DRB102 receptacle (RH firewall) pin 32	630	ST185-AB-cabin	FLEX
	w_shc_etbs	20	Black		Throttle shield	DRB102 receptacle (RH firewall) pin 47	630	ST185-AB-cabin	ETBS

SP-SHIELD-A-APS - At the pedal branch - SP-SHIELD-A-APS: pedal drains

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_shd_a_aps	20	Black		Shield drain	6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7	450	ST185-AB-cabin	SHLD
	cab_aps1_sh		Shield		Pedal 1 cable shield	cab_aps1 screen (floats at the device end)	840	ST185-AB-cabin	APS
	cab_aps_sh		Shield		Pedal 2 cable shield	cab_aps2 screen (floats at the device end)	840	ST185-AB-cabin	APS

SP-SHIELD-A-WSF - At the front WS inline - SP-SHIELD-A-WSF: front wheel-speed drains

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_shd_a_wsf	20	Black		Shield drain	6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7	230	ST185-AB-cabin	SHLD
	cab_fspur_sh		Shield		WS-F screen drain	to front VR conditioner pin 5	500	ST185-AB-cabin	WSSF
	cab_fspur_fr_sh_ac		Shield		FR wheel spd shield	cab_fspur_fr_ac screen (floats at the device end)	500	ST185-AB-cabin	WSSF

SP-SHIELD-B - 6 mm behind the ECU-B plug - SP-SHIELD-B: drains to B17

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_shc_knock1	20	Black		Knock sensor shield	DRB102 receptacle (RH firewall) pin 73	630	ST185-AB-cabin	KNOCK
	w_shc_etbm	20	Black		ETB motor shield	DRB102 receptacle (RH firewall) pin 50	630	ST185-AB-cabin	ETBM
	w_shc_vss	20	Black		Gearbox speed shield	DRB102 receptacle (RH firewall) pin 69	630	ST185-AB-cabin	VSS
	w_drain_ecu_b	20	Black		Shield drain B	ECU-B (Link G4X plug B, centre dash) pin 17	200	ST185-AB-cabin	SHLD
	w_shd_b_wsr	20	Black		Shield drain B	At the rear WS inline - SP-SHIELD-B-WSR: rear wheel-speed drains	230	ST185-AB-cabin	SHLD

SP-SHIELD-B-WSR - At the rear WS inline - SP-SHIELD-B-WSR: rear wheel-speed drains

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_rspur_sh		Shield		WS-R screen drain	IX-WS-R inline to rear VRC (dash above ECU) pin 5	500	ST185-AB-cabin	WSSR
	w_shd_b_wsr	20	Black		Shield drain B	6 mm behind the ECU-B plug - SP-SHIELD-B: drains to B17	230	ST185-AB-cabin	SHLD

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

SHIELD DRAINS - screens to A7 / B17 (R207), CAN screen to the CAN trunk (part 1 of 2)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	13 OF 51	432 x 279 mm	DG	2026-10-29

SHIELD DRAINS - screens to A7 / B17 (R207), CAN screen to the CAN trunk (part 2 of 2)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

DRB102-CABIN - DRB102 receptacle (RH firewall)

housing DRB12-102PAE-L018 + DRBF-1A | contacts 0460-202-1631 | pins per TE drawing DRB12-102PXE-L018 rev C3

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
32	w_shc_flex	20	Black		Flex fuel shield	6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7	630	ST185-AB-cabin	FLEX
47	w_shc_etbs	20	Black		Throttle shield	6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7	630	ST185-AB-cabin	ETBS
50	w_shc_etbm	20	Black		ETB motor shield	6 mm behind the ECU-B plug - SP-SHIELD-B: drains to B17	630	ST185-AB-cabin	ETBM
57	w_shc_turbo	20	Black		Turbo speed shield	6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7	630	ST185-AB-cabin	TURBO
66	w_shc_cam	20	Black		Cam sensor shield	6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7	630	ST185-AB-cabin	CAM
69	w_shc_vss	20	Black		Gearbox speed shield	6 mm behind the ECU-B plug - SP-SHIELD-B: drains to B17	630	ST185-AB-cabin	VSS
73	w_shc_knock1	20	Black		Knock sensor shield	6 mm behind the ECU-B plug - SP-SHIELD-B: drains to B17	630	ST185-AB-cabin	KNOCK
76	w_shc_crank	20	Black		Crank sensor shield	6 mm behind the ECU-A plug - SP-SHIELD-A: drains to A7	630	ST185-AB-cabin	CRANKC
81	cab_canbh_bc_sh		Shield		CAN-Lambda shield	CAN trunk, bulkhead B branch - SP-CAN-SH-BH: CAN screen to CAN-Lambda	680	ST185-AB-cabin	CAN

IX-WS-F-S - to front VR conditioner

housing DT06-6S + W6S | contacts 0462-201-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
5	cab_fspur_sh		Shield		WS-F screen drain	At the front WS inline - SP-SHIELD-A-WSF: front wheel-speed drains	500	ST185-AB-cabin	WSSF

IX-WS-R-S - IX-WS-R inline to rear VRC (dash above ECU)

housing DT06-6S + W6S | contacts 0462-201-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
5	cab_rspur_sh		Shield		WS-R screen drain	At the rear WS inline - SP-SHIELD-B-WSR: rear wheel-speed drains	500	ST185-AB-cabin	WSSR

CAB-APS1 cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_aps1_sh		Shield		Pedal 1 cable shield	At the pedal branch - SP-SHIELD-A-APS: pedal drains	840	ST185-AB-cabin	APS

CAB-APS2 cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_aps_sh		Shield		Pedal 2 cable shield	At the pedal branch - SP-SHIELD-A-APS: pedal drains	840	ST185-AB-cabin	APS

CAB-FSPUR-FR-AC cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_fspur_fr_sh_ac		Shield		FR wheel spd shield	At the front WS inline - SP-SHIELD-A-WSF: front wheel-speed drains	500	ST185-AB-cabin	WSSF

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
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K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

SHIELD DRAINS - screens to A7 / B17 (R207), CAN screen to the CAN trunk (part 2 of 2)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	**L**	**14 OF 51**	**432 x 279 mm**	**DG**	**2026-10-29**

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS) (part 1 of 7)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

SP-CANH-LAM - CAN H join - bulkhead, Lambda, 120 ohm end

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	wc_h_lam_be	20	White		CAN H to CAN-Lambda	BULKHEAD pin 79	1157	ST185-AB-engine	CAN
	wc_term3	20	Green		Lambda 120R CAN H	r_term_end pin 1	200	ST185-AB-engine	CAN
	wc_h_lam_pin	20	White		CAN H to CAN-Lambda	Link CAN-Lambda pin 4	200	ST185-AB-engine	CAN

SP-CANL-LAM - CAN L join - bulkhead, Lambda, 120 ohm end

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	wc_l_lam_be	20	Green		CAN L to CAN-Lambda	BULKHEAD pin 80	1157	ST185-AB-engine	CAN
	wc_term4	20	Green		Lambda 120R CAN L	r_term_end pin 2	200	ST185-AB-engine	CAN
	wc_l_lam_pin	20	Green		CAN L to CAN-Lambda	Link CAN-Lambda pin 3	200	ST185-AB-engine	CAN

SP-CHASSIS-ENG - Coil ground splice - ignition coils 1-4 to bulkhead pin 4 (POWER box NEG bar)

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_cop1g	18	Black		Coil 1 ground	COP 1 (1ZZ) pin 4	493	ST185-AB-engine	COP1
	w_cop2g	18	Black		Coil 2 ground	COP 2 (1ZZ) pin 4	560	ST185-AB-engine	COP2
	w_cop3g	18	Black		Coil 3 ground	COP 3 (1ZZ) pin 4	457	ST185-AB-engine	COP3
	w_cop4g	18	Black		Coil 4 ground	COP 4 (1ZZ) pin 4	352	ST185-AB-engine	COP4
	w_cop_gnd_e_ae	12	Black		Coil ground	BULKHEAD pin 4	1198	ST185-AB-engine	COPG

SP-COP-ENG - Coil +12 V splice - BH-A pin 1 (LOGIC COP 15A) to coils 1-4

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_cop_pwr_e_ae	12	Red		EFI relay k_efi -> LOGIC COP 15 A fuse	BULKHEAD pin 1	1198	ST185-AB-engine	COP
	w112	16	Red		EFI relay k_efi -> LOGIC COP 15 A fuse	COP 1 (1ZZ) pin 1	493	ST185-AB-engine	COP
	w113	16	Red		EFI relay k_efi -> LOGIC COP 15 A fuse	COP 2 (1ZZ) pin 1	560	ST185-AB-engine	COP
	w114	16	Red		EFI relay k_efi -> LOGIC COP 15 A fuse	COP 3 (1ZZ) pin 1	457	ST185-AB-engine	COP
	w115	16	Red		EFI relay k_efi -> LOGIC COP 15 A fuse	COP 4 (1ZZ) pin 1	352	ST185-AB-engine	COP

SP-GNDOUT-B-ENG - Sensor return splice - oil temp and charge temp, tied to the BH-B pin 29 splice

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w10	20	Blue		Gnd charge temp	Charge-Pipe IAT Sensor pin 2	1690	ST185-AB-engine	CPIAT
	w8	20	Blue		Gnd oil temp	Oil Temp Sensor pin 2	1391	ST185-AB-engine	OILT
	w_rail_gnd_b_e	20	Blue		Gnd oil/charge temp	Sensor return splice - BH-B pin 29 (ECU B22) to oil and coolant pressure	200	ST185-AB-engine	GNDOUT

SP-GNDOUT-B-ENG2 - Sensor return splice - BH-B pin 29 (ECU B22) to oil and coolant pressure

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_gnd_b_e_c19	20	Blue		Gnd oil/clnt/temps	BULKHEAD pin 29	403	ST185-AB-engine	GNDOUT
	w36	20	Blue		Gnd coolant press	Coolant Pressure Sensor pin 1	810	ST185-AB-engine	CLNTP
	w30	20	Blue		Gnd oil pressure	Oil Pressure Sensor pin 1	2045	ST185-AB-engine	OILP
	w_rail_gnd_b_e	20	Blue		Gnd oil/charge temp	Sensor return splice - oil temp and charge temp, tied to the BH-B pin 29 splice	200	ST185-AB-engine	GNDOUT

SP-GNDOUT-B-ENG3 - Sensor return splice - BH-B pin 15 (ECU B22) to fuel pressure and gearbox speed

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_gnd_b_e_c21	20	Blue		Gnd fuel press + VSS	BULKHEAD pin 15	1094	ST185-AB-engine	GNDOUT
	w33	20	Blue		Gnd fuel pressure	Fuel Pressure Sensor pin 1	610	ST185-AB-engine	FUELP
	w_vss_gnd	20	Yellow		Gnd gearbox speed	Gearbox VSS (Toyota 83181-20040, 3-wire) - actual road speed, plug 90980-11143 pin 2	684	ST185-AB-engine	VSS

REVISIONS

REV	DATE	BY	DESCRIPTION
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ST185-harness - every wiring harness of the car, one drawing set

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS) (part 1 of 7)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	15 OF 51	432 x 279 mm	DG	2026-10-29

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS) (part 2 of 7)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

SP-GNDOUT-ENG - Sensor return splice - BH-A pin 34 (ECU A24) to coolant temp, intake temp, throttle

splice: crimp splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_rail_sp_gndout_e	20	Blue		Gnd IAT/TPS/coolant	BULKHEAD pin 34	403	ST185-AB-engine	GNDOUT
	w4	20	Blue		Gnd coolant temp	ECT Sensor pin 2	1812	ST185-AB-engine	ECT
	w127	20	Yellow		Gnd throttle sensor	Bosch 74.5mm ETB pin 2	1716	ST185-AB-engine	ETBS
	w6	20	Blue		Gnd intake temp	IAT Sensor (Manifold) pin 2	1285	ST185-AB-engine	IAT

SP-GNDOUT-ENG2 - Sensor return splice - BH-A pin 18 (ECU A24) to MAP and flex sensor

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_gnd_a_e_c44	20	Blue		Gnd flex + MAP	BULKHEAD pin 18	1270	ST185-AB-engine	GNDOUT
	w47	20	Yellow		Gnd flex fuel	Flex Fuel Sensor pin B	315	ST185-AB-engine	FLEX
	w27	20	Blue		Gnd MAP sensor	MAP Sensor pin 1	356	ST185-AB-engine	MAP

SP-INJ-ENG - Injector +12 V splice - BH-A pin 2 (LOGIC INJ 15A) to injectors 1-4

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_inj_pwr_e_ae	12	Red		EFI relay k_efi -> LOGIC INJ 15 A fuse	BULKHEAD pin 2	980	ST185-AB-engine	INJ
	w211	16	Red		EFI relay k_efi -> LOGIC INJ 15 A fuse	Injector 1 pin 1	475	ST185-AB-engine	INJ
	w212	16	Red		EFI relay k_efi -> LOGIC INJ 15 A fuse	Injector 2 pin 1	388	ST185-AB-engine	INJ
	w213	16	Red		EFI relay k_efi -> LOGIC INJ 15 A fuse	Injector 3 pin 1	307	ST185-AB-engine	INJ
	w214	16	Red		EFI relay k_efi -> LOGIC INJ 15 A fuse	Injector 4 pin 1	242	ST185-AB-engine	INJ

SP-LAM12-B - CAN-Lambda +12 V splice - 12 AWG from BH-B pin 95 to the 16 AWG lead

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_lam12_be14	12	Red		EFI relay k_efi -> LOGIC LAM 7.5 A fuse	BULKHEAD pin 95	1157	ST185-AB-engine	LAM
	w_lam12_be	16	Red		EFI relay k_efi -> LOGIC LAM 7.5 A fuse	Link CAN-Lambda pin 1	200	ST185-AB-engine	LAM

SP-LAMG-B - CAN-Lambda ground splice - 12 AWG from BH-B pin 96 to the 16 AWG lead

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_lamg_be	16	Black		CAN-Lambda ground	Link CAN-Lambda pin 2	200	ST185-AB-engine	LAM
	w_lamg_be14	12	Black		CAN-Lambda ground	BULKHEAD pin 96	1157	ST185-AB-engine	LAM

SP-SW12-ENG - Switched +12 V splice - BH-A pin 30 to boost solenoid and flex sensor

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_rail_sp_sw12_e	18	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	BULKHEAD pin 30	403	ST185-AB-engine	SW12
	w219	20	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	Boost Control Solenoid pin 1	1840	ST185-AB-engine	SW12
	w217	20	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	Flex Fuel Sensor pin A	1182	ST185-AB-engine	SW12

R-TERM-END - r_term_end

inline resistor 120R 1/4W

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	wc_term3	20	Green		Lambda 120R CAN H	CAN H join - bulkhead, Lambda, 120 ohm end	200	ST185-AB-engine	CAN
2	wc_term4	20	Green		Lambda 120R CAN L	CAN L join - bulkhead, Lambda, 120 ohm end	200	ST185-AB-engine	CAN

REVISIONS

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K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS) (part 2 of 7)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	16 OF 51	432 x 279 mm	DG	2026-10-29

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS) (part 3 of 7)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

DRB102-ENGINE - BULKHEAD

housing DRB16-102SAE-L018 | contacts 0462-210-1231 | pins per TE drawing DRB12-102PXE-L018 rev C3

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_cop_pwr_e_ae	12	Red		EFI relay k_efi -> LOGIC COP 15 A fuse	Coil +12 V splice - BH-A pin 1 (LOGIC COP 15A) to coils 1-4	1198	ST185-AB-engine	COP
2	w_inj_pwr_e_ae	12	Red		EFI relay k_efi -> LOGIC INJ 15 A fuse	Injector +12 V splice - BH-A pin 2 (LOGIC INJ 15A) to injectors 1-4	980	ST185-AB-engine	INJ
3	w_strl_e	12	Red		Starter solenoid S	Starter solenoid S (50) terminal post, spade	2433	ST185-AB-engine	STR
4	w_cop_gnd_e_ae	12	Black		Coil ground to the POWER box NEG bar	Coil ground splice - ignition coils 1-4 to bulkhead pin 4 (POWER box NEG bar)	1198	ST185-AB-engine	COPG
5					spare, sealing plug				
6					spare, sealing plug				
7					spare, sealing plug				
8					spare, sealing plug				
9	w120_e	20	White		Coil 1 trigger (IGT)	COP 1 (1ZZ) pin 3	1491	ST185-AB-engine	COP1
10	w121_e	20	White		Coil 2 trigger (IGT)	COP 2 (1ZZ) pin 3	1558	ST185-AB-engine	COP2
11	w122_e	20	White		Coil 3 trigger (IGT)	COP 3 (1ZZ) pin 3	1455	ST185-AB-engine	COP3
12	w123_e	20	White		Coil 4 trigger (IGT)	COP 4 (1ZZ) pin 3	1350	ST185-AB-engine	COP4
13	w_5v_brail_e_c46	20	Orange		+5 V fuel pressure	Fuel Pressure Sensor pin 2	1504	ST185-AB-engine	5V
14	w32_e	20	White		Fuel pressure signal	Fuel Pressure Sensor pin 3	1504	ST185-AB-engine	FUELP
15	w_gnd_b_e_c21	20	Blue		Gnd fuel press + VSS	Sensor return splice - BH-B pin 15 (ECU B22) to fuel pressure and gearbox speed	1094	ST185-AB-engine	GNDOUT
16	w206	20	Orange		+5 V MAP sensor	MAP Sensor pin 2	1426	ST185-AB-engine	MAP
17	w26_e	20	White		MAP sensor signal	MAP Sensor pin 3	1426	ST185-AB-engine	MAP
18	w_gnd_a_e_c44	20	Blue		Gnd flex + MAP	Sensor return splice - BH-A pin 18 (ECU A24) to MAP and flex sensor	1270	ST185-AB-engine	GNDOUT
19					spare, sealing plug				
20					spare, sealing plug				
21	w108_e	20	White		Injector 1 drive (-)	Injector 1 pin 2	1255	ST185-AB-engine	INJ1
22	w109_e	20	White		Injector 2 drive (-)	Injector 2 pin 2	1168	ST185-AB-engine	INJ2
23	w110_e	20	White		Injector 3 drive (-)	Injector 3 pin 2	1087	ST185-AB-engine	INJ3
24	w111_e	20	White		Injector 4 drive (-)	Injector 4 pin 2	1022	ST185-AB-engine	INJ4
25	w_5v_brail_e_c47	20	Orange		+5 V coolant press	Coolant Pressure Sensor pin 2	1013	ST185-AB-engine	5V
26	w35_e	20	White		Coolant press signal	Coolant Pressure Sensor pin 3	1013	ST185-AB-engine	CLNTP
27	w_5v_brail_e_c45	20	Orange		+5 V oil pressure	Oil Pressure Sensor pin 2	2248	ST185-AB-engine	5V
28	w29_e	20	White		Oil pressure signal	Oil Pressure Sensor pin 3	2248	ST185-AB-engine	OILP
29	w_gnd_b_e_c19	20	Blue		Gnd oil/clnt/temps	Sensor return splice - BH-B pin 29 (ECU B22) to oil and coolant pressure	403	ST185-AB-engine	GNDOUT
30	w_rail_sp_sw12_e	18	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	Switched +12 V splice - BH-A pin 30 to boost solenoid and flex sensor	403	ST185-AB-engine	SW12
31	w48_e	20	Blue		Flex fuel signal	Flex Fuel Sensor pin C	1385	ST185-AB-engine	FLEX
32	cab_flex_sh		Shield		Flex fuel shield	cab_flex screen (floats at the device end)	1385	ST185-AB-engine	FLEX
33	w202_e	20	White		Intake temp signal	IAT Sensor (Manifold) pin 1	1488	ST185-AB-engine	IAT
34	w_rail_sp_gndout_e	20	Blue		Gnd IAT/TPS/coolant	Sensor return splice - BH-A pin 34 (ECU A24) to coolant temp, intake temp, throttle	403	ST185-AB-engine	GNDOUT
35					spare, sealing plug				
36					spare, sealing plug				
37					spare, sealing plug				
38					spare, sealing plug				
39					spare, sealing plug				
40					spare, sealing plug				
41	w126	20	Red		+5 V throttle sensor	Bosch 74.5mm ETB pin 3	1919	ST185-AB-engine	ETBS
42	w128_e	20	Green		Throttle pos 1 main	Bosch 74.5mm ETB pin 6	1919	ST185-AB-engine	ETBS
43					spare, sealing plug				
44	w129_e	20	Blue		Throttle pos 2 sub	Bosch 74.5mm ETB pin 5	1919	ST185-AB-engine	ETBS
45	w124_e_ae	18	Red		ETB motor +	Bosch 74.5mm ETB pin 4	1919	ST185-AB-engine	ETBM
46	w215_e_ae	18	Black		ETB motor -	Bosch 74.5mm ETB pin 1	1919	ST185-AB-engine	ETBM
47	cab_etbs_sh		Shield		Throttle shield	cab_etbs screen (floats at the device end)	1919	ST185-AB-engine	ETBS
48					spare, sealing plug				
49					spare, sealing plug				
50	cab_etbm_ae_sh		Shield		ETB motor shield	cab_etbm_ae screen (floats at the device end)	1919	ST185-AB-engine	ETBM
51					spare, sealing plug				
52	w218	20	Red		+5 V turbo speed	Turbo Speed Sensor pin A	2167	ST185-AB-engine	TURBO
53	w16_e	20	Blue		Turbo speed signal	Turbo Speed Sensor pin B	2167	ST185-AB-engine	TURBO
54					spare, sealing plug				
55					spare, sealing plug				
56	w17	20	Yellow		Gnd turbo speed	Turbo Speed Sensor pin C	2167	ST185-AB-engine	TURBO
57	cab_turbo_sh		Shield		Turbo speed shield	cab_turbo screen (floats at the device end)	2167	ST185-AB-engine	TURBO
58	w103_e	20	White		Boost solenoid (-)	Boost Control Solenoid pin 2	2043	ST185-AB-engine	BOOST
59	w204_e	20	White		Charge temp signal	Charge-Pipe IAT Sensor pin 1	1893	ST185-AB-engine	CPIAT
60					spare, sealing plug				
61	w203_e	20	White		Oil temp signal	Oil Temp Sensor pin 1	1594	ST185-AB-engine	OILT
62					spare, sealing plug				
63	w_rail_sp_8v_e	20	Red		+8 V to cam sensor	Cam Hall Sensor (RacerX) pin 1	1152	ST185-AB-engine	8V
64	cab_cam_c2	20	Blue		Cam sensor signal	Cam Hall Sensor (RacerX) pin 2	1152	ST185-AB-engine	CAM
65	w24	20	Yellow		Gnd cam sensor	Cam Hall Sensor (RacerX) pin 3	1152	ST185-AB-engine	CAM
66	cab_cam_sh		Shield		Cam sensor shield	cab_cam screen (floats at the device end)	1152	ST185-AB-engine	CAM
67	w_vss_12v	20	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	Gearbox VSS (Toyota 83181-20040, 3-wire) - actual road speed, plug 90980-11143 pin 1	1578	ST185-AB-engine	SW12
68	w_vss_sig_e	20	Blue		Gearbox speed signal	Gearbox VSS (Toyota 83181-20040, 3-wire) - actual road speed, plug 90980-11143 pin 3	1578	ST185-AB-engine	VSS
69	cab_vss_sh		Shield		Gearbox speed shield	cab_vss screen (floats at the device end)	1578	ST185-AB-engine	VSS
70	w201_e	20	White		Coolant temp signal	ECT Sensor pin 1	2015	ST185-AB-engine	ECT
71	cab_knock1_c1	20	Blue		Knock sensor signal	Knock Sensor 1 pin 1	1051	ST185-AB-engine	KNOCK1
72	cab_knock1_c2	20	White		Knock sensor (-)	Knock Sensor 1 pin 2	1051	ST185-AB-engine	KNOCK1
73	cab_knock1_sh		Shield		Knock sensor shield	cab_knock1 screen (floats at the device end)	1051	ST185-AB-engine	KNOCK1
74	cab_crank_c1	20	White		Crank sensor signal	Crank VR Sensor (36-2) pin 1	1410	ST185-AB-engine	CRANK
75	w2	20	Blue		Gnd crank sensor	Crank VR Sensor (36-2) pin 2	1410	ST185-AB-engine	CRANK
76	cab_crank_sh		Shield		Crank sensor shield	cab_crank screen (floats at the device end)	1410	ST185-AB-engine	CRANK
77	w_rev_feed_e	16	Red		Back-up sw +12 V in	Back-up SW B1 pin 2	2341	ST185-AB-engine	REV
78	w_sw7_e	16	Yellow		Reverse sw +12 V out	Back-up SW B1 pin 1	2341	ST185-AB-engine	REV
79	wc_h_lam_be	20	White		CAN H to CAN-Lambda	CAN H join - bulkhead, Lambda, 120 ohm end	1157	ST185-AB-engine	CAN
80	wc_l_lam_be	20	Green		CAN L to CAN-Lambda	CAN L join - bulkhead, Lambda, 120 ohm end	1157	ST185-AB-engine	CAN
81	cab_lamcan_be_sh		Shield		CAN-Lambda shield	cab_can_lam screen (floats at the device end)	1157	ST185-AB-engine	CAN
82					spare, sealing plug				
86					spare, sealing plug				
87					spare, sealing plug				
88					spare, sealing plug				
89					spare, sealing plug				
90					spare, sealing plug				
91					spare, sealing plug				
92					spare, sealing plug				
93					spare, sealing plug				
94					spare, sealing plug				
95	w_lam12_be14	12	Red		EFI relay k_efi -> LOGIC LAM 7.5 A fuse	CAN-Lambda +12 V splice - 12 AWG from BH-B pin 95 to the 16 AWG lead	1157	ST185-AB-engine	LAM
96	w_lamg_be14	12	Black		CAN-Lambda ground	CAN-Lambda ground splice - 12 AWG from BH-B pin 96 to the 16 AWG lead	1157	ST185-AB-engine	LAM
97					spare, sealing plug				
98					spare, sealing plug				
99					spare, sealing plug				
100					spare, sealing plug				
101					spare, sealing plug				
102					spare, sealing plug				

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS) (part 3 of 7)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	17 OF 51	432 x 386 mm	DG	2026-10-29

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS) (part 4 of 7)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

BOOST-SOL - Boost Control Solenoid

housing 12052641 | contacts 12084200 | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w219	20	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	Switched +12 V splice - BH-A pin 30 to boost solenoid and flex sensor	1840	ST185-AB-engine	SW12
2	w103_e	20	White		Boost solenoid (-)	BULKHEAD pin 58	2043	ST185-AB-engine	BOOST

CAM - Cam Hall Sensor (RacerX)

housing TBD | contacts TBD | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_rail_sp_8v_e	20	Red		+8 V to cam sensor	BULKHEAD pin 63	1152	ST185-AB-engine	8V
2	cab_cam_c2	20	Blue		Cam sensor signal	BULKHEAD pin 64	1152	ST185-AB-engine	CAM
3	w24	20	Yellow		Gnd cam sensor	BULKHEAD pin 65	1152	ST185-AB-engine	CAM

CAN-LAMBDA - Link CAN-Lambda

housing DTM06-4S + WM-4S | contacts 0462-005-2031 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_lam12_be	16	Red		EFI relay k_efi -> LOGIC LAM 7.5 A fuse	CAN-Lambda +12 V splice - 12 AWG from BH-B pin 95 to the 16 AWG lead	200	ST185-AB-engine	LAM
2	w_lamg_be	16	Black		CAN-Lambda ground	CAN-Lambda ground splice - 12 AWG from BH-B pin 96 to the 16 AWG lead	200	ST185-AB-engine	LAM
3	wc_l_lam_pin	20	Green		CAN L to CAN-Lambda	CAN L join - bulkhead, Lambda, 120 ohm end	200	ST185-AB-engine	CAN
4	wc_h_lam_pin	20	White		CAN H to CAN-Lambda	CAN H join - bulkhead, Lambda, 120 ohm end	200	ST185-AB-engine	CAN

CHARGE-IAT - Charge-Pipe IAT Sensor

housing 90980-11062 | contacts 8100-0461 | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w204_e	20	White		Charge temp signal	BULKHEAD pin 59	1893	ST185-AB-engine	CPIAT
2	w10	20	Blue		Gnd charge temp	Sensor return splice - oil temp and charge temp, tied to the BH-B pin 29 splice	1690	ST185-AB-engine	CPIAT

CLNT-P - Coolant Pressure Sensor

housing TBD | contacts TBD | pins per device reference docs/devices/SENSOR-AND-ACTUATOR-REFERENCE.md (Coolant pressure pin table)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w36	20	Blue		Gnd coolant press	Sensor return splice - BH-B pin 29 (ECU B22) to oil and coolant pressure	810	ST185-AB-engine	CLNTP
2	w_5v_brail_e_c47	20	Orange		+5 V coolant press	BULKHEAD pin 25	1013	ST185-AB-engine	5V
3	w35_e	20	White		Coolant press signal	BULKHEAD pin 26	1013	ST185-AB-engine	CLNTP

COIL-1 - COP 1 (1ZZ)

housing 90980-11885 | contacts 8100-0461 | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w112	16	Red		EFI relay k_efi -> LOGIC COP 15 A fuse	Coil +12 V splice - BH-A pin 1 (LOGIC COP 15A) to coils 1-4	493	ST185-AB-engine	COP
2					spare, not used				
3	w120_e	20	White		Coil 1 trigger (IGT)	BULKHEAD pin 9	1491	ST185-AB-engine	COP1
4	w_cop1g	18	Black		Coil 1 ground	Coil ground splice - ignition coils 1-4 to bulkhead pin 4 (POWER box NEG bar)	493	ST185-AB-engine	COP1

COIL-2 - COP 2 (1ZZ)

housing 90980-11885 | contacts 8100-0461 | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w113	16	Red		EFI relay k_efi -> LOGIC COP 15 A fuse	Coil +12 V splice - BH-A pin 1 (LOGIC COP 15A) to coils 1-4	560	ST185-AB-engine	COP
2					spare, not used				
3	w121_e	20	White		Coil 2 trigger (IGT)	BULKHEAD pin 10	1558	ST185-AB-engine	COP2
4	w_cop2g	18	Black		Coil 2 ground	Coil ground splice - ignition coils 1-4 to bulkhead pin 4 (POWER box NEG bar)	560	ST185-AB-engine	COP2

COIL-3 - COP 3 (1ZZ)

housing 90980-11885 | contacts 8100-0461 | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w114	16	Red		EFI relay k_efi -> LOGIC COP 15 A fuse	Coil +12 V splice - BH-A pin 1 (LOGIC COP 15A) to coils 1-4	457	ST185-AB-engine	COP
2					spare, not used				
3	w122_e	20	White		Coil 3 trigger (IGT)	BULKHEAD pin 11	1455	ST185-AB-engine	COP3
4	w_cop3g	18	Black		Coil 3 ground	Coil ground splice - ignition coils 1-4 to bulkhead pin 4 (POWER box NEG bar)	457	ST185-AB-engine	COP3

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
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K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
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ST185-harness - every wiring harness of the car, one drawing set

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS) (part 4 of 7)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	18 OF 51	432 x 279 mm	DG	2026-10-29

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS) (part 5 of 7)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

COIL-4 - COP 4 (1ZZ)

housing 90980-11885 | contacts 8100-0461 | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w115	16	Red		EFI relay k_efi -> LOGIC COP 15 A fuse	Coil +12 V splice - BH-A pin 1 (LOGIC COP 15A) to coils 1-4	352	ST185-AB-engine	COP
2					spare, not used				
3	w123_e	20	White		Coil 4 trigger (IGT)	BULKHEAD pin 12	1350	ST185-AB-engine	COP4
4	w_cop4g	18	Black		Coil 4 ground	Coil ground splice - ignition coils 1-4 to bulkhead pin 4 (POWER box NEG bar)	352	ST185-AB-engine	COP4

CRANK - Crank VR Sensor (36-2)

housing DT06-2S-CE05 + W2S | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	cab_crank_c1	20	White		Crank sensor signal	BULKHEAD pin 74	1410	ST185-AB-engine	CRANK
2	w2	20	Blue		Gnd crank sensor	BULKHEAD pin 75	1410	ST185-AB-engine	CRANK

ECT Sensor

housing 90980-11062 | contacts 8100-0461 | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w201_e	20	White		Coolant temp signal	BULKHEAD pin 70	2015	ST185-AB-engine	ECT
2	w4	20	Blue		Gnd coolant temp	Sensor return splice - BH-A pin 34 (ECU A24) to coolant temp, intake temp, throttle	1812	ST185-AB-engine	ECT

ETB - Bosch 74.5mm ETB

housing D 261 205 358-01 | contacts TBD | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w215_e_ae	18	Black		ETB motor -	BULKHEAD pin 46	1919	ST185-AB-engine	ETBM
2	w127	20	Yellow		Gnd throttle sensor	Sensor return splice - BH-A pin 34 (ECU A24) to coolant temp, intake temp, throttle	1716	ST185-AB-engine	ETBS
3	w126	20	Red		+5 V throttle sensor	BULKHEAD pin 41	1919	ST185-AB-engine	ETBS
4	w124_e_ae	18	Red		ETB motor +	BULKHEAD pin 45	1919	ST185-AB-engine	ETBM
5	w129_e	20	Blue		Throttle pos 2 sub	BULKHEAD pin 44	1919	ST185-AB-engine	ETBS
6	w128_e	20	Green		Throttle pos 1 main	BULKHEAD pin 42	1919	ST185-AB-engine	ETBS

FLEX - Flex Fuel Sensor

housing 13519047 | contacts 15326426 | pins per device reference docs/devices/SENSOR-AND-ACTUATOR-REFERENCE.md (Flex fuel pin table)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
A	w217	20	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	Switched +12 V splice - BH-A pin 30 to boost solenoid and flex sensor	1182	ST185-AB-engine	SW12
B	w47	20	Yellow		Gnd flex fuel	Sensor return splice - BH-A pin 18 (ECU A24) to MAP and flex sensor	315	ST185-AB-engine	FLEX
C	w48_e	20	Blue		Flex fuel signal	BULKHEAD pin 31	1385	ST185-AB-engine	FLEX

FUEL-P - Fuel Pressure Sensor

housing 354404 | contacts in kit 354404 | pins per device reference docs/devices/SENSOR-AND-ACTUATOR-REFERENCE.md (Fuel pressure pin table)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w33	20	Blue		Gnd fuel pressure	Sensor return splice - BH-B pin 15 (ECU B22) to fuel pressure and gearbox speed	610	ST185-AB-engine	FUELP
2	w_5v_brail_e_c46	20	Orange		+5 V fuel pressure	BULKHEAD pin 13	1504	ST185-AB-engine	5V
3	w32_e	20	White		Fuel pressure signal	BULKHEAD pin 14	1504	ST185-AB-engine	FUELP

IAT Sensor (Manifold)

housing 90980-11062 | contacts 8100-0461 | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w202_e	20	White		Intake temp signal	BULKHEAD pin 33	1488	ST185-AB-engine	IAT
2	w6	20	Blue		Gnd intake temp	Sensor return splice - BH-A pin 34 (ECU A24) to coolant temp, intake temp, throttle	1285	ST185-AB-engine	IAT

INJ-1 - Injector 1

housing 15419715 | contacts 12191819 | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w211	16	Red		EFI relay k_efi -> LOGIC INJ 15 A fuse	Injector +12 V splice - BH-A pin 2 (LOGIC INJ 15A) to injectors 1-4	475	ST185-AB-engine	INJ
2	w108_e	20	White		Injector 1 drive (-)	BULKHEAD pin 21	1255	ST185-AB-engine	INJ1

REVISIONS

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ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS) (part 5 of 7)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	19 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS) (part 6 of 7)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

INJ-2 - Injector 2

housing 15419715 | contacts 12191819 | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w212	16	Red		EFI relay k_efi -> LOGIC INJ 15 A fuse	Injector +12 V splice - BH-A pin 2 (LOGIC INJ 15A) to injectors 1-4	388	ST185-AB-engine	INJ
2	w109_e	20	White		Injector 2 drive (-)	BULKHEAD pin 22	1168	ST185-AB-engine	INJ2

INJ-3 - Injector 3

housing 15419715 | contacts 12191819 | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w213	16	Red		EFI relay k_efi -> LOGIC INJ 15 A fuse	Injector +12 V splice - BH-A pin 2 (LOGIC INJ 15A) to injectors 1-4	307	ST185-AB-engine	INJ
2	w110_e	20	White		Injector 3 drive (-)	BULKHEAD pin 23	1087	ST185-AB-engine	INJ3

INJ-4 - Injector 4

housing 15419715 | contacts 12191819 | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w214	16	Red		EFI relay k_efi -> LOGIC INJ 15 A fuse	Injector +12 V splice - BH-A pin 2 (LOGIC INJ 15A) to injectors 1-4	242	ST185-AB-engine	INJ
2	w111_e	20	White		Injector 4 drive (-)	BULKHEAD pin 24	1022	ST185-AB-engine	INJ4

KNOCK-1 - Knock Sensor 1

housing 1 928 403 874 | contacts BDK 2.8 | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	cab_knock1_c1	20	Blue		Knock sensor signal	BULKHEAD pin 71	1051	ST185-AB-engine	KNOCK1
2	cab_knock1_c2	20	White		Knock sensor (-)	BULKHEAD pin 72	1051	ST185-AB-engine	KNOCK1

MAP Sensor

housing 354404 | contacts in kit 354404 | pins per device reference docs/devices/SENSOR-AND-ACTUATOR-REFERENCE.md (MAP pin table)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w27	20	Blue		Gnd MAP sensor	Sensor return splice - BH-A pin 18 (ECU A24) to MAP and flex sensor	356	ST185-AB-engine	MAP
2	w206	20	Orange		+5 V MAP sensor	BULKHEAD pin 16	1426	ST185-AB-engine	MAP
3	w26_e	20	White		MAP sensor signal	BULKHEAD pin 17	1426	ST185-AB-engine	MAP

OIL-P - Oil Pressure Sensor

housing 354404 | contacts in kit 354404 | pins per device reference docs/devices/SENSOR-AND-ACTUATOR-REFERENCE.md (Oil pressure pin table)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w30	20	Blue		Gnd oil pressure	Sensor return splice - BH-B pin 29 (ECU B22) to oil and coolant pressure	2045	ST185-AB-engine	OILP
2	w_5v_brail_e_c45	20	Orange		+5 V oil pressure	BULKHEAD pin 27	2248	ST185-AB-engine	5V
3	w29_e	20	White		Oil pressure signal	BULKHEAD pin 28	2248	ST185-AB-engine	OILP

OIL-T - Oil Temp Sensor

housing 90980-11062 | contacts 8100-0461 | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w203_e	20	White		Oil temp signal	BULKHEAD pin 61	1594	ST185-AB-engine	OILT
2	w8	20	Blue		Gnd oil temp	Sensor return splice - oil temp and charge temp, tied to the BH-B pin 29 splice	1391	ST185-AB-engine	OILT

REV-SW - Back-up SW B1

housing 90980-11250 | contacts 8100-0461 | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_sw7_e	16	Yellow		Reverse sw +12 V out	BULKHEAD pin 78	2341	ST185-AB-engine	REV
2	w_rev_feed_e	16	Red		Back-up sw +12 V in	BULKHEAD pin 77	2341	ST185-AB-engine	REV

TURBO-SPD - Turbo Speed Sensor

housing DT06-3S-CE05 + W3S | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
A	w218	20	Red		+5 V turbo speed	BULKHEAD pin 52	2167	ST185-AB-engine	TURBO
B	w16_e	20	Blue		Turbo speed signal	BULKHEAD pin 53	2167	ST185-AB-engine	TURBO
C	w17	20	Yellow		Gnd turbo speed	BULKHEAD pin 56	2167	ST185-AB-engine	TURBO

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS) (part 6 of 7)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	20 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS) (part 7 of 7)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

VSS - Gearbox VSS (Toyota 83181-20040, 3-wire) - actual road speed, plug 90980-11143

housing 90980-11143 | contacts 8100-0461 | pins per Toyota 3-wire speed sensor pinout (wilbo666 wiki; Daniel 2026-09-28)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_vss_12v	20	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	BULKHEAD pin 67	1578	ST185-AB-engine	SW12
2	w_vss_gnd	20	Yellow		Gnd gearbox speed	Sensor return splice - BH-B pin 15 (ECU B22) to fuel pressure and gearbox speed	684	ST185-AB-engine	VSS
3	w_vss_sig_e	20	Blue		Gearbox speed signal	BULKHEAD pin 68	1578	ST185-AB-engine	VSS

STARTER-TRIGGER - Starter solenoid S (50) terminal post, spade

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_strl_e	12	Red		Starter solenoid S	BULKHEAD pin 3	2433	ST185-AB-engine	STR

CAB-CAM cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_cam_sh		Shield		Cam sensor shield	BULKHEAD pin 66	1152	ST185-AB-engine	CAM

CAB-CAN-LAM cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_lamcan_be_sh		Shield		CAN-Lambda shield	BULKHEAD pin 81	1157	ST185-AB-engine	CAN

CAB-CRANK cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_crank_sh		Shield		Crank sensor shield	BULKHEAD pin 76	1410	ST185-AB-engine	CRANK

CAB-ETBM-AE cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_etbm_ae_sh		Shield		ETB motor shield	BULKHEAD pin 50	1919	ST185-AB-engine	ETBM

CAB-ETBS cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_etbs_sh		Shield		Throttle shield	BULKHEAD pin 47	1919	ST185-AB-engine	ETBS

CAB-FLEX cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_flex_sh		Shield		Flex fuel shield	BULKHEAD pin 32	1385	ST185-AB-engine	FLEX

CAB-KNOCK1 cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_knock1_sh		Shield		Knock sensor shield	BULKHEAD pin 73	1051	ST185-AB-engine	KNOCK1

CAB-TURBO cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_turbo_sh		Shield		Turbo speed shield	BULKHEAD pin 57	2167	ST185-AB-engine	TURBO

CAB-VSS cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_vss_sh		Shield		Gearbox speed shield	BULKHEAD pin 69	1578	ST185-AB-engine	VSS

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

ENGINE BAY - DRB102 ENGINE HALF TO EVERY ENGINE DEVICE (ECU HARNESS) (part 7 of 7)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	21 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

ANTI-THEFT - ALARM MODULE, SIREN, TRIGGERS, STATUS LED (part 1 of 3)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

ECU-B (Link G4X plug B, centre dash)

housing 4-1437290-0 | contacts 3-1447221-4 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
27	w_at_immob2	20	White		Anti-theft input	Alarm module - main harness (DT06-12SA) pin 9	380	ST185-AntiTheft	IMMOB

SP-12V-SHK - Shock sensor +12 V splice - LOGIC ALMS 2A fuse to shock sensor pin A

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_at_12v_shk	20	Red		AMI BUS1 50 A fuse -> LOGIC ALMS 2 A fuse	Shock / tilt sensor * pin A	200	ST185-AntiTheft	ALMS

SP-AT-12V - Alarm +12 V splice - LOGIC ALM 5A fuse to alarm pin 1

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_at_12v	20	Red		AMI BUS1 50 A fuse -> LOGIC ALM 5 A fuse	Alarm module - main harness (DT06-12SA) pin 1	200	ST185-AntiTheft	ALM

SP-DOOR - Door trigger splice - door courtesy switches to alarm pin 6

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_at_door	20	Yellow		Door trigger	Alarm module - main harness (DT06-12SA) pin 6	200	ST185-AntiTheft	DOOR
	w_at_door_tie	20	Yellow		Door trigger	OEM door courtesy switch line (R-W) at J/B No.1 connector 1B pin 11 (left kick panel); 0 V wit...	1220	ST185-AntiTheft	DOOR

SP-HOOD - Hood trigger splice - hood pin switch to alarm pin 7

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_at_hood	20	Yellow		Hood trigger	Alarm module - main harness (DT06-12SA) pin 7	200	ST185-AntiTheft	HOOD
	w_at_hood_tie	20	Yellow		Hood trigger	Hood pin switch lead (new switch, RH front via loom C, spot TBD)	2260	ST185-AntiTheft	HOOD

SP-IG - Alarm ignition splice - LOGIC ALMIG 2A fuse to alarm pin 3

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_at_ig	20	Red		EFI relay k_efi -> LOGIC ALMIG 2 A fuse	Alarm module - main harness (DT06-12SA) pin 3	200	ST185-AntiTheft	ALMIG

SP-PARK - Park flash splice - alarm pin 5 to the park lamp relay

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_at_park	20	White		Park lamp flash	Alarm module - main harness (DT06-12SA) pin 5	200	ST185-AntiTheft	PARK
	w_at_park_tie	20	White		Park lamp flash	Park lamp flash relay lead (cabin relay, part and spot TBD)	1220	ST185-AntiTheft	PARK

SP-TRUNK - Trunk trigger splice - trunk pin switch to alarm pin 8

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_at_trunk	20	Yellow		Trunk trigger	Alarm module - main harness (DT06-12SA) pin 8	200	ST185-AntiTheft	TRUNK
	w_at_trunk_tie	20	Yellow		Trunk trigger	Trunk pin switch lead (new switch, spot TBD)	4420	ST185-AntiTheft	TRUNK

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

ANTI-THEFT - ALARM MODULE, SIREN, TRIGGERS, STATUS LED (part 1 of 3)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	**L**	**22 OF 51**	**432 x 279 mm**	**DG**	**2026-10-29**

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

ANTI-THEFT - ALARM MODULE, SIREN, TRIGGERS, STATUS LED (part 2 of 3)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

ALARM - Alarm module - main harness (DT06-12SA)

housing DT06-12SA + W12S | contacts 0462-201-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_at_12v	20	Red		AMI BUS1 50 A fuse -> LOGIC ALM 5 A fuse	Alarm +12 V splice - LOGIC ALM 5A fuse to alarm pin 1	200	ST185-AntiTheft	ALM
2	w_at_gnd_a1	20	Black		Alarm ground	Cabin ground stud near the alarm (spot and EWD ID TBD, M6)	200	ST185-AntiTheft	ALM
3	w_at_ig	20	Red		EFI relay k_efi -> LOGIC ALMIG 2 A fuse	Alarm ignition splice - LOGIC ALMIG 2A fuse to alarm pin 3	200	ST185-AntiTheft	ALMIG
4	w_at_siren	20	Red		Siren +	Siren (engine bay, center firewall behind the engine) - leads cross via harness C's own RH fe...	1660	ST185-AntiTheft	SIREN
5	w_at_park	20	White		Park lamp flash	Park flash splice - alarm pin 5 to the park lamp relay	200	ST185-AntiTheft	PARK
6	w_at_door	20	Yellow		Door trigger	Door trigger splice - door courtesy switches to alarm pin 6	200	ST185-AntiTheft	DOOR
7	w_at_hood	20	Yellow		Hood trigger	Hood trigger splice - hood pin switch to alarm pin 7	200	ST185-AntiTheft	HOOD
8	w_at_trunk	20	Yellow		Trunk trigger	Trunk trigger splice - trunk pin switch to alarm pin 8	200	ST185-AntiTheft	TRUNK
9	w_at_immob2	20	White		Anti-theft input	ECU-B (Link G4X plug B, centre dash) pin 27	380	ST185-AntiTheft	IMMOB
10	w_at_led	20	Red		Status LED +	Status LED (dash) pin 1	750	ST185-AntiTheft	ALMLED
11	w_at_valet	20	Yellow		Valet switch	Valet / program switch (hidden) pin 1	450	ST185-AntiTheft	VALET
12	w_at_shk_sig	20	White		Shock trigger	Shock / tilt sensor * pin C	200	ST185-AntiTheft	ALMS

SHOCK-SENSOR - Shock / tilt sensor *

housing DT06-3S + W3S | contacts 0462-201-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
A	w_at_12v_shk	20	Red		AMI BUS1 50 A fuse -> LOGIC ALMS 2 A fuse	Shock sensor +12 V splice - LOGIC ALMS 2A fuse to shock sensor pin A	200	ST185-AntiTheft	ALMS
B	w_at_gnd_b1	20	Black		Shock sensor ground	Cabin ground stud near the alarm (spot and EWD ID TBD, M6)	200	ST185-AntiTheft	ALMS
C	w_at_shk_sig	20	White		Shock trigger	Alarm module - main harness (DT06-12SA) pin 12	200	ST185-AntiTheft	ALMS

SIREN - Siren (engine bay, center firewall behind the engine) - leads cross via harness C's own RH fender grommet, decided 2026-10-10, not the FR wheel-speed grommet

housing DT06-2S-CE05 + W2S | contacts 0462-201-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_at_siren	20	Red		Siren +	Alarm module - main harness (DT06-12SA) pin 4	1660	ST185-AntiTheft	SIREN
2	w_at_gnd_a2	20	Black		Siren ground	Cabin ground stud near the alarm (spot and EWD ID TBD, M6)	1660	ST185-AntiTheft	SIREN

STATUS-LED - Status LED (dash)

housing DT06-2S + W2S | contacts 0462-201-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_at_led	20	Red		Status LED +	Alarm module - main harness (DT06-12SA) pin 10	750	ST185-AntiTheft	ALMLED
2	w_at_gnd_b2	20	Black		Status LED gnd	Cabin ground stud near the alarm (spot and EWD ID TBD, M6)	750	ST185-AntiTheft	ALMLED

VALET-SW - Valet / program switch (hidden)

housing DT06-2S + W2S | contacts 0462-201-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_at_valet	20	Yellow		Valet switch	Alarm module - main harness (DT06-12SA) pin 11	450	ST185-AntiTheft	VALET
2	w_at_gnd_c1	20	Black		Valet switch ground	Cabin ground stud near the alarm (spot and EWD ID TBD, M6)	450	ST185-AntiTheft	VALET

T-AT-DOOR - OEM door courtesy switch line (R-W) at J/B No.1 connector 1B pin 11 (left kick panel); 0 V with any door open

terminal flying lead (soldered, no terminal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_at_door_tie	20	Yellow		Door trigger	Door trigger splice - door courtesy switches to alarm pin 6	1220	ST185-AntiTheft	DOOR

T-AT-GND - Cabin ground stud near the alarm (spot and EWD ID TBD, M6)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_at_gnd_a1	20	Black		Alarm ground	Alarm module - main harness (DT06-12SA) pin 2	200	ST185-AntiTheft	ALM
	w_at_gnd_a2	20	Black		Siren ground	Siren (engine bay, center firewall behind the engine) - leads cross via harness C's own RH fe...	1660	ST185-AntiTheft	SIREN

T-AT-GND-B - Cabin ground stud near the alarm (spot and EWD ID TBD, M6)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_at_gnd_b2	20	Black		Status LED gnd	Status LED (dash) pin 2	750	ST185-AntiTheft	ALMLED
	w_at_gnd_b1	20	Black		Shock sensor ground	Shock / tilt sensor * pin B	200	ST185-AntiTheft	ALMS

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
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ST185-harness - every wiring harness of the car, one drawing set

ANTI-THEFT - ALARM MODULE, SIREN, TRIGGERS, STATUS LED (part 2 of 3)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	23 OF 51	432 x 279 mm	DG	2026-10-29

ANTI-THEFT - ALARM MODULE, SIREN, TRIGGERS, STATUS LED (part 3 of 3)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

T-AT-GND-C - Cabin ground stud near the alarm (spot and EWD ID TBD, M6)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_at_gnd_c1	20	Black		Valet switch ground	Valet / program switch (hidden) pin 2	450	ST185-AntiTheft	VALET

T-AT-HOOD - Hood pin switch lead (new switch, RH front via loom C, spot TBD)

terminal flying lead (soldered, no terminal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_at_hood_tie	20	Yellow		Hood trigger	Hood trigger splice - hood pin switch to alarm pin 7	2260	ST185-AntiTheft	HOOD

T-AT-PARK - Park lamp flash relay lead (cabin relay, part and spot TBD)

terminal flying lead (soldered, no terminal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_at_park_tie	20	White		Park lamp flash	Park flash splice - alarm pin 5 to the park lamp relay	1220	ST185-AntiTheft	PARK

T-AT-TRUNK - Trunk pin switch lead (new switch, spot TBD)

terminal flying lead (soldered, no terminal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_at_trunk_tie	20	Yellow		Trunk trigger	Trunk trigger splice - trunk pin switch to alarm pin 8	4420	ST185-AntiTheft	TRUNK

REVISIONS

REV	DATE	BY	DESCRIPTION
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J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
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ST185-harness - every wiring harness of the car, one drawing set

ANTI-THEFT - ALARM MODULE, SIREN, TRIGGERS, STATUS LED (part 3 of 3)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	24 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

CAN TRUNK - ECU, CSB3, CLUSTER, REALDASH PI, BULKHEAD, CAN-LAMBDA (part 1 of 3)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

ECU-C - ECU Comms/Tuning Port

housing DTM06-6S + WM-6S | contacts 0462-201-2031 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_shd_can_trunk	20	Black		CAN screen trunk	CAN trunk, cluster branch - SP-CAN-SH-CLU: CAN screen	650	ST185-CAN	CAN
2					spare, sealing plug				
3	wc_h_ecu	20	White		CAN 1 high	CAN H join - ECU, CSB3, cluster, Pi, bulkhead	250	ST185-CAN	CAN
4	wc_l_ecu	20	Green		CAN 1 low	CAN L join - ECU, CSB3, cluster, Pi, bulkhead	250	ST185-CAN	CAN
5					spare, sealing plug				
6					spare, sealing plug				

SP-CAN-SH-BH - CAN trunk, bulkhead B branch - SP-CAN-SH-BH: CAN screen to CAN-Lambda

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_shd_can_bh	20	Black		CAN screen trunk	CAN trunk, cluster branch - SP-CAN-SH-CLU: CAN screen	600	ST185-CAN	CAN

SP-CAN-SH-CLU - CAN trunk, cluster branch - SP-CAN-SH-CLU: CAN screen

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_shd_can_trunk	20	Black		CAN screen trunk	ECU Comms/Tuning Port pin 1	650	ST185-CAN	CAN
	cab_can_ecu_sh		Shield		ECU CAN shield	cab_can_ecu screen (floats at the device end)	650	ST185-CAN	CAN
	cab_can_clu_sh		Shield		Cluster CAN shield	cab_can_clu screen (floats at the device end)	200	ST185-CAN	CAN
	w_shd_can_csb	20	Black		CAN screen trunk	CAN trunk, CSB3 branch - SP-CAN-SH-CSB: CAN screen	1020	ST185-CAN	CAN
	w_shd_can_bh	20	Black		CAN screen trunk	CAN trunk, bulkhead B branch - SP-CAN-SH-BH: CAN screen to CAN-Lambda	600	ST185-CAN	CAN

SP-CAN-SH-CSB - CAN trunk, CSB3 branch - SP-CAN-SH-CSB: CAN screen

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_can_csb_sh		Shield		CSB3 CAN shield	cab_can_csb_csb screen (floats at the device end)	200	ST185-CAN	CAN
	w_shd_can_csb	20	Black		CAN screen trunk	CAN trunk, cluster branch - SP-CAN-SH-CLU: CAN screen	1020	ST185-CAN	CAN
	w_shd_can_rd	20	Black		CAN screen trunk	CAN trunk, RealDash Pi branch - SP-CAN-SH-RD: CAN screen	870	ST185-CAN	CAN

SP-CAN-SH-RD - CAN trunk, RealDash Pi branch - SP-CAN-SH-RD: CAN screen

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_can_rd_sh		Shield		Pi CAN shield	cab_can_rd screen (floats at the device end)	200	ST185-CAN	CAN
	w_shd_can_rd	20	Black		CAN screen trunk	CAN trunk, CSB3 branch - SP-CAN-SH-CSB: CAN screen	870	ST185-CAN	CAN

SP-CANH - CAN H join - ECU, CSB3, cluster, Pi, bulkhead

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	wc_h_ecu	20	White		CAN 1 high	ECU Comms/Tuning Port pin 3	250	ST185-CAN	CAN
	wc_term1	20	Green		ECU 120R CAN H	r_term_ecu pin 1	200	ST185-CAN	CAN
	wc_h_csb	20	White		CAN H to CSB3	ECUMaster CSB3 - Power/CAN/Switch I/O pin 5	620	ST185-CAN	CAN
	wc_h_clu	20	White		CAN H to cluster	Cluster CAN board CANH + CANL pins (old meter spot)	600	ST185-CAN	CAN
	wc_h_rd	20	White		CAN H to RealDash Pi	Pi USB-CAN adapter CANH + CANL pins (center dash)	450	ST185-CAN	CAN

SP-CANL - CAN L join - ECU, CSB3, cluster, Pi, bulkhead

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	wc_l_ecu	20	Green		CAN 1 low	ECU Comms/Tuning Port pin 4	250	ST185-CAN	CAN
	wc_term2	20	Green		ECU 120R CAN L	r_term_ecu pin 2	200	ST185-CAN	CAN
	wc_l_csb	20	Green		CAN L to CSB3	ECUMaster CSB3 - Power/CAN/Switch I/O pin 6	620	ST185-CAN	CAN
	wc_l_clu	20	Green		CAN L to cluster	Cluster CAN board CANH + CANL pins (old meter spot)	600	ST185-CAN	CAN
	wc_l_rd	20	Green		CAN L to RealDash Pi	Pi USB-CAN adapter CANH + CANL pins (center dash)	450	ST185-CAN	CAN

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

CAN TRUNK - ECU, CSB3, CLUSTER, REALDASH PI, BULKHEAD, CAN-LAMBDA (part 1 of 3)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	25 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

CAN TRUNK - ECU, CSB3, CLUSTER, REALDASH PI, BULKHEAD, CAN-LAMBDA (part 2 of 3)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

SP-DEV - Dash 12 V split - cluster + RealDash Pi, center dash

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_dev_clu	16	Red		EFI relay k_efi -> LOGIC DEV 15 A fuse	Cluster buck converter IN+ pin (old meter spot)	850	ST185-CAN	DEV
	w_dev_rd	16	Red		EFI relay k_efi -> LOGIC DEV 15 A fuse	Pi 12-to-5 V supply IN+ pin (center dash)	200	ST185-CAN	DEV

R-TERM-ECU - r_term_ecu

inline resistor 120R 1/4W

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	wc_term1	20	Green		ECU 120R CAN H	CAN H join - ECU, CSB3, cluster, Pi, bulkhead	200	ST185-CAN	CAN
2	wc_term2	20	Green		ECU 120R CAN L	CAN L join - ECU, CSB3, cluster, Pi, bulkhead	200	ST185-CAN	CAN

CSB3 - ECUMaster CSB3 - Power/CAN/Switch I/O

housing HD36-24-33SE | contacts 0462-201-2031 | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
5	wc_h_csb	20	White		CAN H to CSB3	CAN H join - ECU, CSB3, cluster, Pi, bulkhead	620	ST185-CAN	CAN
6	wc_l_csb	20	Green		CAN L to CSB3	CAN L join - ECU, CSB3, cluster, Pi, bulkhead	620	ST185-CAN	CAN

POWER-NEG-BAR - POWER box NEG bar M8 stud (glove box)

terminal 320575

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_gnd_clu	16	Black		Cluster ground	Cluster buck converter IN- pin (old meter spot)	1130	ST185-CAN	DEV
	w_gnd_rd	16	Black		RealDash Pi ground	Pi 12-to-5 V supply IN- pin (center dash)	980	ST185-CAN	DEV

T-CLUSTER-CAN - Cluster CAN board CANH + CANL pins (old meter spot)

terminal flying lead (soldered, no terminal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	wc_h_clu	20	White		CAN H to cluster	CAN H join - ECU, CSB3, cluster, Pi, bulkhead	600	ST185-CAN	CAN
	wc_l_clu	20	Green		CAN L to cluster	CAN L join - ECU, CSB3, cluster, Pi, bulkhead	600	ST185-CAN	CAN

T-CLUSTER-GND-LEAD - Cluster buck converter IN- pin (old meter spot)

terminal flying lead (soldered, no terminal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_gnd_clu	16	Black		Cluster ground	POWER box NEG bar M8 stud (glove box)	1130	ST185-CAN	DEV

T-CLUSTER-PWR - Cluster buck converter IN+ pin (old meter spot)

terminal flying lead (soldered, no terminal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_dev_clu	16	Red		EFI relay k_efi -> LOGIC DEV 15 A fuse	Dash 12 V split - cluster + RealDash Pi, center dash	850	ST185-CAN	DEV

T-REALDASH-CAN - Pi USB-CAN adapter CANH + CANL pins (center dash)

terminal flying lead (soldered, no terminal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	wc_h_rd	20	White		CAN H to RealDash Pi	CAN H join - ECU, CSB3, cluster, Pi, bulkhead	450	ST185-CAN	CAN
	wc_l_rd	20	Green		CAN L to RealDash Pi	CAN L join - ECU, CSB3, cluster, Pi, bulkhead	450	ST185-CAN	CAN

T-REALDASH-GND-LEAD - Pi 12-to-5 V supply IN- pin (center dash)

terminal flying lead (soldered, no terminal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_gnd_rd	16	Black		RealDash Pi ground	POWER box NEG bar M8 stud (glove box)	980	ST185-CAN	DEV

T-REALDASH-PWR - Pi 12-to-5 V supply IN+ pin (center dash)

terminal flying lead (soldered, no terminal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_dev_rd	16	Red		EFI relay k_efi -> LOGIC DEV 15 A fuse	Dash 12 V split - cluster + RealDash Pi, center dash	200	ST185-CAN	DEV

REVISIONS

REV	DATE	BY	DESCRIPTION
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ST185-harness - every wiring harness of the car, one drawing set

CAN TRUNK - ECU, CSB3, CLUSTER, REALDASH PI, BULKHEAD, CAN-LAMBDA (part 2 of 3)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	26 OF 51	432 x 279 mm	DG	2026-10-29

CAN TRUNK - ECU, CSB3, CLUSTER, REALDASH PI, BULKHEAD, CAN-LAMBDA (part 3 of 3)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

CAB-CAN-CLU cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_can_clu_sh		Shield		Cluster CAN shield	CAN trunk, cluster branch - SP-CAN-SH-CLU: CAN screen	200	ST185-CAN	CAN

CAB-CAN-CSB-CSB cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_can_csb_sh		Shield		CSB3 CAN shield	CAN trunk, CSB3 branch - SP-CAN-SH-CSB: CAN screen	200	ST185-CAN	CAN

CAB-CAN-ECU cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_can_ecu_sh		Shield		ECU CAN shield	CAN trunk, cluster branch - SP-CAN-SH-CLU: CAN screen	650	ST185-CAN	CAN

CAB-CAN-RD cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_can_rd_sh		Shield		Pi CAN shield	CAN trunk, RealDash Pi branch - SP-CAN-SH-RD: CAN screen	200	ST185-CAN	CAN

REVISIONS

REV	DATE	BY	DESCRIPTION
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ST185-harness - every wiring harness of the car, one drawing set

CAN TRUNK - ECU, CSB3, CLUSTER, REALDASH PI, BULKHEAD, CAN-LAMBDA (part 3 of 3)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	27 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

CSB3 - SWITCH INPUTS, CRUISE LADDER, OPTO BOARD (part 1 of 2)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

SP-CRUISE-LADDER - Cruise ladder splice - stalk pin 5, 1k pull-up and CSB3 A2

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_an2	20	Yellow		Cruise ladder	OEM cruise stalk switch C14 pin 5, W wire pin 5	1220	ST185-CSB3	CRUISE
	w_cru_sig	20	Yellow		Cruise ladder	r_cruise pin 2	200	ST185-CSB3	CRUISE
	w_cru_ladder_out	20	Yellow		Cruise ladder	ECUMaster CSB3 - Power/CAN/Switch I/O pin 8	200	ST185-CSB3	CRUISE

SP-CSB-GND - CSB3 ground splice - CSB3 pin 2, opto board COM and stalk ground

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_csbg	20	Black		CSB3 ground	ECUMaster CSB3 - Power/CAN/Switch I/O pin 2	200	ST185-CSB3	CSB
	w_cru_gnd	20	Blue		Cruise stalk ground	OEM cruise stalk switch C14 pin 17, BR wire pin 17	1220	ST185-CSB3	CSB
	w_opto_inc	20	Black		Opto COM ground	Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3 pin COM1	200	ST185-CSB3	CSB

R-CRUISE - r_cruise

inline resistor 1k 1/4W

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_cru_sig	20	Yellow		Cruise ladder	Cruise ladder splice - stalk pin 5, 1k pull-up and CSB3 A2	200	ST185-CSB3	CRUISE

CSB3 - ECUMaster CSB3 - Power/CAN/Switch I/O

housing HD36-24-33SE | contacts 0462-201-2031 | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_csb12	20	Red		EFI relay k_efi -> LOGIC CSB 5 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	1050	ST185-CSB3	CSB
2	w_csbg	20	Black		CSB3 ground	CSB3 ground splice - CSB3 pin 2, opto board COM and stalk ground	200	ST185-CSB3	CSB
3	w_opto_vcc	20	Orange		Opto +5 V	Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3 pin VCC	200	ST185-CSB3	OPTO
4	w_opto_outc	20	Black		Opto out ground	Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3 pin NEG	200	ST185-CSB3	OPTO
7					spare, sealing plug				
8	w_cru_ladder_out	20	Yellow		Cruise ladder	Cruise ladder splice - stalk pin 5, 1k pull-up and CSB3 A2	200	ST185-CSB3	CRUISE
9					spare, sealing plug				
10					spare, sealing plug				
11					spare, sealing plug				
12					spare, sealing plug				
13					spare, sealing plug				
14					spare, sealing plug				
15					spare, sealing plug				
16					spare, sealing plug				
17	w_sw2	20	Yellow		Cruise MAIN on	OEM cruise stalk switch C14 pin 15, Y-B wire pin 15	1220	ST185-CSB3	CRUISE
18					spare, sealing plug				
19					spare, sealing plug				
20	w_sw5	20	Yellow		Clutch pedal sw	OEM cruise control clutch switch C17 pin 2, B wire pin 2	1460	ST185-CSB3	CLUTCH
21	w_opto_out1	20	Yellow		Brake to CSB3 S7	Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3 pin OUT1	200	ST185-CSB3	BRAKE
22	w_opto_out2	20	Yellow		Reverse to CSB3 S8	Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3 pin OUT2	200	ST185-CSB3	REV
23					spare, sealing plug				
24					spare, sealing plug				
25					spare, sealing plug				
27					spare, sealing plug				
28					spare, sealing plug				
29					spare, sealing plug				
30					spare, sealing plug				
31					spare, sealing plug				
32					spare, sealing plug				
33					spare, sealing plug				

LOGIC - B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS R75): Bus 1 (permanent) and Bus 2 (switched, k_efi 30) mini fuse strips, micro relays M1-M10

housing ST185-FB-LOGIC rev 5 | contacts 1-968849-1 | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
BUS2-5	w_csb12	20	Red		EFI relay k_efi -> LOGIC CSB 5 A fuse	ECUMaster CSB3 - Power/CAN/Switch I/O pin 1	1050	ST185-CSB3	CSB

REVISIONS

REV	DATE	BY	DESCRIPTION
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ST185-harness - every wiring harness of the car, one drawing set

CSB3 - SWITCH INPUTS, CRUISE LADDER, OPTO BOARD (part 1 of 2)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	28 OF 51	432 x 279 mm	DG	2026-10-29

CSB3 - SWITCH INPUTS, CRUISE LADDER, OPTO BOARD (part 2 of 2)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

OPTO-8 - Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3

housing TBD | contacts screw terminal (on the board) | pins per terminal names printed on the board

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
COM1	w_opto_inc	20	Black		Opto COM ground	CSB3 ground splice - CSB3 pin 2, opto board COM and stalk ground	200	ST185-CSB3	CSB
COM2					spare, not used				
IN1	w_sw6	16	Yellow		Brake on +12 V	OEM stop lamp switch S9 pin 1 (pedal), G-W wire pin 1	1460	ST185-CSB3	BRAKE
IN3	w_hl_sense_in	20	Red		Lights on +12 V	OEM dimmer rheostat R10 pin 1 (left of the steering wheel), green TAIL pin 1	1220	ST185-CSB3	HLSENSE
IN4					spare, not used				
IN5					spare, not used				
IN6					spare, not used				
IN7					spare, not used				
IN8					spare, not used				
NEG	w_opto_outc	20	Black		Opto out ground	ECUMaster CSB3 - Power/CAN/Switch I/O pin 4	200	ST185-CSB3	OPTO
OUT1	w_opto_out1	20	Yellow		Brake to CSB3 S7	ECUMaster CSB3 - Power/CAN/Switch I/O pin 21	200	ST185-CSB3	BRAKE
OUT2	w_opto_out2	20	Yellow		Reverse to CSB3 S8	ECUMaster CSB3 - Power/CAN/Switch I/O pin 22	200	ST185-CSB3	REV
OUT3	w_hl_sense_out	20	Black		Headlight sense to cluster	Centre cluster headlight-sense input (J8 pin 15, GPIO28, active-low) pin 15	1020	ST185-CSB3	HLSENSE
OUT4					spare, not used				
OUT5					spare, not used				
OUT6					spare, not used				
OUT7					spare, not used				
OUT8					spare, not used				
VCC	w_opto_vcc	20	Orange		Opto +5 V	ECUMaster CSB3 - Power/CAN/Switch I/O pin 3	200	ST185-CSB3	OPTO

T-CLUSTER-HL - Centre cluster headlight-sense input (J8 pin 15, GPIO28, active-low)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
15	w_hl_sense_out	20	Black		Headlight sense to cluster	Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3 pin OUT3	1020	ST185-CSB3	HLSENSE

OEM cruise control clutch switch C17 pin 2, B wire

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_sw5	20	Yellow		Clutch pedal sw	ECUMaster CSB3 - Power/CAN/Switch I/O pin 20	1460	ST185-CSB3	CLUTCH

OEM cruise stalk switch C14 pin 15, Y-B wire

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
15	w_sw2	20	Yellow		Cruise MAIN on	ECUMaster CSB3 - Power/CAN/Switch I/O pin 17	1220	ST185-CSB3	CRUISE

OEM cruise stalk switch C14 pin 17, BR wire

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
17	w_cru_gnd	20	Blue		Cruise stalk ground	CSB3 ground splice - CSB3 pin 2, opto board COM and stalk ground	1220	ST185-CSB3	CSB

OEM cruise stalk switch C14 pin 5, W wire

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
5	w_an2	20	Yellow		Cruise ladder	Cruise ladder splice - stalk pin 5, 1k pull-up and CSB3 A2	1220	ST185-CSB3	CRUISE

OEM dimmer rheostat R10 pin 1 (left of the steering wheel), green TAIL

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_hl_sense_in	20	Red		Lights on +12 V	Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3 pin IN3	1220	ST185-CSB3	HLSENSE

OEM stop lamp switch S9 pin 1 (pedal), G-W wire

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_sw6	16	Yellow		Brake on +12 V	Optocoupler board, 8 channel (Dan's, generic EL814) - inline at the CSB3 pin IN1	1460	ST185-CSB3	BRAKE

REVISIONS

REV	DATE	BY	DESCRIPTION
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ST185-harness - every wiring harness of the car, one drawing set

CSB3 - SWITCH INPUTS, CRUISE LADDER, OPTO BOARD (part 2 of 2)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	29 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP (part 1 of 6)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

ECU-A - Link G4X plug A (centre dash)

housing 4-1437290-1 | contacts 3-1447221-4 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
20	w81	20	White	□□	ETB relay coil (-)	B2-M4 ETB Power Relay (seated in the LOGIC box M4) - ACTIVE-LOW - ECU sinks coil- (85)...	680	ST185-CabinPower	ETB
28	w76	18	White	■■	EFI relay coil (-)	B1-R2 EFI Main Relay (seated in the POWER box R2) - ACTIVE-LOW - ECU sinks coil- (85); ...	680	ST185-CabinPower	EFI

ECU-B (Link G4X plug B, centre dash)

housing 4-1437290-0 | contacts 3-1447221-3 | pins per Link G4X XtremeX QSG p.5 pin numbering (1-34)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
5	w82_ac	18	Red	□□	ETB motor +12 V	B2-M4 ETB Power Relay (seated in the LOGIC box M4) - ACTIVE-LOW - ECU sinks coil- (85)...	680	ST185-CabinPower	ETB

BUS-SW12 - SW12 bus plug DT06-12 (glove box, by LOGIC box)

housing DT06-12SA + W12S; DT04-12PA-P021 (bussed cap) | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_cp_sw12_in	16	Red	■	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	200	ST185-CabinPower	SW12
3	w_k_fp_86_cp	18	Red	■	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	B3-K1 Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ...	2650	ST185-CabinPower	SW12
4	w_k_fan_86_cp	18	Red	■	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	B1-R4 Radiator fan relay (seated in the POWER box R4) pin 86	200	ST185-CabinPower	SW12
5	w_k_etb_86_cp	20	Red	■	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	B2-M4 ETB Power Relay (seated in the LOGIC box M4) - ACTIVE-LOW - ECU sinks coil- (85)...	200	ST185-CabinPower	SW12
7	w_k_fan2_86_cp	18	Red	■	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	B1-R5 Condenser fan relay (seated in the POWER box R5) pin 86	200	ST185-CabinPower	SW12
8	w_k_eps_86_cp	18	Red	■	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	B4 EPS pump relay (HCR150, glove box, beside the POWER box) - coil- (85) switched to gro...	200	ST185-CabinPower	SW12
9	w_k_eps_en_86_cp	18	Red	■	EFI relay k_efi -> LOGIC SW12 7.5 A fuse	B2-M6 EPS Enable Relay (seated in the LOGIC box M6) - ACTIVE-LOW - ECU sinks coil- (85...	200	ST185-CabinPower	SW12
10					spare, sealing plug				
11					spare, sealing plug				
12					spare, sealing plug				

SP-12V-SHK - Shock sensor +12 V splice - LOGIC ALMS 2A fuse to shock sensor pin A

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_at_12v_shk_tie_cp	20	Red	■□□	AMI BUS1 50 A fuse -> LOGIC ALMS 2 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	860	ST185-CabinPower	ALMS

SP-ACC12 - B2-BUS2-8 ACC 10 A to k_acc 30 and 86

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_fb_k_acc_30_cp	16	Red	■■	A/C clutch +12 V in	B2-M5 A/C clutch relay (k_acc) - amp A15 pin 8 sinks coil- (85); coil+ (86) fused +12 V from B...	200	ST185-CabinPower	ACC
	w_acc_fuse	16	Red	■■	ACC 10 A out	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	200	ST185-CabinPower	ACC

SP-AT-12V - Alarm +12 V splice - LOGIC ALM 5A fuse to alarm pin 1

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_at_12v_tie_cp	20	Red	■□■	AMI BUS1 50 A fuse -> LOGIC ALM 5 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	860	ST185-CabinPower	ALM

SP-DEV - Dash 12 V split - cluster + RealDash Pi, center dash

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_dev_cp	16	Red	■■■	EFI relay k_efi -> LOGIC DEV 15 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	980	ST185-CabinPower	DEV

SP-HEAD-COIL - Headlight coil splice - HEAD LH and RH relay coils to cowl plug IE1 pin 2

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_hl_85_cp	18	White	■■■	Head LH coil (-)	B2-M1 Headlight LH relay (seated in the LOGIC box M1) pin 85	200	ST185-CabinPower	HEAD
	w_hr_85_cp	18	White	■■■	Head RH coil (-)	B2-M2 Headlight RH relay (seated in the LOGIC box M2) pin 85	200	ST185-CabinPower	HEAD
	w_head_coil_cp	18	White	■■■	Headlight coil (-)	OEM cowl plug IE1 pin 2 (LH kick), R-Y wire pin 2	1580	ST185-CabinPower	HEAD

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP (part 1 of 6)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	30 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP (part 2 of 6)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

SP-HL - HEAD LH splice - HL 15A fuse to HEAD LH relay pins 30 and 86

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_hl_cp	14	Red		AMI BUS1 50 A fuse -> LOGIC HL 15 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	200	ST185-CabinPower	HL
	w_hl_30_cp	14	Red		AMI BUS1 50 A fuse -> LOGIC HL 15 A fuse	B2-M1 Headlight LH relay (seated in the LOGIC box M1) pin 30	200	ST185-CabinPower	HL
	w_hl_86_cp	16	Red		AMI BUS1 50 A fuse -> LOGIC HL 15 A fuse	B2-M1 Headlight LH relay (seated in the LOGIC box M1) pin 86	200	ST185-CabinPower	HL

SP-HORN - Horn feed split - LH + RH horns, front centre

splice: crimp splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_pmu_horn_cp	14	Red		LOGIC HZ 15 A fuse -> horn relay k_horn	B2-M3 Horn relay (seated in the LOGIC box M3) pin 87	2300	ST185-CabinPower	HZ

SP-HR - HEAD RH splice - HR 15A fuse to HEAD RH relay pins 30 and 86

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_hr_cp	14	Red		AMI BUS1 50 A fuse -> LOGIC HR 15 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	200	ST185-CabinPower	HR
	w_hr_30_cp	14	Red		AMI BUS1 50 A fuse -> LOGIC HR 15 A fuse	B2-M2 Headlight RH relay (seated in the LOGIC box M2) pin 30	200	ST185-CabinPower	HR
	w_hr_86_cp	16	Red		AMI BUS1 50 A fuse -> LOGIC HR 15 A fuse	B2-M2 Headlight RH relay (seated in the LOGIC box M2) pin 86	200	ST185-CabinPower	HR

SP-HZ - HAZ-HORN splice - HZ 15A fuse to the hazard feed and the horn relay

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_hz_cp	14	Red		AMI BUS1 50 A fuse -> LOGIC HZ 15 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	200	ST185-CabinPower	HZ
	w_haz_cp	14	Red		AMI BUS1 50 A fuse -> LOGIC HZ 15 A fuse	OEM IE1 pin 12 (left kick, cowl side) pin 12	1580	ST185-CabinPower	HZ
	w_horn_30_cp	16	Red		AMI BUS1 50 A fuse -> LOGIC HZ 15 A fuse	B2-M3 Horn relay (seated in the LOGIC box M3) pin 30	200	ST185-CabinPower	HZ
	w_horn_86_cp	16	Red		AMI BUS1 50 A fuse -> LOGIC HZ 15 A fuse	B2-M3 Horn relay (seated in the LOGIC box M3) pin 86	200	ST185-CabinPower	HZ

SP-IG - Alarm ignition splice - LOGIC ALMIG 2A fuse to alarm pin 3

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_at_ig_tie_cp	20	Red		EFI relay k_efi -> LOGIC ALMIG 2 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	860	ST185-CabinPower	ALMIG

DRB102-CABIN - DRB102 receptacle (RH firewall)

housing DRB12-102PAE-L018 + DRBF-1A | contacts 0460-220-1231 | pins per TE drawing DRB12-102PXE-L018 rev C3

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_cop_pwr_ac	14	Red		EFI relay k_efi -> LOGIC COP 15 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	450	ST185-CabinPower	COP
2	w_inj_pwr_ac	14	Red		EFI relay k_efi -> LOGIC INJ 15 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	450	ST185-CabinPower	INJ
3	w_strl_c	12	Red		Starter solenoid S	OEM R/B No.4 starter relay pin 4 (RH kick), starter out pin 4	700	ST185-CabinPower	STR
95	w_lam12_cp	14	Red		EFI relay k_efi -> LOGIC LAM 7.5 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	450	ST185-CabinPower	LAM

K-ACC - B2-M5 A/C clutch relay (k_acc) - amp A15 pin 8 sinks coil- (85); coil+ (86) fused +12 V from B2-BUS2-8

housing 6-1419137-4 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
30	w_fb_k_acc_30_cp	16	Red		A/C clutch +12 V in	SP-ACC12 - B2-BUS2-8 ACC 10 A to k_acc 30 and 86	200	ST185-CabinPower	ACC
87a					spare, not used				

K-EFI - B1-R2 EFI Main Relay (seated in the POWER box R2) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) permanent 12 V (LOGIC EFI86 5 A)

housing 1-1414147-0 | contacts 280756-4 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
30	w77_cp	10	Red		EFI relay k_efi (LOGIC Bus 2 feed)	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	200	ST185-CabinPower	EFI
85	w76	18	White		EFI relay coil (-)	Link G4X plug A (centre dash) pin 28	680	ST185-CabinPower	EFI
86	w_k_efi_86_cp	18	Red		AMI BUS1 50 A fuse -> LOGIC EFI86 5 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	200	ST185-CabinPower	EFI
87	w_fb_k_efi_87_cp	12	Red		AMI EFI 30 A fuse	B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse...	200	ST185-CabinPower	EFI

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP (part 2 of 6)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	31 OF 51	432 x 279 mm	DG	2026-10-29

CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP (part 3 of 6)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

K-EPS - B4 EPS pump relay (HCR150, glove box, beside the POWER box) - coil- (85) switched to ground by the MR-S pump pin B5 (R22); coil+ (86) switched 12 V - HCR 150 coil is a 2-pole plug (TE terminals 1/2, not marked 85/86), res...

housing V23132-A2001-B200 | contacts TBD | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
30	w_fb_k_eps_30_ac	8	Red		AMI EPS 60 A fuse	B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse...	200	ST185-CabinPower	EPS
86	w_k_eps_86_cp	18	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	SW12 bus plug DT06-12 (glove box, by LOGIC box) pin 8	200	ST185-CabinPower	SW12
87	w_eps_pwr_bc	8	Red		AMI EPS 60 A fuse -> EHPS relay k_eps	MR-S EHPS pump plug A pin 1	1500	ST185-CabinPower	EPS
87a					spare, not used				

K-EPS-EN - B2-M6 EPS Enable Relay (seated in the LOGIC box M6) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) switched 12 V

housing 6-1419137-4 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
30	w_fb_k_eps_en_30_cp	18	Red		EFI relay k_efi -> LOGIC EPSEN 7.5 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	200	ST185-CabinPower	EPSEN
86	w_k_eps_en_86_cp	18	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	SW12 bus plug DT06-12 (glove box, by LOGIC box) pin 9	200	ST185-CabinPower	SW12
87	w_k_eps_en_87_cp	18	Red		LOGIC EPSEN 7.5 A fuse -> EPS enable relay k_eps_...	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	200	ST185-CabinPower	EPSEN
87a					spare, not used				

K-ETB - B2-M4 ETB Power Relay (seated in the LOGIC box M4) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) switched 12 V

housing 4-1904124-2 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
30	w_fb_k_etb_30_cp	16	Red		AMI BUS1 50 A fuse -> LOGIC ETB 10 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	200	ST185-CabinPower	ETB
85	w81	20	White		ETB relay coil (-)	Link G4X plug A (centre dash) pin 20	680	ST185-CabinPower	ETB
86	w_k_etb_86_cp	20	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	SW12 bus plug DT06-12 (glove box, by LOGIC box) pin 5	200	ST185-CabinPower	SW12
87	w82_ac	18	Red		ETB motor +12 V	ECU-B (Link G4X plug B, centre dash) pin 5	680	ST185-CabinPower	ETB
87a					spare, not used				

K-FAN - B1-R4 Radiator fan relay (seated in the POWER box R4)

housing 1-1414147-0 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
86	w_k_fan_86_cp	18	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	SW12 bus plug DT06-12 (glove box, by LOGIC box) pin 4	200	ST185-CabinPower	SW12
87	w_fb_k_fan_87_cp	8	Red		AMI FAN 40 A fuse	B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse...	200	ST185-CabinPower	FAN

K-FAN2 - B1-R5 Condenser fan relay (seated in the POWER box R5)

housing 1-1393304-0 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
86	w_k_fan2_86_cp	18	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	SW12 bus plug DT06-12 (glove box, by LOGIC box) pin 7	200	ST185-CabinPower	SW12
87	w_fb_k_fan2_87_ac	8	Red		AMI FAN2 30 A fuse	B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse...	200	ST185-CabinPower	FAN2

K-FP - B3-K1 Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) switched 12 V - HCR 150 coil is a 2-pole Superseal plug (TE terminals 1/2, not marked 85/86), resistor-su...

housing V23132-A2001-B200 | contacts 282110-1 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
86	w_k_fp_86_cp	18	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	SW12 bus plug DT06-12 (glove box, by LOGIC box) pin 3	2650	ST185-CabinPower	SW12

K-HL - B2-M1 Headlight LH relay (seated in the LOGIC box M1)

housing 5-1393292-8 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
30	w_hl_30_cp	14	Red		AMI BUS1 50 A fuse -> LOGIC HL 15 A fuse	HEAD LH splice - HL 15A fuse to HEAD LH relay pins 30 and 86	200	ST185-CabinPower	HL
85	w_hl_85_cp	18	White		Head LH coil (-)	Headlight coil splice - HEAD LH and RH relay coils to cowl plug IE1 pin 2	200	ST185-CabinPower	HEAD
86	w_hl_86_cp	16	Red		AMI BUS1 50 A fuse -> LOGIC HL 15 A fuse	HEAD LH splice - HL 15A fuse to HEAD LH relay pins 30 and 86	200	ST185-CabinPower	HL
87a					spare, not used				

K-HORN - B2-M3 Horn relay (seated in the LOGIC box M3)

housing 4-1904124-3 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
30	w_horn_30_cp	16	Red		AMI BUS1 50 A fuse -> LOGIC HZ 15 A fuse	HAZ-HORN splice - HZ 15A fuse to the hazard feed and the horn relay	200	ST185-CabinPower	HZ
85	w_horn_85_cp	18	White		Horn relay coil (-)	OEM cowl plug IE1 pin 19 (LH kick), G wire pin 19	1580	ST185-CabinPower	HZ
86	w_horn_86_cp	16	Red		AMI BUS1 50 A fuse -> LOGIC HZ 15 A fuse	HAZ-HORN splice - HZ 15A fuse to the hazard feed and the horn relay	200	ST185-CabinPower	HZ
87	w_pmu_horn_cp	14	Red		LOGIC HZ 15 A fuse -> horn relay k_horn	Horn feed split - LH + RH horns, front centre	2300	ST185-CabinPower	HZ
87a					spare, not used				

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
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ST185-harness - every wiring harness of the car, one drawing set

CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP (part 3 of 6)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	32 OF 51	432 x 279 mm	DG	2026-10-29

CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP (part 4 of 6)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

K-HR - B2-M2 Headlight RH relay (seated in the LOGIC box M2)

housing 5-1393292-8 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
30	w_hr_30_cp	14	Red		AMI BUS1 50 A fuse -> LOGIC HR 15 A fuse	HEAD RH splice - HR 15A fuse to HEAD RH relay pins 30 and 86	200	ST185-CabinPower	HR
85	w_hr_85_cp	18	White		Head RH coil (-)	Headlight coil splice - HEAD LH and RH relay coils to cowl plug IE1 pin 2	200	ST185-CabinPower	HEAD
86	w_hr_86_cp	16	Red		AMI BUS1 50 A fuse -> LOGIC HR 15 A fuse	HEAD RH splice - HR 15A fuse to HEAD RH relay pins 30 and 86	200	ST185-CabinPower	HR
87a					spare, not used				

LOGIC - B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS R75): Bus 1 (permanent) and Bus 2 (switched, k_efi 30) mini fuse strips, micro relays M1-M10

housing ST185-FB-LOGIC rev 5 | contacts 2R5 | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
BUS1	w_hc_fb	8	Red		AMI BUS1 50 A fuse	B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse...	200	ST185-CabinPower	BUS1
BUS1-1	w_k_efi_86_cp	18	Red		AMI BUS1 50 A fuse -> LOGIC EFI86 5 A fuse	B1-R2 EFI Main Relay (seated in the POWER box R2) - ACTIVE-LOW - ECU sinks coil- (85); ...	200	ST185-CabinPower	EFI
BUS1-10	w_fb_k_etb_30_cp	16	Red		AMI BUS1 50 A fuse -> LOGIC ETB 10 A fuse	B2-M4 ETB Power Relay (seated in the LOGIC box M4) - ACTIVE-LOW - ECU sinks coil- (85)...	200	ST185-CabinPower	ETB
BUS1-2	w_hl_cp	14	Red		AMI BUS1 50 A fuse -> LOGIC HL 15 A fuse	HEAD LH splice - HL 15A fuse to HEAD LH relay pins 30 and 86	200	ST185-CabinPower	HL
BUS1-3	w_hr_cp	14	Red		AMI BUS1 50 A fuse -> LOGIC HR 15 A fuse	HEAD RH splice - HR 15A fuse to HEAD RH relay pins 30 and 86	200	ST185-CabinPower	HR
BUS1-4	w_hz_cp	14	Red		AMI BUS1 50 A fuse -> LOGIC HZ 15 A fuse	HAZ-HORN splice - HZ 15A fuse to the hazard feed and the horn relay	200	ST185-CabinPower	HZ
BUS1-7	w_at_12v_tie_cp	20	Red		AMI BUS1 50 A fuse -> LOGIC ALM 5 A fuse	Alarm +12 V splice - LOGIC ALM 5A fuse to alarm pin 1	860	ST185-CabinPower	ALM
BUS1-8	w_at_12v_shk_tie_cp	20	Red		AMI BUS1 50 A fuse -> LOGIC ALMS 2 A fuse	Shock sensor +12 V splice - LOGIC ALMS 2A fuse to shock sensor pin A	860	ST185-CabinPower	ALMS
BUS2	w77_cp	10	Red		EFI relay k_efi (LOGIC Bus 2 feed)	B1-R2 EFI Main Relay (seated in the POWER box R2) - ACTIVE-LOW - ECU sinks coil- (85); ...	200	ST185-CabinPower	EFI
BUS2-1	w_lam12_cp	14	Red		EFI relay k_efi -> LOGIC LAM 7.5 A fuse	DRB102 receptacle (RH firewall) pin 95	450	ST185-CabinPower	LAM
BUS2-12 IN	w_k_eps_en_87_cp	18	Red		LOGIC EPSEN 7.5 A fuse -> EPS enable relay k_eps_...	B2-M6 EPS Enable Relay (seated in the LOGIC box M6) - ACTIVE-LOW - ECU sinks coil- (85...	200	ST185-CabinPower	EPSEN
BUS2-13 IN	w_start_in_cp	20	Yellow		Start sw ST1 feed	OEM ST1 lead B-W at old R/B No.4 pin 1, RH kick panel pin 1	450	ST185-CabinPower	B30
BUS2-2	w_cp_sw12_in	16	Red		EFI relay k_efi -> LOGIC SW12 7.5 A fuse	SW12 bus plug DT06-12 (glove box, by LOGIC box) pin 1	200	ST185-CabinPower	SW12
BUS2-3	w_cop_pwr_ac	14	Red		EFI relay k_efi -> LOGIC COP 15 A fuse	DRB102 receptacle (RH firewall) pin 1	450	ST185-CabinPower	COP
BUS2-4	w_inj_pwr_ac	14	Red		EFI relay k_efi -> LOGIC INJ 15 A fuse	DRB102 receptacle (RH firewall) pin 2	450	ST185-CabinPower	INJ
BUS2-6	w_dev_cp	16	Red		EFI relay k_efi -> LOGIC DEV 15 A fuse	Dash 12 V split - cluster + RealDash Pi, center dash	980	ST185-CabinPower	DEV
BUS2-7	w_fb_k_eps_en_30_cp	18	Red		EFI relay k_efi -> LOGIC EPSEN 7.5 A fuse	B2-M6 EPS Enable Relay (seated in the LOGIC box M6) - ACTIVE-LOW - ECU sinks coil- (85...	200	ST185-CabinPower	EPSEN
BUS2-8	w_acc_fuse	16	Red		EFI relay k_efi -> LOGIC ACC 10 A fuse	SP-ACC12 - B2-BUS2-8 ACC 10 A to k_acc 30 and 86	200	ST185-CabinPower	ACC
BUS2-9	w_at_ig_tie_cp	20	Red		EFI relay k_efi -> LOGIC ALMIG 2 A fuse	Alarm ignition splice - LOGIC ALMIG 2A fuse to alarm pin 3	860	ST185-CabinPower	ALMIG

MRS-A - MR-S EHPS pump plug A

housing 90980-12068 | contacts TBD | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_eps_pwr_bc	8	Red		AMI EPS 60 A fuse -> EHPS relay k_eps	B4 EPS pump relay (HCR150, glove box, beside the POWER box) - coil- (85) switched to gro...	1500	ST185-CabinPower	EPS

POWER - B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fusebox rev 7, DECISIONS R75, R91): the tinned copper + bar IS the AMI bar (7 AMI IN relay feeds, 7 AMI OUT; no 1/0 link cable), NEG ...

housing ST185-FB-POWER rev 7 | contacts 2R5 | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
AMI IN-1	w_rb4_feed	8	Red		AMI HTR 40 A fuse	OEM R/B No.4 40 A HEATER fuse pin 1 (RH kick) pin 1	450	ST185-CabinPower	HTR
AMI IN-2	w_fb_k_efi_87_cp	12	Red		AMI EFI 30 A fuse	B1-R2 EFI Main Relay (seated in the POWER box R2) - ACTIVE-LOW - ECU sinks coil- (85); ...	200	ST185-CabinPower	EFI
AMI IN-4	w_fb_k_fan_87_cp	8	Red		AMI FAN 40 A fuse	B1-R4 Radiator fan relay (seated in the POWER box R4) pin 87	200	ST185-CabinPower	FAN
AMI IN-5	w_fb_k_fan2_87_ac	8	Red		AMI FAN2 30 A fuse	B1-R5 Condenser fan relay (seated in the POWER box R5) pin 87	200	ST185-CabinPower	FAN2
AMI IN-6	w_rb2_feed	10	Red		AMI PWR 30 A fuse	OEM R/B No.2 30 A POWER fuse pin 1 (LH kick) pin 1	1580	ST185-CabinPower	PWR
AMI OUT-1	w_hc_fb	8	Red		AMI BUS1 50 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	200	ST185-CabinPower	BUS1
AMI OUT-5	w_pdb_jb1	6	Red		AMI JB1 60 A fuse	OEM J/B No.1 1l pin 1 (left kick) pin 1	1580	ST185-CabinPower	JB1
AMI OUT-6	w_fb_k_eps_30_ac	8	Red		AMI EPS 60 A fuse	B4 EPS pump relay (HCR150, glove box, beside the POWER box) - coil- (85) switched to gro...	200	ST185-CabinPower	EPS

FOG-CAP-1 - Cut and capped fog lead (circuit removed)

terminal heat-shrink cap

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_fog_rb3_1	18	White		Fog circuit removed	OEM R/B No.3 fog relay pin 1 (fog lamp feed), RH kick / dash pin 1	200	ST185-CabinPower	FOG

FOG-CAP-4 - Cut and capped fog lead (circuit removed)

terminal heat-shrink cap

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_fog_rb3_4	18	White		Fog circuit removed	OEM R/B No.3 fog relay pin 4 (20 A FR FOG in), RH kick / dash pin 4	200	ST185-CabinPower	FOG

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP (part 4 of 6)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	33 OF 51	432 x 279 mm	DG	2026-10-29

CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP (part 5 of 6)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

FOG-CAP-6 - Cut and capped fog lead (circuit removed)

terminal heat-shrink cap

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_fog_sw	18	White		Fog circuit removed	OEM fog light switch F12 pin 6 (coil-), RH kick / dash pin 6	1580	ST185-CabinPower	FOG

OEM fog light switch F12 pin 6 (coil-), RH kick / dash

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
6	w_fog_sw	18	White		Fog circuit removed	Cut and capped fog lead (circuit removed)	1580	ST185-CabinPower	FOG

OEM IE1 pin 12 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
12	w_haz_cp	14	Red		AMI BUS1 50 A fuse -> LOGIC HZ 15 A fuse	HAZ-HORN splice - HZ 15A fuse to the hazard feed and the horn relay	1580	ST185-CabinPower	HZ

OEM cowl plug IE1 pin 19 (LH kick), G wire

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
19	w_horn_85_cp	18	White		Horn relay coil (-)	B2-M3 Horn relay (seated in the LOGIC box M3) pin 85	1580	ST185-CabinPower	HZ

OEM cowl plug IE1 pin 2 (LH kick), R-Y wire

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_head_coil_cp	18	White		Headlight coil (-)	Headlight coil splice - HEAD LH and RH relay coils to cowl plug IE1 pin 2	1580	ST185-CabinPower	HEAD

OEM J/B No.1 1I pin 1 (left kick)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_pdb_jb1	6	Red		AMI JB1 60 A fuse	B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse...	1580	ST185-CabinPower	JB1

OEM R/B No.2 30 A POWER fuse pin 1 (LH kick)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_rb2_feed	10	Red		POWER fuse feed	B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse...	1580	ST185-CabinPower	PWR

OEM R/B No.3 fog relay pin 1 (fog lamp feed), RH kick / dash

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_fog_rb3_1	18	White		Fog circuit removed	Cut and capped fog lead (circuit removed)	200	ST185-CabinPower	FOG

OEM R/B No.3 fog relay pin 4 (20 A FR FOG in), RH kick / dash

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
4	w_fog_rb3_4	18	White		Fog circuit removed	Cut and capped fog lead (circuit removed)	200	ST185-CabinPower	FOG

OEM R/B No.4 40 A HEATER fuse pin 1 (RH kick)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_rb4_feed	8	Red		Heater fuse feed	B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse...	450	ST185-CabinPower	HTR

OEM ST1 lead B-W at old R/B No.4 pin 1, RH kick panel

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_start_in_cp	20	Yellow		Start sw ST1 feed	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	450	ST185-CabinPower	B30

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP (part 5 of 6)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	34 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP (part 6 of 6)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

OEM R/B No.4 starter relay pin 4 (RH kick), starter out

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
4	w_strl_c	12	Red	□■	Starter solenoid S	DRB102 receptacle (RH firewall) pin 3	700	ST185-CabinPower	STR

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
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ST185-harness - every wiring harness of the car, one drawing set

CABIN POWER - POWER + LOGIC BOXES TO EVERY CABIN FEED, RELAY COIL AND OEM TAP (part 6 of 6)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	35 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

CLUSTER LEDS - LED BOARD ON THE METER PLUGS C11 / C12 (part 1 of 2)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

SP-LED-12V - LED +12 V splice - IG +12 V from C12 pin 9 to the park brake, charge and oil LEDs

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_ig_tap	20	Red		LED IG +12 V	OEM cluster plug C12 pin 9, B-O wire pin 9	200	ST185-ClusterLED	LIG
	w_chg_a	20	Red		Charge LED +	Charge LED pin 1	200	ST185-ClusterLED	LCHG
	w_oil_a	20	Red		Oil LED +	Oil pressure LED pin 1	200	ST185-ClusterLED	LOIL
	w_pk_a	20	Red		Park brake LED +	Park brake LED pin 1	200	ST185-ClusterLED	LPK

SP-LED-GND - LED ground splice - turn LH, turn RH and high beam LED returns to the LH kick stud

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_hb_k	20	Black		High beam LED gnd	High beam LED pin 2	200	ST185-ClusterLED	LHB
	w_tl_k	20	Black		Turn LH LED gnd	Turn LH LED pin 2	200	ST185-ClusterLED	LTL
	w_tr_k	20	Black		Turn RH LED gnd	Turn RH LED pin 2	200	ST185-ClusterLED	LTR
	w_led_gnd_out	20	Black		LED common ground	Ground stud, LH kick panel (EWD ground point ID)	1550	ST185-ClusterLED	LGND

D-CHG - Charge LED

inline diode TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_chg_a	20	Red		Charge LED +	LED +12 V splice - IG +12 V from C12 pin 9 to the park brake, charge and oil LEDs	200	ST185-ClusterLED	LCHG
2	w_chg_k	20	White		Charge lamp drive	OEM cluster plug C12 pin 8, Y wire pin 8	200	ST185-ClusterLED	LCHG

D-HB - High beam LED

inline diode TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_hb_a	20	Red		High beam LED +	OEM cluster plug C11 pin 12, R-L wire pin 12	200	ST185-ClusterLED	LHB
2	w_hb_k	20	Black		High beam LED gnd	LED ground splice - turn LH, turn RH and high beam LED returns to the LH kick stud	200	ST185-ClusterLED	LHB

D-OIL - Oil pressure LED

inline diode TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_oil_a	20	Red		Oil LED +	LED +12 V splice - IG +12 V from C12 pin 9 to the park brake, charge and oil LEDs	200	ST185-ClusterLED	LOIL
2	w_oil_k	20	White		Oil LED sink (CSB3)	ECUMaster CSB3 - Power/CAN/Switch I/O pin 26	1020	ST185-ClusterLED	LOIL

D-PK - Park brake LED

inline diode TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_pk_a	20	Red		Park brake LED +	LED +12 V splice - IG +12 V from C12 pin 9 to the park brake, charge and oil LEDs	200	ST185-ClusterLED	LPK
2	w_pk_k	20	Yellow		Park brake sw (-)	OEM cluster plug C12 pin 1, R-G wire pin 1	200	ST185-ClusterLED	LPK

D-TL - Turn LH LED

inline diode TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_tl_a	20	Red		Turn LH LED +	OEM cluster plug C11 pin 2, G-B wire pin 2	200	ST185-ClusterLED	LTL
2	w_tl_k	20	Black		Turn LH LED gnd	LED ground splice - turn LH, turn RH and high beam LED returns to the LH kick stud	200	ST185-ClusterLED	LTL

D-TR - Turn RH LED

inline diode TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_tr_a	20	Red		Turn RH LED +	OEM cluster plug C11 pin 11, G-Y wire pin 11	200	ST185-ClusterLED	LTR
2	w_tr_k	20	Black		Turn RH LED gnd	LED ground splice - turn LH, turn RH and high beam LED returns to the LH kick stud	200	ST185-ClusterLED	LTR

CSB3 - ECUMaster CSB3 - Power/CAN/Switch I/O

housing HD36-24-33SE | contacts 0462-201-2031 | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
26	w_oil_k	20	White		Oil LED sink (CSB3)	Oil pressure LED pin 2	1020	ST185-ClusterLED	LOIL

REVISIONS

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K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

CLUSTER LEDS - LED BOARD ON THE METER PLUGS C11 / C12 (part 1 of 2)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	36 OF 51	432 x 279 mm	DG	2026-10-29

CLUSTER LEDS - LED BOARD ON THE METER PLUGS C11 / C12 (part 2 of 2)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

GND-ID - Ground stud, LH kick panel (EWD ground point ID)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_led_gnd_out	20	Black		LED common ground	LED ground splice - turn LH, turn RH and high beam LED returns to the LH kick stud	1550	ST185-ClusterLED	LGND

OEM cluster plug C11 pin 11, G-Y wire

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
11	w_tr_a	20	Red		Turn RH LED +	Turn RH LED pin 1	200	ST185-ClusterLED	LTR

OEM cluster plug C11 pin 12, R-L wire

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
12	w_hb_a	20	Red		High beam LED +	High beam LED pin 1	200	ST185-ClusterLED	LHB

OEM cluster plug C11 pin 2, G-B wire

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_tl_a	20	Red		Turn LH LED +	Turn LH LED pin 1	200	ST185-ClusterLED	LTL

OEM cluster plug C12 pin 1, R-G wire

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_pk_k	20	Yellow		Park brake sw (-)	Park brake LED pin 2	200	ST185-ClusterLED	LPK

OEM cluster plug C12 pin 8, Y wire

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
8	w_chg_k	20	White		Charge lamp drive	Charge LED pin 2	200	ST185-ClusterLED	LCHG

OEM cluster plug C12 pin 9, B-O wire

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
9	w_ig_tap	20	Red		LED IG +12 V	LED +12 V splice - IG +12 V from C12 pin 9 to the park brake, charge and oil LEDs	200	ST185-ClusterLED	LIG

REVISIONS

REV	DATE	BY	DESCRIPTION
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ST185-harness - every wiring harness of the car, one drawing set

CLUSTER LEDS - LED BOARD ON THE METER PLUGS C11 / C12 (part 2 of 2)

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	37 OF 51	432 x 279 mm	DG	2026-10-29

HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS (part 1 of 9)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

SP-CLR - Clearance feed split - F7 LH + F8 RH

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_clr	16	Red		Clearance lamp feed	OEM J/B No.1 1H pin 6 (left kick) pin 1H-6	3680	ST185-EngineRoom-C	CLR
	w_clr_f7	16	Red		LH clearance +	LH front turn / clearance lamp F7 pin 1 pin 1	900	ST185-EngineRoom-C	CLR
	w_clr_f8	16	Red		RH clearance +	RH front turn / clearance lamp F8 pin 1 pin 1	900	ST185-EngineRoom-C	CLR

SP-HL-HIGH - High-beam return join - H1 + H2 pin 1 to IE1 pin 11

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_hl_high_h1	16	Red		LH high-beam return	LH headlamp H1 pin 1 pin 1	900	ST185-EngineRoom-C	HHI
	w_hl_high_h2	16	Red		RH high-beam return	RH headlamp H2 pin 1 pin 1	900	ST185-EngineRoom-C	HHI
	w_hl_high	14	Red		High-beam return	OEM IE1 pin 11 (left kick, cowl side) pin 11	3680	ST185-EngineRoom-C	HHI

SP-HL-LOW - Low-beam return join - H1 + H2 pin 2 to IE1 pin 18

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_hl_low_h1	16	Red		LH low-beam return	LH headlamp H1 pin 2 pin 2	900	ST185-EngineRoom-C	HLO
	w_hl_low_h2	16	Red		RH low-beam return	RH headlamp H2 pin 2 pin 2	900	ST185-EngineRoom-C	HLO
	w_hl_low	14	Red		Low-beam return	OEM IE1 pin 18 (left kick, cowl side) pin 18	3680	ST185-EngineRoom-C	HLO

SP-HORN - Horn feed split - LH + RH horns, front centre

splice: crimp splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_horn_lh	16	Red		LOGIC HZ 15 A fuse -> horn relay k_horn	Horn LH (EWD H3) - custom DT 2-way pin 1	200	ST185-EngineRoom-C	HZ
	w_horn_rh	16	Red		LOGIC HZ 15 A fuse -> horn relay k_horn	Horn RH (EWD H4) - custom DT 2-way pin 1	200	ST185-EngineRoom-C	HZ

SP-RTR-CLOSE - Retract close-limit join - R2 + R3 pin 3 to IE1 pin 9

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_rtr_close	18	Yellow		Retract close limit	OEM IE1 pin 9 (left kick, cowl side) pin 9	3680	ST185-EngineRoom-C	RTM
	w_rtr_close_r2	18	Yellow		Retract close limit	LH retract motor R2 pin 3 pin 3	900	ST185-EngineRoom-C	RTM
	w_rtr_close_r3	18	Yellow		Retract close limit	RH retract motor R3 pin 3 pin 3	900	ST185-EngineRoom-C	RTM

SP-RTR-GND-R2 - LH retract motor ground - 16 AWG pigtail to 12 AWG, at R2

splice: crimp splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_rtr_gnd_r2	16	Black		LH retract ground	LH retract motor R2 pin 5 pin 5	200	ST185-EngineRoom-C	RTM
	w_rtr_gnd_r2_t	12	Black		LH retract ground	Ground stud EB, LH front fender (EWD ground point EB)	200	ST185-EngineRoom-C	RTM

SP-RTR-GND-R3 - RH retract motor ground - 16 AWG pigtail to 12 AWG, at R3

splice: crimp splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_rtr_gnd_r3	16	Black		RH retract ground	RH retract motor R3 pin 5 pin 5	200	ST185-EngineRoom-C	RTM
	w_rtr_gnd_r3_t	12	Black		RH retract ground	Ground stud EB, LH front fender (EWD ground point EB)	1600	ST185-EngineRoom-C	RTM

SP-RTR-OPEN - Retract open-limit join - R2 + R3 pin 1 to IE1 pin 16

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_rtr_open	18	Yellow		Retract open limit	OEM IE1 pin 16 (left kick, cowl side) pin 16	3680	ST185-EngineRoom-C	RTM
	w_rtr_open_r2	18	Yellow		Retract open limit	LH retract motor R2 pin 1 pin 1	900	ST185-EngineRoom-C	RTM
	w_rtr_open_r3	18	Yellow		Retract open limit	RH retract motor R3 pin 1 pin 1	900	ST185-EngineRoom-C	RTM

SP-RTR-R2-4 - LH retract motor feed - 12 AWG to 16 AWG pigtail, at R2

splice: crimp splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_rtr_r2_4	12	Red		LH retract motor 4	OEM IE1 pin 14 (left kick, cowl side) pin 14	4380	ST185-EngineRoom-C	RTM
	w_rtr_r2_4_p	16	Red		LH retract motor 4	LH retract motor R2 pin 4 pin 4	200	ST185-EngineRoom-C	RTM

REVISIONS

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I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
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HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS (part 1 ...

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	38 OF 51	432 x 279 mm	DG	2026-10-29

HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS (part 2 of 9)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

SP-RTR-R3-4 - RH retract motor feed - 12 AWG to 16 AWG pigtail, at R3

splice: crimp splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_rtr_r3_4	12	Red		RH retract motor 4	OEM IE1 pin 7 (left kick, cowl side) pin 7	2980	ST185-EngineRoom-C	RTM
	w_rtr_r3_4_p	16	Red		RH retract motor 4	RH retract motor R3 pin 4 pin 4	200	ST185-EngineRoom-C	RTM

DRB102-ENGINE - BULKHEAD

housing DRB16-102SAE-L018 | contacts 0462-201-1631 | pins per TE drawing DRB12-102PXE-L018 rev C3

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
83	w_hv_mrs_en_e	20	Red		EFI relay k_efi -> LOGIC MRS 7.5 A fuse	MR-S EHPS pump plug C pin 1	1055	ST185-EngineRoom-C	EPSEN
84	w_mrs_e	20	White		EHPS pump speed	MR-S EHPS pump plug B pin 2	1055	ST185-EngineRoom-C	MRS
85	w_eps_rly_e	18	White		EHPS relay coil (-)	MR-S EHPS pump plug B pin 5	1055	ST185-EngineRoom-C	EPS

ALT-PLUG - Alternator G2 - round 3-pin plug (Toyota 90980-10341; 1993 EWD p.74-75 G2 (B) pinout)

housing 90980-10341 | contacts TBD | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_alt_s_c	20	White		Alt S battery sense	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	2660	ST185-EngineRoom-C	ALTS
2	w_alt_l_c	20	White		Alt L charge lamp	OEM IG2 pin 16 (under the ECU, cowl side) pin 16	3140	ST185-EngineRoom-C	ALTL

FAN-COND - Uprated condenser fan (+)

housing DTHD06-1-8S | contacts 0462-203-08141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_fan2_pwr	8	Red		Cond fan +12 V	B1-R5 Condenser fan relay (seated in the POWER box R5) pin 30	1900	ST185-EngineRoom-C	FAN2

FAN-COND-GND - Uprated condenser fan (GND)

housing DTHD06-1-8S | contacts 0462-203-08141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_fan2_gnd	8	Black		Cond fan ground	Ground stud EA, RH front fender (EWD ground point EA)	500	ST185-EngineRoom-C	FAN2

FAN-RAD - Uprated radiator fan (+)

housing DTHD06-1-8S | contacts 0462-203-08141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_fan_pwr	8	Red		AMI FAN 40 A fuse -> rad fan relay k_fan	B1-R4 Radiator fan relay (seated in the POWER box R4) pin 30	2200	ST185-EngineRoom-C	FAN

FAN-RAD-GND - Uprated radiator fan (GND)

housing DTHD06-1-8S | contacts 0462-203-08141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_fan_gnd	8	Black		Rad fan ground	Ground stud EA, RH front fender (EWD ground point EA)	800	ST185-EngineRoom-C	FAN

HORN-LH - Horn LH (EWD H3) - custom DT 2-way

housing DT06-2S-CE05 + W2S | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_horn_lh	16	Red		LOGIC HZ 15 A fuse -> horn relay k_horn	Horn feed split - LH + RH horns, front centre	200	ST185-EngineRoom-C	HZ
2	w_horn_lh_g	16	Black		Horn LH ground	Ground stud EA, RH front fender (EWD ground point EA)	900	ST185-EngineRoom-C	HZ

HORN-RH - Horn RH (EWD H4) - custom DT 2-way

housing DT06-2S-CE05 + W2S | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_horn_rh	16	Red		LOGIC HZ 15 A fuse -> horn relay k_horn	Horn feed split - LH + RH horns, front centre	200	ST185-EngineRoom-C	HZ
2	w_horn_rh_g	16	Black		Horn RH ground	Ground stud EA, RH front fender (EWD ground point EA)	900	ST185-EngineRoom-C	HZ

K-ACC - B2-M5 A/C clutch relay (k_acc) - amp A15 pin 8 sinks coil- (85); coil+ (86) fused +12 V from B2-BUS2-8

housing 6-1419137-4 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
87	w_acc_clutch	16	Red		A/C clutch +12 V	A/C compressor A4 pin 4 pin 4	1350	ST185-EngineRoom-C	ACC

REVISIONS

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DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	39 OF 51	432 x 279 mm	DG	2026-10-29

HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS (part 3 of 9)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

K-FAN - B1-R4 Radiator fan relay (seated in the POWER box R4)

housing 1-1414147-0 | contacts 280755-4 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
30	w_fan_pwr	8	Red		AMI FAN 40 A fuse -> rad fan relay k_fan	Uprated radiator fan (+) pin 1	2200	ST185-EngineRoom-C	FAN

K-FAN2 - B1-R5 Condenser fan relay (seated in the POWER box R5)

housing 1-1393304-0 | contacts 280755-4 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
30	w_fan2_pwr	8	Red		Cond fan +12 V	Uprated condenser fan (+) pin 1	1900	ST185-EngineRoom-C	FAN2

K-HL - B2-M1 Headlight LH relay (seated in the LOGIC box M1)

housing 5-1393292-8 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
87	w_pmu_hl	14	Red		LOGIC HL 15 A fuse -> HEAD LH relay k_hl	LH headlamp H1 pin 3 pin 3	3000	ST185-EngineRoom-C	HL

K-HR - B2-M2 Headlight RH relay (seated in the LOGIC box M2)

housing 5-1393292-8 | contacts 42281-1 | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
87	w_pmu_hr	14	Red		LOGIC HR 15 A fuse -> HEAD RH relay k_hr	RH headlamp H2 pin 3 pin 3	1600	ST185-EngineRoom-C	HR

LOGIC - B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS R75): Bus 1 (permanent) and Bus 2 (switched, k_efi 30) mini fuse strips, micro relays M1-M10

housing ST185-FB-LOGIC rev 5 | contacts 1-968851-1 | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
BUS1-5	w_pmu_dome	14	Red		AMI BUS1 50 A fuse -> LOGIC DM 20 A fuse	OEM J/B No.1 1H pin 2 (left kick) pin 2	1580	ST185-EngineRoom-C	DM
BUS1-6	w_rb2_wl	14	Red		AMI BUS1 50 A fuse -> LOGIC WL 15 A fuse	OEM W-L lead at old R/B2 pin 2 (left kick) pin 2	1580	ST185-EngineRoom-C	WL
BUS1-9	w_alt_s_c	20	White		Alt S battery sense	Alternator G2 - round 3-pin plug (Toyota 90980-10341; 1993 EWD p.74-75 G2 (B) pinout) pin 1	2660	ST185-EngineRoom-C	ALTS

MRS-A - MR-S EHPS pump plug A

housing 90980-12068 | contacts TBD | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_eps_gnd	8	Black		EHPS pump ground	Engine block ground stud, lug 3 (Route 1 point 5)	1750	ST185-EngineRoom-C	EPS

MRS-B - MR-S EHPS pump plug B

housing 90980-10897 | contacts TBD | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1					spare, not used				
2	w_mrs_e	20	White		EHPS pump speed	BULKHEAD pin 84	1055	ST185-EngineRoom-C	MRS
3					spare, not used				
4					spare, not used				
5	w_eps_rly_e	18	White		EHPS relay coil (-)	BULKHEAD pin 85	1055	ST185-EngineRoom-C	EPS
6					spare, not used				

MRS-C - MR-S EHPS pump plug C

housing 90980-10942 | contacts TBD | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_hv_mrs_en_e	20	Red		EFI relay k_efi -> LOGIC MRS 7.5 A fuse	BULKHEAD pin 83	1055	ST185-EngineRoom-C	EPSEN
2					spare, not used				

POWER - B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fusebox rev 7, DECISIONS R75, R91): the tinned copper + bar IS the AMI bar (7 AMI IN relay feeds, 7 AMI OUT; no 1/0 link cable), NEG ...

housing ST185-FB-POWER rev 7 | contacts 2R5 | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
AMI OUT-2	w_pmu_rtr	10	Red		AMI RTR 30 A fuse	OEM IE1 pin 3 (left kick, cowl side) pin 3	1580	ST185-EngineRoom-C	RTR
AMI OUT-3	w_pdb_am1	8	Red		AMI AM1 40 A fuse	OEM IE1 pin 10 (left kick, cowl side) pin 10	1580	ST185-EngineRoom-C	AM1
AMI OUT-4	w_pdb_am2	10	Red		AMI AM2 30 A fuse	OEM IE1 pin 17 (left kick, cowl side) pin 17	1580	ST185-EngineRoom-C	AM2

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DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	40 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS (part 4 of 9)

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POWER-STUDS - POWER box bus zone (rev 7, glove box)

housing ST185-FB-POWER rev 7 BUS ZONE | contacts TBD | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
NEG	w_gnd_id	1/0	Black		POWER box gnd to ID	Ground stud ID, left kick panel (EWD ground point ID)	1580	ST185-EngineRoom-C	BATN
NEG	w_gnd_ig	1/0	Black		POWER box gnd to IG	Ground bolt IG, right kick (EWD ground point IG)	450	ST185-EngineRoom-C	BATN

GND-EA - Ground stud EA, RH front fender (EWD ground point EA)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_fan2_gnd	8	Black		Cond fan ground	Uprated condenser fan (GND) pin 1	500	ST185-EngineRoom-C	FAN2
	w_horn_lh_g	16	Black		Horn LH ground	Horn LH (EWD H3) - custom DT 2-way pin 2	900	ST185-EngineRoom-C	HZ
	w_horn_rh_g	16	Black		Horn RH ground	Horn RH (EWD H4) - custom DT 2-way pin 2	900	ST185-EngineRoom-C	HZ
	w_fan_gnd	8	Black		Rad fan ground	Uprated radiator fan (GND) pin 1	800	ST185-EngineRoom-C	FAN
	w_f8_gnd	16	Black		RH front lamp gnd	RH front turn / clearance lamp F8 pin 2 pin 2	200	ST185-EngineRoom-C	CLR

GND-EB - Ground stud EB, LH front fender (EWD ground point EB)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_f7_gnd	16	Black		LH front lamp gnd	LH front turn / clearance lamp F7 pin 2 pin 2	200	ST185-EngineRoom-C	CLR
	w_rtr_gnd_r2_t	12	Black		LH retract ground	LH retract motor ground - 16 AWG pigtail to 12 AWG, at R2	200	ST185-EngineRoom-C	RTM
	w_rtr_gnd_r3_t	12	Black		RH retract ground	RH retract motor ground - 16 AWG pigtail to 12 AWG, at R3	1600	ST185-EngineRoom-C	RTM
	w_gnd_eb	4	Black		Block strap to EB	Engine block ground stud, lug 2 (Route 1 point 5)	4050	ST185-EngineRoom-C	BATN

GND-ID - Ground stud ID, left kick panel (EWD ground point ID)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_gnd_id	1/0	Black		POWER box gnd to ID	POWER box bus zone (rev 7, glove box) pin NEG	1580	ST185-EngineRoom-C	BATN

GND-IG - Ground bolt IG, right kick (EWD ground point IG)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_gnd_ig	1/0	Black		POWER box gnd to IG	POWER box bus zone (rev 7, glove box) pin NEG	450	ST185-EngineRoom-C	BATN

T-ENG-BLOCK-2 - Engine block ground stud, lug 2 (Route 1 point 5)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_gnd_eb	4	Black		Block strap to EB	Ground stud EB, LH front fender (EWD ground point EB)	4050	ST185-EngineRoom-C	BATN

T-ENG-BLOCK-3 - Engine block ground stud, lug 3 (Route 1 point 5)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_eps_gnd	8	Black		EHPS pump ground	MR-S EHPS pump plug A pin 2	1750	ST185-EngineRoom-C	EPS

A/C compressor A4 pin 4

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
4	w_acc_clutch	16	Red		LOGIC ACC 10 A fuse -> A/C clutch relay k_acc	B2-M5 A/C clutch relay (k_acc) - amp A15 pin 8 sinks coil- (85); coil+ (86) fused +12 V from B...	1350	ST185-EngineRoom-C	ACC

A/C pressure switch A5 pin 1

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_acp_1	18	Yellow		A/C press sw feed	OEM R/B No.4 10 A A/C fuse pin 2 (RH kick), V-R out pin 2	1600	ST185-EngineRoom-C	ACPS

LH front turn / clearance lamp F7 pin 1

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_clr_f7	16	Red		LH clearance +	Clearance feed split - F7 LH + F8 RH	900	ST185-EngineRoom-C	CLR

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LH front turn / clearance lamp F7 pin 2

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_f7_gnd	16	Black	☐	LH front lamp gnd	Ground stud EB, LH front fender (EWD ground point EB)	200	ST185-EngineRoom-C	CLR

LH front turn / clearance lamp F7 pin 3

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
3	w_turn_f7	18	Red	■■■	LH front turn +	OEM J/B No.1 1H pin 4 (left kick) pin 1H-4	4380	ST185-EngineRoom-C	TURN

RH front turn / clearance lamp F8 pin 1

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_clr_f8	16	Red	☐	RH clearance +	Clearance feed split - F7 LH + F8 RH	900	ST185-EngineRoom-C	CLR

RH front turn / clearance lamp F8 pin 2

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_f8_gnd	16	Black	☐	RH front lamp gnd	Ground stud EA, RH front fender (EWD ground point EA)	200	ST185-EngineRoom-C	CLR

RH front turn / clearance lamp F8 pin 3

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
3	w_turn_f8	18	Red	■■■	RH front turn +	OEM J/B No.1 1H pin 8 (left kick) pin 1H-8	2980	ST185-EngineRoom-C	TURN

Front wiper motor F9 pin 1 (cowl, driver side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_wip_hi	14	Red	■■	Wiper HIGH	OEM IE1 pin 20 (left kick, cowl side) pin 20	800	ST185-EngineRoom-C	WIP

Front wiper motor F9 pin 2 (cowl, driver side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_wip_lo	14	Red	■■	Wiper LOW	OEM IE1 pin 13 (left kick, cowl side) pin 13	800	ST185-EngineRoom-C	WIP

Front wiper motor F9 pin 3 (cowl, driver side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
3	w_wip_park	14	Yellow	■■	Wiper park sw	OEM IE1 pin 6 (left kick, cowl side) pin 6	800	ST185-EngineRoom-C	WIP

Front wiper motor F9 pin 4 (cowl, driver side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
4	w_wip_b	14	Red	■■	Wiper +B	OEM J/B No.1 1H pin 5 (left kick) pin 1H-5	800	ST185-EngineRoom-C	WIP

LH headlamp H1 pin 1

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_hl_high_h1	16	Red	□■	LH high-beam return	High-beam return join - H1 + H2 pin 1 to IE1 pin 11	900	ST185-EngineRoom-C	HHI

LH headlamp H1 pin 2

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_hl_low_h1	16	Red	□□	LH low-beam return	Low-beam return join - H1 + H2 pin 2 to IE1 pin 18	900	ST185-EngineRoom-C	HLO

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS (part 5 ...

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	42 OF 51	432 x 279 mm	DG	2026-10-29

HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS (part 6 of 9)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

LH headlamp H1 pin 3

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
3	w_pmu_hl	14	Red		LOGIC HL 15 A fuse -> HEAD LH relay k_hl	B2-M1 Headlight LH relay (seated in the LOGIC box M1) pin 87	3000	ST185-EngineRoom-C	HL

RH headlamp H2 pin 1

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_hl_high_h2	16	Red		RH high-beam return	High-beam return join - H1 + H2 pin 1 to IE1 pin 11	900	ST185-EngineRoom-C	HHI

RH headlamp H2 pin 2

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_hl_low_h2	16	Red		RH low-beam return	Low-beam return join - H1 + H2 pin 2 to IE1 pin 18	900	ST185-EngineRoom-C	HLO

RH headlamp H2 pin 3

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
3	w_pmu_hr	14	Red		LOGIC HR 15 A fuse -> HEAD RH relay k_hr	B2-M2 Headlight RH relay (seated in the LOGIC box M2) pin 87	1600	ST185-EngineRoom-C	HR

OEM IE1 pin 10 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
10	w_pdb_am1	8	Red		AMI AM1 40 A fuse	B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse...	1580	ST185-EngineRoom-C	AM1

OEM IE1 pin 11 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
11	w_hl_high	14	Red		High-beam return	High-beam return join - H1 + H2 pin 1 to IE1 pin 11	3680	ST185-EngineRoom-C	HHI

OEM IE1 pin 13 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
13	w_wip_lo	14	Red		Wiper LOW	Front wiper motor F9 pin 2 (cowl, driver side) pin 2	800	ST185-EngineRoom-C	WIP

OEM IE1 pin 14 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
14	w_rtr_r2_4	12	Red		LH retract motor 4	LH retract motor feed - 12 AWG to 16 AWG pigtail, at R2	4380	ST185-EngineRoom-C	RTM

OEM IE1 pin 15 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
15	w_rtr_r2_2	16	Red		LH retract motor 2	LH retract motor R2 pin 2 pin 2	4380	ST185-EngineRoom-C	RTM

OEM IE1 pin 16 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
16	w_rtr_open	18	Yellow		Retract open limit	Retract open-limit join - R2 + R3 pin 1 to IE1 pin 16	3680	ST185-EngineRoom-C	RTM

OEM IE1 pin 17 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
17	w_pdb_am2	10	Red		AMI AM2 30 A fuse	B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse...	1580	ST185-EngineRoom-C	AM2

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS (part 6 ...

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	43 OF 51	432 x 279 mm	DG	2026-10-29

HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS (part 7 of 9)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

OEM IE1 pin 18 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
18	w_hl_low	14	Red	□□	Low-beam return	Low-beam return join - H1 + H2 pin 2 to IE1 pin 18	3680	ST185-EngineRoom-C	HLO

OEM IE1 pin 20 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
20	w_wip_hi	14	Red	■□	Wiper HIGH	Front wiper motor F9 pin 1 (cowl, driver side) pin 1	800	ST185-EngineRoom-C	WIP

OEM IE1 pin 3 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
3	w_pmu_rtr	10	Red	■■■	AMI RTR 30 A fuse	B1 POWER box with the bus bar in it - custom fuse + relay box, glove box outboard (cad/fuse...	1580	ST185-EngineRoom-C	RTR

OEM IE1 pin 6 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
6	w_wip_park	14	Yellow	■□	Wiper park sw	Front wiper motor F9 pin 3 (cowl, driver side) pin 3	800	ST185-EngineRoom-C	WIP

OEM IE1 pin 7 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
7	w_rtr_r3_4	12	Red	■	RH retract motor 4	RH retract motor feed - 12 AWG to 16 AWG pigtail, at R3	2980	ST185-EngineRoom-C	RTM

OEM IE1 pin 8 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
8	w_rtr_r3_2	16	Red	■	RH retract motor 2	RH retract motor R3 pin 2 pin 2	2980	ST185-EngineRoom-C	RTM

OEM IE1 pin 9 (left kick, cowl side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
9	w_rtr_close	18	Yellow	■	Retract close limit	Retract close-limit join - R2 + R3 pin 3 to IE1 pin 9	3680	ST185-EngineRoom-C	RTM

OEM IG2 pin 16 (under the ECU, cowl side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
16	w_alt_l_c	20	White	■□	Alt L charge lamp	Alternator G2 - round 3-pin plug (Toyota 90980-10341; 1993 EWD p.74-75 G2 (B) pinout) pin 2	3140	ST185-EngineRoom-C	ALTL

OEM J/B No.1 1H pin 4 (left kick)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1H-4	w_turn_f7	18	Red	■■	LH front turn +	LH front turn / clearance lamp F7 pin 3 pin 3	4380	ST185-EngineRoom-C	TURN

OEM J/B No.1 1H pin 5 (left kick)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1H-5	w_wip_b	14	Red	■□	Wiper +B	Front wiper motor F9 pin 4 (cowl, driver side) pin 4	800	ST185-EngineRoom-C	WIP

OEM J/B No.1 1H pin 6 (left kick)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1H-6	w_clr	16	Red	□	Clearance lamp feed	Clearance feed split - F7 LH + F8 RH	3680	ST185-EngineRoom-C	CLR

REVISIONS

REV	DATE	BY	DESCRIPTION
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K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
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HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS (part 7 ...

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	44 OF 51	432 x 279 mm	DG	2026-10-29

HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS (part 8 of 9)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

OEM J/B No.1 1H pin 8 (left kick)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1H-8	w_turn_f8	18	Red		RH front turn +	RH front turn / clearance lamp F8 pin 3 pin 3	2980	ST185-EngineRoom-C	TURN

OEM J/B No.1 1H pin 2 (left kick)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_pmu_dome	14	Red		AMI BUS1 50 A fuse -> LOGIC DM 20 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	1580	ST185-EngineRoom-C	DM

LH retract motor R2 pin 1

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_rtr_open_r2	18	Yellow		Retract open limit	Retract open-limit join - R2 + R3 pin 1 to IE1 pin 16	900	ST185-EngineRoom-C	RTM

LH retract motor R2 pin 2

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_rtr_r2_2	16	Red		LH retract motor 2	OEM IE1 pin 15 (left kick, cowl side) pin 15	4380	ST185-EngineRoom-C	RTM

LH retract motor R2 pin 3

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
3	w_rtr_close_r2	18	Yellow		Retract close limit	Retract close-limit join - R2 + R3 pin 3 to IE1 pin 9	900	ST185-EngineRoom-C	RTM

LH retract motor R2 pin 4

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
4	w_rtr_r2_4_p	16	Red		LH retract motor 4	LH retract motor feed - 12 AWG to 16 AWG pigtail, at R2	200	ST185-EngineRoom-C	RTM

LH retract motor R2 pin 5

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
5	w_rtr_gnd_r2	16	Black		LH retract ground	LH retract motor ground - 16 AWG pigtail to 12 AWG, at R2	200	ST185-EngineRoom-C	RTM

RH retract motor R3 pin 1

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_rtr_open_r3	18	Yellow		Retract open limit	Retract open-limit join - R2 + R3 pin 1 to IE1 pin 16	900	ST185-EngineRoom-C	RTM

RH retract motor R3 pin 2

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_rtr_r3_2	16	Red		RH retract motor 2	OEM IE1 pin 8 (left kick, cowl side) pin 8	2980	ST185-EngineRoom-C	RTM

RH retract motor R3 pin 3

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
3	w_rtr_close_r3	18	Yellow		Retract close limit	Retract close-limit join - R2 + R3 pin 3 to IE1 pin 9	900	ST185-EngineRoom-C	RTM

RH retract motor R3 pin 4

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
4	w_rtr_r3_4_p	16	Red		RH retract motor 4	RH retract motor feed - 12 AWG to 16 AWG pigtail, at R3	200	ST185-EngineRoom-C	RTM

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
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HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS (part 8 ...

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	45 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS (part 9 of 9)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

RH retract motor R3 pin 5

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
5	w_rtr_gnd_r3	16	Black		RH retract ground	RH retract motor ground - 16 AWG pigtail to 12 AWG, at R3	200	ST185-EngineRoom-C	RTM

OEM W-L lead at old R/B2 pin 2 (left kick)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_rb2_wl	14	Red		AMI BUS1 50 A fuse -> LOGIC WL 15 A fuse	B2 LOGIC box - custom fuse + relay box, glove box inboard (cad/fusebox rev 5, DECISIONS ...	1580	ST185-EngineRoom-C	WL

OEM R/B No.4 10 A A/C fuse pin 2 (RH kick), V-R out

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_acp_1	18	Yellow		A/C press sw feed	A/C pressure switch A5 pin 1 pin 1	1600	ST185-EngineRoom-C	ACPS

OEM washer W1 pin 1 cowl lead (firewall, cabin side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_wash_neg	14	Red		Washer switch side	Washer motor W1 pin 1 (RH front corner, low) pin 1	1300	ST185-EngineRoom-C	WSH

Washer motor W1 pin 1 (RH front corner, low)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_wash_neg	14	Red		Washer switch side	OEM washer W1 pin 1 cowl lead (firewall, cabin side) pin 1	1300	ST185-EngineRoom-C	WSH

OEM washer W1 pin 2 cowl lead (firewall, cabin side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_wash_pos	14	Red		Washer motor +12 V	Washer motor W1 pin 2 (RH front corner, low) pin 2	1300	ST185-EngineRoom-C	WSH

Washer motor W1 pin 2 (RH front corner, low)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
2	w_wash_pos	14	Red		Washer motor +12 V	OEM washer W1 pin 2 cowl lead (firewall, cabin side) pin 2	1300	ST185-EngineRoom-C	WSH

REVISIONS

REV	DATE	BY	DESCRIPTION
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HARNESS C - GLOVE BOX OUT THROUGH THE RH FENDER: FANS, HORNS, LIGHTS, RETRACTS, A/C, EPS (part 9 ...

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	46 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

HEAVY DC - BATTERY, KILL SWITCH, RADLOK PAIR, STARTER, ALTERNATOR, FUEL PUMP RELAY BLOCK (part 1 of 2)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

ALT-PLUG - Alternator G2 - round 3-pin plug (Toyota 90980-10341; 1993 EWD p.74-75 G2 (B) pinout)

housing 90980-10341 | contacts TBD | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
3	w_ig_ks_out	16	Red		Alt IG excite	Kill switch - Longacre 52-45780, 4 terminal (trunk, passenger rear corner, Route 1) pin ALT2	5110	ST185-HeavyDC	ALTIG

BATT-RELAY-BLOCK - Battery relay block - fuel pump relay + its fuse, trunk, passenger side between the rear wall and the battery (cad/fusebox rev 5, DECISIONS R75, Z4)

housing ST185-FB-BATT rev 5 | contacts TBD | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
BATT	w_fp_feed	10	White		Battery relay box +	Kill switch - Longacre 52-45780, 4 terminal (trunk, passenger rear corner, Route 1) pin MAIN2	200	ST185-HeavyDC	FP
FP	w_fp_30	10	Red		FP 40 A fuse (battery relay block)	B3-K1 Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ...	200	ST185-HeavyDC	FP

JUMP-POST - Engine-bay jump post

housing TBD | contacts TBD | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_jump	2	Red		Jump post feed	Starter B+ stud, lug 3 (Route 1 point 2)	410	ST185-HeavyDC	BAT

K-FP - B3-K1 Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) switched 12 V - HCR 150 coil is a 2-pole Superseal plug (TE terminals 1/2, not marked 85/86), resistor-su...

housing V23132-A2001-B200 | contacts TBD | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
30	w_fp_30	10	Red		FP 40 A fuse (battery relay block)	Battery relay block - fuel pump relay + its fuse, trunk, passenger side between the rear wall an...	200	ST185-HeavyDC	FP
87a					spare, not used				

KILL-SW - Kill switch - Longacre 52-45780, 4 terminal (trunk, passenger rear corner, Route 1)

housing 52-45780 | contacts TBD | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
ALT1	w_ig_ks_in	16	Red		Alt IG to kill sw	OEM IG2 pin 15 (under the ECU, cowl side) pin 15	3130	ST185-HeavyDC	ALTIG
ALT2	w_ig_ks_out	16	Red		Alt IG excite	Alternator G2 - round 3-pin plug (Toyota 90980-10341; 1993 EWD p.74-75 G2 (B) pinout) pin 3	5110	ST185-HeavyDC	ALTIG
MAIN1	w_trunk_pos	1/0	Red		battery +	Battery positive terminal post, trunk RH rear (Route 1 point 1)	200	ST185-HeavyDC	BAT
MAIN2	w_fp_feed	10	White		Battery relay box +	Battery relay block - fuel pump relay + its fuse, trunk, passenger side between the rear wall an...	200	ST185-HeavyDC	FP
MAIN2	w_ks_h1	1/0	Red		battery +	POWER box bus zone (rev 7, glove box) pin BAT	2650	ST185-HeavyDC	BAT

POWER-STUDS - POWER box bus zone (rev 7, glove box)

housing ST185-FB-POWER rev 7 BUS ZONE | contacts TBD | pins per ST185 fuse-box design (cad/fusebox/rev7/wiring_plan.json, R204)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
ALT	w_h3_radlok_pos	1/0	Red		Starter feed (H3)	RADLOK + firewall pin 1	600	ST185-HeavyDC	BAT
BAT	w_ks_h1	1/0	Red		battery +	Kill switch - Longacre 52-45780, 4 terminal (trunk, passenger rear corner, Route 1) pin MAIN2	2650	ST185-HeavyDC	BAT
NEG	w_h6a_radlok_neg	1/0	Black		Engine ground (H6a)	RADLOK - firewall pin 1	600	ST185-HeavyDC	BATN
NEG	w_trunk_neg	1/0	Black		Battery - main	Battery negative terminal post, trunk RH rear (Route 1 point 1)	2650	ST185-HeavyDC	BATN

RADLOK-NEG-CAB - RADLOK - firewall

housing RL00801-50BK | contacts INTEGRAL | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_h6a_radlok_neg	1/0	Black		Engine ground (H6a)	POWER box bus zone (rev 7, glove box) pin NEG	600	ST185-HeavyDC	BATN
1	w_rln	1/0	Black		Engine ground (H6a)	RADLOK - engine pin 1	2760	ST185-HeavyDC	BATN

RADLOK-NEG-ENG - RADLOK - engine

housing RL00801-50BK | contacts INTEGRAL | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_rln_blk	1/0	Black		Engine ground (H6a)	Engine block ground stud, lug 1 (Route 1 point 5)	890	ST185-HeavyDC	BATN
1	w_rln	1/0	Black		Engine ground (H6a)	RADLOK - firewall pin 1	2760	ST185-HeavyDC	BATN

RADLOK-POS-CAB - RADLOK + firewall

housing RL00801-50RE | contacts INTEGRAL | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_h3_radlok_pos	1/0	Red		Starter feed (H3)	POWER box bus zone (rev 7, glove box) pin ALT	600	ST185-HeavyDC	BAT
1	w_rl_pos	1/0	Red		Starter feed (H3)	RADLOK + engine pin 1	2760	ST185-HeavyDC	BAT

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

HEAVY DC - BATTERY, KILL SWITCH, RADLOK PAIR, STARTER, ALTERNATOR, FUEL PUMP RELAY BLOCK (part 1 o...

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	47 OF 51	432 x 279 mm	DG	2026-10-29

HEAVY DC - BATTERY, KILL SWITCH, RADLOK PAIR, STARTER, ALTERNATOR, FUEL PUMP RELAY BLOCK (part 2 of 2)

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

RADLOK-POS-ENG - RADLOK + engine

housing RL00801-50RE | contacts INTEGRAL | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_rlp_str	1/0	Red		Starter feed (H3)	Starter B+ stud, lug 1 (Route 1 point 2)	1700	ST185-HeavyDC	BAT
1	w_rl_pos	1/0	Red		Starter feed (H3)	RADLOK + firewall pin 1	2760	ST185-HeavyDC	BAT

BATT-POS - Battery positive terminal post, trunk RH rear (Route 1 point 1)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_trunk_pos	1/0	Red		battery +	Kill switch - Longacre 52-45780, 4 terminal (trunk, passenger rear corner, Route 1) pin MAIN1	200	ST185-HeavyDC	BAT

T-ALT-B - Alternator B+ stud, rear RH of engine (Route 1 point 3)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_alt	2	Red		Alternator B+	Starter B+ stud, lug 2 (Route 1 point 2)	2000	ST185-HeavyDC	BAT

T-ENG-BLOCK - Engine block ground stud, lug 1 (Route 1 point 5)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_rln_blk	1/0	Black		Engine ground (H6a)	RADLOK - engine pin 1	890	ST185-HeavyDC	BATN

T-STARTER-B - Starter B+ stud, lug 1 (Route 1 point 2)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_rlp_str	1/0	Red		Starter feed (H3)	RADLOK + engine pin 1	1700	ST185-HeavyDC	BAT

T-STARTER-B-2 - Starter B+ stud, lug 2 (Route 1 point 2)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_alt	2	Red		Alternator B+	Alternator B+ stud, rear RH of engine (Route 1 point 3)	2000	ST185-HeavyDC	BAT

T-STARTER-B-3 - Starter B+ stud, lug 3 (Route 1 point 2)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_jump	2	Red		Jump post feed	Engine-bay jump post pin 1	410	ST185-HeavyDC	BAT

T-TRUNK-NEG - Battery negative terminal post, trunk RH rear (Route 1 point 1)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_trunk_neg	1/0	Black		Battery - main	POWER box bus zone (rev 7, glove box) pin NEG	2650	ST185-HeavyDC	BATN

OEM IG2 pin 15 (under the ECU, cowl side)

factory lead tap (EWD locator in the signal)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
15	w_ig_ks_in	16	Red		Alt IG to kill sw	Kill switch - Longacre 52-45780, 4 terminal (trunk, passenger rear corner, Route 1) pin ALT1	3130	ST185-HeavyDC	ALTIG

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
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K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
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ST185-harness - every wiring harness of the car, one drawing set

HEAVY DC - BATTERY, KILL SWITCH, RADLOK PAIR, STARTER, ALTERNATOR, FUEL PUMP RELAY BLOCK (part 2 o...

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	48 OF 51	432 x 279 mm	DG	2026-10-29

REAR FUEL - FUEL PUMP AND LEVEL SENDER

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

FUEL-LEVEL - Fuel Level Sender - DT 2-way (Dan rev 2, 2026-10-10 - no OEM connector)

housing DT04-2P + W2P | contacts 0460-202-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_fl_sig_r	20	White		Fuel level signal	IX-FUEL inline to fuel sender (trunk LH) pin 1	500	ST185-RearFuel	FLVL
2	w45_r	20	Blue		Fuel level ground	IX-FUEL inline to fuel sender (trunk LH) pin 2	500	ST185-RearFuel	FLVL

FUEL-PUMP - Fuel Pump

housing TBD | contacts TBD | pins per TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_fpump_r	10	Red		Fuel pump +12 V	B3-K1 Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ...	1500	ST185-RearFuel	FP
2	w_fpg	12	Black		Fuel pump ground	Ground stud, trunk LH side (EWD ID TBD, M6 assumed)	200	ST185-RearFuel	FP

IX-FUEL-S - IX-FUEL inline to fuel sender (trunk LH)

housing DT06-2S + W2S | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	w_fl_sig_r	20	White		Fuel level signal	Fuel Level Sender - DT 2-way (Dan rev 2, 2026-10-10 - no OEM connector) pin 1	500	ST185-RearFuel	FLVL
2	w45_r	20	Blue		Fuel level ground	Fuel Level Sender - DT 2-way (Dan rev 2, 2026-10-10 - no OEM connector) pin 2	500	ST185-RearFuel	FLVL

K-FP - B3-K1 Fuel Pump Relay, TE HCR150 (seated in the battery relay block K1) - ACTIVE-LOW - ECU sinks coil- (85); coil+ (86) switched 12 V - HCR 150 coil is a 2-pole Superseal plug (TE terminals 1/2, not marked 85/86), resistor-su...

housing V23132-A2001-B200 | contacts TBD | pins per ISO 7588 relay terminal numbers

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
87	w_fpump_r	10	Red		Fuel pump +12 V	Fuel Pump pin 1	1500	ST185-RearFuel	FP

T-TRUNK-FP-GND - Ground stud, trunk LH side (EWD ID TBD, M6 assumed)

terminal TBD

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	w_fpg	12	Black		Fuel pump ground	Fuel Pump pin 2	200	ST185-RearFuel	FP

REVISIONS

REV	DATE	BY	DESCRIPTION
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ST185-harness - every wiring harness of the car, one drawing set

REAR FUEL - FUEL PUMP AND LEVEL SENDER

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	49 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

FRONT WHEEL SPEED - FL / FR SENSORS TO THE FRONT VRC

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

SP-WS-F-SHIELD - Sensor screen join - FL + FR screens to the VRC IN pin 3 drain

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_fl_sh		Shield		FL sensor shield	cab_fl screen (floats at the device end)	200	ST185-WheelSpeed-Front	WSS_FL
	cab_fr_sh		Shield		FR sensor shield	cab_fr screen (floats at the device end)	200	ST185-WheelSpeed-Front	WSS_FR
	w_ws_f_shield	20	Black		Front sensor shields	VRC front IN plug pin 3	200	ST185-WheelSpeed-Front	VRC_F_DRAIN

IX-WS-F-P - IX-WS-F

housing DT04-6P + W6P | contacts 0460-202-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	cab_fout_5v	20	Red		VRC-F +5V	VRC front OUT plug pin 1	200	ST185-WheelSpeed-Front	VRC_F_5V
2	cab_fout_gnd	20	Yellow		VRC-F GND	VRC front OUT plug pin 2	200	ST185-WheelSpeed-Front	VRC_F_GND
3	cab_fout_l	20	Green		FL conditioned out	VRC front OUT plug pin 4	200	ST185-WheelSpeed-Front	WSS_FL
4	cab_fout_r	20	Blue		FR conditioned out	VRC front OUT plug pin 5	200	ST185-WheelSpeed-Front	WSS_FR
5	cab_fout_sh		Shield		WS-F screen drain	VRC front OUT plug pin 3	200	ST185-WheelSpeed-Front	VRC_F_DRAIN
6					spare, sealing plug				

VRC-F-IN - VRC front IN plug

housing LP-16-J05PE-02-001 | contacts solder cup (LP-16, S16) | pins per CNLINKO LP-16 5-pin, pins 1-5 (Dan 2026-10-19)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	cab_fl_p	20	Blue		FL wheel speed +	ABS Wheel Speed FL pin 1	1670	ST185-WheelSpeed-Front	WSS_FL
2	cab_fl_n	20	White		FL wheel speed -	ABS Wheel Speed FL pin 2	1670	ST185-WheelSpeed-Front	WSS_FL
3	w_ws_f_shield	20	Black		Front sensor shields	Sensor screen join - FL + FR screens to the VRC IN pin 3 drain	200	ST185-WheelSpeed-Front	VRC_F_DRAIN
4	cab_fr_p	20	Blue		FR wheel speed +	ABS Wheel Speed FR pin 1	1560	ST185-WheelSpeed-Front	WSS_FR
5	cab_fr_n	20	White		FR wheel speed -	ABS Wheel Speed FR pin 2	1560	ST185-WheelSpeed-Front	WSS_FR

VRC-F-OUT - VRC front OUT plug

housing LP-16-J05PE-02-001 | contacts solder cup (LP-16, S16) | pins per CNLINKO LP-16 5-pin, pins 1-5 (Dan 2026-10-19)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	cab_fout_5v	20	Red		VRC-F +5V	IX-WS-F pin 1	200	ST185-WheelSpeed-Front	VRC_F_5V
2	cab_fout_gnd	20	Yellow		VRC-F GND	IX-WS-F pin 2	200	ST185-WheelSpeed-Front	VRC_F_GND
3	cab_fout_sh		Shield		WS-F screen drain	IX-WS-F pin 5	200	ST185-WheelSpeed-Front	VRC_F_DRAIN
4	cab_fout_l	20	Green		FL conditioned out	IX-WS-F pin 3	200	ST185-WheelSpeed-Front	WSS_FL
5	cab_fout_r	20	Blue		FR conditioned out	IX-WS-F pin 4	200	ST185-WheelSpeed-Front	WSS_FR

WSS-FL - ABS Wheel Speed FL

housing DT06-2S-CE05 + W2S | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	cab_fl_p	20	Blue		FL wheel speed +	VRC front IN plug pin 1	1670	ST185-WheelSpeed-Front	WSS_FL
2	cab_fl_n	20	White		FL wheel speed -	VRC front IN plug pin 2	1670	ST185-WheelSpeed-Front	WSS_FL

WSS-FR - ABS Wheel Speed FR

housing DT06-2S-CE05 + W2S | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	cab_fr_p	20	Blue		FR wheel speed +	VRC front IN plug pin 4	1560	ST185-WheelSpeed-Front	WSS_FR
2	cab_fr_n	20	White		FR wheel speed -	VRC front IN plug pin 5	1560	ST185-WheelSpeed-Front	WSS_FR

CAB-FL cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_fl_sh		Shield		FL sensor shield	Sensor screen join - FL + FR screens to the VRC IN pin 3 drain	200	ST185-WheelSpeed-Front	WSS_FL

CAB-FR cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_fr_sh		Shield		FR sensor shield	Sensor screen join - FL + FR screens to the VRC IN pin 3 drain	200	ST185-WheelSpeed-Front	WSS_FR

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
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K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

FRONT WHEEL SPEED - FL / FR SENSORS TO THE FRONT VRC

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	50 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale

REAR WHEEL SPEED - RL / RR SENSORS TO THE REAR VRC

Length = layout path + end allowance (estimate until measured). Built with = the harness that builds that wire (a wire passing through is built with its owner). A pin with no wire takes a sealing plug.

SP-WS-R-SHIELD - Sensor screen join - RL + RR screens to the VRC IN pin 3 drain

splice: solder splice

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_rl_sh		Shield		RL sensor shield	cab_rl screen (floats at the device end)	200	ST185-WheelSpeed-Rear	WSS_RL
	cab_rr_sh		Shield		RR sensor shield	cab_rr screen (floats at the device end)	200	ST185-WheelSpeed-Rear	WSS_RR
	w_ws_r_shield	20	Black		Rear sensor shields	VRC rear IN plug pin 3	200	ST185-WheelSpeed-Rear	VRC_R_DRAIN

IX-WS-R-P - IX-WS-R

housing DT04-6P + W6P | contacts 0460-202-1631 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	cab_rout_5v	20	Red		VRC-R +5V	VRC rear OUT plug pin 1	200	ST185-WheelSpeed-Rear	VRC_R_5V
2	cab_rout_gnd	20	Yellow		VRC-R GND	VRC rear OUT plug pin 2	200	ST185-WheelSpeed-Rear	VRC_R_GND
3	cab_rout_l	20	Green		Rear left wheel spd	VRC rear OUT plug pin 4	200	ST185-WheelSpeed-Rear	WSS_RL
4	cab_rout_r	20	Blue		Rear right wheel spd	VRC rear OUT plug pin 5	200	ST185-WheelSpeed-Rear	WSS_RR
5	cab_rout_sh		Shield		WS-R screen drain	VRC rear OUT plug pin 3	200	ST185-WheelSpeed-Rear	VRC_R_DRAIN
6					spare, sealing plug				

VRC-R-IN - VRC rear IN plug

housing LP-16-J05PE-02-001 | contacts solder cup (LP-16, S16) | pins per CNLINKO LP-16 5-pin, pins 1-5 (Dan 2026-10-19)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	cab_rl_p	20	Blue		RL wheel speed +	ABS Wheel Speed RL pin 1	3570	ST185-WheelSpeed-Rear	WSS_RL
2	cab_rl_n	20	White		RL wheel speed -	ABS Wheel Speed RL pin 2	3570	ST185-WheelSpeed-Rear	WSS_RL
3	w_ws_r_shield	20	Black		Rear sensor shields	Sensor screen join - RL + RR screens to the VRC IN pin 3 drain	200	ST185-WheelSpeed-Rear	VRC_R_DRAIN
4	cab_rr_p	20	Blue		RR wheel speed +	ABS Wheel Speed RR pin 1	4470	ST185-WheelSpeed-Rear	WSS_RR
5	cab_rr_n	20	White		RR wheel speed -	ABS Wheel Speed RR pin 2	4470	ST185-WheelSpeed-Rear	WSS_RR

VRC-R-OUT - VRC rear OUT plug

housing LP-16-J05PE-02-001 | contacts solder cup (LP-16, S16) | pins per CNLINKO LP-16 5-pin, pins 1-5 (Dan 2026-10-19)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	cab_rout_5v	20	Red		VRC-R +5V	IX-WS-R pin 1	200	ST185-WheelSpeed-Rear	VRC_R_5V
2	cab_rout_gnd	20	Yellow		VRC-R GND	IX-WS-R pin 2	200	ST185-WheelSpeed-Rear	VRC_R_GND
3	cab_rout_sh		Shield		WS-R screen drain	IX-WS-R pin 5	200	ST185-WheelSpeed-Rear	VRC_R_DRAIN
4	cab_rout_l	20	Green		Rear left wheel spd	IX-WS-R pin 3	200	ST185-WheelSpeed-Rear	WSS_RL
5	cab_rout_r	20	Blue		Rear right wheel spd	IX-WS-R pin 4	200	ST185-WheelSpeed-Rear	WSS_RR

WSS-RL - ABS Wheel Speed RL

housing DT06-2S-CE05 + W2S | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	cab_rl_p	20	Blue		RL wheel speed +	VRC rear IN plug pin 1	3570	ST185-WheelSpeed-Rear	WSS_RL
2	cab_rl_n	20	White		RL wheel speed -	VRC rear IN plug pin 2	3570	ST185-WheelSpeed-Rear	WSS_RL

WSS-RR - ABS Wheel Speed RR

housing DT06-2S-CE05 + W2S | contacts 0462-201-16141 | pins per Deutsch DT / DTM cavity numbers (moulded on the housing)

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
1	cab_rr_p	20	Blue		RR wheel speed +	VRC rear IN plug pin 4	4470	ST185-WheelSpeed-Rear	WSS_RR
2	cab_rr_n	20	White		RR wheel speed -	VRC rear IN plug pin 5	4470	ST185-WheelSpeed-Rear	WSS_RR

CAB-RL cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_rl_sh		Shield		RL sensor shield	Sensor screen join - RL + RR screens to the VRC IN pin 3 drain	200	ST185-WheelSpeed-Rear	WSS_RL

CAB-RR cable screen (floats at the device end)

cable screen

PIN	WIRE	AWG	COLOUR	BANDS	SIGNAL	OTHER END	LEN MM	BUILT WITH	CIRCUIT
	cab_rr_sh		Shield		RR sensor shield	Sensor screen join - RL + RR screens to the VRC IN pin 3 drain	200	ST185-WheelSpeed-Rear	WSS_RR

REVISIONS

REV	DATE	BY	DESCRIPTION
I	2026-10-27	DG	Crank and cam each run in one shielded Raychem cable (2-core / 3-core). The CAN-Lambda bra...
J	2026-10-28	DG	ECU engine harness layout tree built from Route 16 (Dan's device spots and branches, DECISIO...
K	2026-10-29	DG	Ignition coil ground goes straight to the glove-box POWER box NEG bar through bulkhead cavity ...
L	2026-10-29	DG	Every connector part-way along a line or run breaks out at least 6 in (152 mm) off it; the last conn...

ST185-harness - every wiring harness of the car, one drawing set

REAR WHEEL SPEED - RL / RR SENSORS TO THE REAR VRC

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-PIN	L	51 OF 51	432 x 279 mm	DG	2026-10-29

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale